Socio-Cognitive Engineering · SCE 2026

Meaningful Connections

Kinnect, a social connection-maker for care home residents with dementia

Group 04

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Chapter 1

Foundation

1.1 Operational Demands

1.1.1 Situated Activities

SA-01 Moving into the care home

Evidence basis

Provisional; based on published literature and group reasoning.

- Sury et al. (2013): moving into residential care is a period of loss and disorientation for people with dementia.
- Lai & Arthur (2003): wandering and exit-seeking in people with dementia.
- Victor (2012): loneliness is widespread in residential care homes.

Goal

Settle into the home, learn the layout and daily rhythm, recognise a few familiar faces, and regain a sense of personal control.

Performing Actor(s)

- ST-01: Resident - moves into the home
- ST-02: Fellow resident - may welcome the newcomer
- ST-03: Relative - arranges the move and shares residents history
- ST-04: Care worker - receives the newcomer and shows them around
- ST-05: Activity coordinator - records the life story, assigns dining seats, and introduces residents
- ST-06: Medical and therapy staff - assess cognitive and mobility needs at admission

Trigger

Living at home becomes unsustainable after a fall, a hospital admission, or caregiver exhaustion, and a room in the care home opens up.

Constituent Tasks

1. Pack belongings and say goodbye to home
2. Arrive at the care home and unpack
3. Learn the way between the bedroom, bathroom, dining room, and communal lounge
4. Learn the daily schedule and house routines
5. Meet care staff and fellow residents (SA-03)
6. Share residents history and personal preferences
7. Find a regular place at table and a comfortable daily rhythm

Context

Admissions are often unplanned and decided by relatives or clinicians, leaving the resident little say over timing or room choice. The newcomer arrives knowing nobody. Staff rotate across shifts, and many fellow residents live with more advanced dementia and speak little. Corridors and doors look alike, exits are locked by keypad-codes, and the unfamiliar layout creates confusion. Meal times and house routines are fixed by the institution.

Frequency and duration

Once per admission; settling in takes weeks to months.

Tools and artifacts

Bedroom, name plate on the door, intake form, life-story book, care plan, house rules, weekly activity programme, coded doors.

Existing workarounds

- Bringing familiar furniture, ornaments, and family photographs from home
- Visiting the care home before admission day
- Recording a life story together with family members
- Assigning a primary care worker as a consistent point of contact
- Frequent family visits during the first days
- Staff-led introductions to table neighbours (SA-03)
- Maintaining familiar habits, such as reading the morning paper or having coffee at a set hour

Observed breakdowns

- The resident does not understand why they are in the home, or believes the stay is only for a few days

- The resident asks to go home and searches for an exit
- The resident cannot find their bedroom, the bathroom, or the dining room
- Grief over the house, spouse, pets, and neighbours left behind
- Resistance to care routines and resentment toward family or staff
- Having nobody to talk to during the day (PS-01)
- Meeting too many unfamiliar faces at once to remember any of them (PS-01)
- Being seated at meals among residents who rarely speak
- Retreating to the bedroom and skipping communal meals or activities (SA-02)
- The residents history is never collected, or stays in a file that floor staff never read
- Family visits gradually become less frequent
- Social withdrawal goes unnoticed amid routine physical care

SA-02 Waiting for a reason to leave the room

Evidence basis

Provisional; based on published literature and group reasoning.

- Jansen et al. (2017): nursing-home residents spend large portions of the waking day isolated in their private bedrooms.
- Victor (2012): loneliness in care homes is widespread.
- Clare (2003): people aware of cognitive decline often withdraw socially to protect themselves from embarrassment.

Goal

Find a worthwhile, comfortable reason to leave the bedroom and spend time around others.

Performing Actor(s)

- ST-01: Resident - stays in the bedroom and decides whether to come out
- ST-03: Relative - visits occasionally
- ST-04: Care worker - checks in briefly and brings care or meals
- ST-05: Activity coordinator - invites the resident to communal activities

Trigger

An empty schedule and no particular person to meet in the common areas, this becomes a daily habit over weeks or months.

Constituent Tasks

1. Wake, wash, and dress
2. Pass the hours watching television, looking out the window, or dozing
3. Listen for a knock at the door, a meal delivery, or a visitor
4. Weigh an invitation from staff to join the lounge or an activity
5. Step out to the common areas, or remain in the room

Context

The resident has lived in the home for months or years. Their private bedroom feels safe, quiet, and within their control, whereas the communal lounge is noisy and full of strangers. Staff check in briefly between care tasks. Earlier acquaintances in the home may have passed away, moved unit, or faded from memory.

Frequency and duration

Daily, often spanning most waking hours.

Tools and artifacts

Bedroom, armchair, television, window, call bell, meal tray, printed activity programme.

Existing workarounds

- Serving meals in the communal dining room
- Staff repeating invitations gently rather than asking only once
- Offering a familiar group activity, such as music or bingo
- Giving the resident a small, valued role, such as folding napkins or tending indoor plants
- Visits and outings with family members

Observed breakdowns

- No compelling reason arises, and the entire day passes alone in the bedroom
- Staff invitations focus on scheduled group events rather than meeting a compatible peer
- The resident declines repeated invitations until staff stop asking (PS-02)
- Daytime napping disrupts nighttime sleep
- Mood, confidence, and physical mobility decline gradually without triggering an alert (PS-02)
- The resident forgets the names of the few peers they once knew

SA-03 Greeting someone new

Evidence basis

Provisional; based on published literature and group reasoning.

- Orange et al. (1996): conversational breakdowns become more frequent in Alzheimer's disease, and repair succeeds less often.
- Clare (2003): awareness of cognitive slips leads residents to guard against losing face.

Goal

Exchange names and a brief, pleasant conversation that leaves both residents willing to speak again.

Performing Actor(s)

- ST-01: Resident - greets or responds to a peer
- ST-02: Fellow resident - greets or responds to ST-01
- ST-03: Relative - occasionally introduces residents during a visit
- ST-04: Care worker - introduces residents in passing
- ST-05: Activity coordinator - arranges seating and introduces compatible peers

Trigger

Encountering a newcomer (SA-01), sitting beside a new neighbour at meals, passing a peer in the lounge, or being introduced by staff.

Constituent Tasks

1. Notice the other resident and decide to speak
2. Offer a greeting
3. Exchange names
4. Mention a personal detail, such as one's hometown, former trade, or family
5. Ask a question, listen, and reply
6. Discover a shared topic or memory
7. Part on warm terms ("See you at dinner")
8. Recognise the person next time and recall their name

Context

Seating in the dining room and lounge is largely arranged by staff. Short-term memory loss erodes names, faces, and even one's own recent stories, while hearing or vision loss makes cues easy to miss. Making small talk with a stranger takes effort, and an awkward silence feels like a personal failure. Especially in shared rooms where others can overhear.

Frequency and duration

Daily; lasting from a few seconds to several minutes.

Tools and artifacts

Communal lounge, dining tables, seating plan, door name plates, life-story books, photographs.

Existing workarounds

- Staff introduce and re-introduce residents by name, adding one shared detail to spark talk
- Staff connect residents who come from the same region, trade, or congregation
- Seating a newcomer beside an outgoing resident at meals
- Door name plates or table place cards
- Relying on immediate, low-memory topics such as the weather, the meal, or the view outside
- Using habitual courtesies ("Good morning", a nod, or a handshake)
- Looking through a photo album or life-story book together

Observed breakdowns

- Both residents feel too shy to speak first
- Greeting a familiar neighbour as a stranger, or mistaking a stranger for someone else
- Forgetting the other person's name and feeling too embarrassed to ask again
- Avoiding a friendly acquaintance specifically to hide having forgotten their name (PS-03)
- Losing the thread mid-sentence and falling silent
- Running out of things to say until one person walks away
- Repeating the same question within minutes, which the other resident notices
- Mishearing a remark in a noisy room and merely nodding along
- Striking a tone that feels overly familiar or stiffly formal
- A quiet greeting goes unheard and is mistaken for deliberate rejection
- Conversation dies as soon as the staff member who introduced them walks away
- Being ignored when trying to join an existing group conversation
- Two residents who share a hometown, trade, or interest sit yards apart for months without ever finding out
- Staff mention a private detail about a resident without asking permission first

- Introductions depend on which caregiver is on shift rather than on residents' compatibility or wishes
- A flurry of first-day introductions is never followed up (PS-01)
- A warm first conversation never leads to a second meeting (PS-03)
- Avoiding a peer after an awkward exchange

SA-04 Inviting someone to spend time together

Evidence basis

Provisional; based on published literature and group reasoning.

- de Medeiros et al. (2012): residents with dementia in long-term care form and maintain meaningful friendships.
- Victor (2012): loneliness in care homes is common.
- Woods et al. (2018): structured reminiscence supports communication, mood, and quality of life.
- Astell et al. (2010): shared touchscreen media (photographs, music, film clips) scaffolds conversation in dementia.
- Särkämö et al. (2014): singing and listening to familiar music improve mood and episodic memory.
- Gonzalez & Kirkevold (2014): garden and horticultural activities increase social engagement and reduce agitation.
- Camp & Skrajner (2004): residents with milder dementia can lead structured small-group games for peers.
- Spector et al. (2003): themed cognitive stimulation activities improve communication and quality of life.

Goal

Spend enjoyable time with a fellow resident over a shared activity, while leaving both people free to decline or finish without losing face.

Performing Actor(s)

- ST-01: Resident - extends or considers an invitation
- ST-02: Fellow resident - accepts or declines
- ST-03: Relative - occasionally includes a fellow resident during a visit
- ST-04: Care worker - suggests a shared activity, brings coffee or materials, or helps residents move to the lounge
- ST-05: Activity coordinator - pairs compatible residents and provides materials or icebreakers
- ST-06: Medical and therapy staff - note any clinical or mobility restrictions in the care plan

Trigger

Feeling lonely, noticing a peer sitting alone, discovering a shared interest during an introduction (SA-03), or receiving a gentle suggestion from staff.

Constituent Tasks

1. Choose a fellow resident to spend time with
2. Find an approachable moment to speak
3. Gauge whether the other person seems receptive
4. Suggest a shared activity and a time
5. Accept or decline the invitation
6. Remember the plan, meet in the common area, and begin
7. Bring the activity to a natural close, with the option to meet again

Context

Invitations in a care home usually happen in communal rooms where others can overhear, making a direct refusal feel rude and an acceptance feel pressured. Having a concrete activity or topic in front of both residents eases the burden of continuous conversation and makes pauses comfortable. Many shared activities benefit from light staff support to set out materials or bring two residents to the same table.

Ordinary activities that work well for two residents include:

- Sharing coffee or tea, playing cards or board games, working on a jigsaw or crossword, reading the newspaper, or helping lay tables
- Reminiscing over old photographs, music, or objects from a familiar era (Woods et al., 2018), including on a shared touchscreen (Astell et al., 2010)
- Listening to or singing familiar songs from their youth (Särkämö et al., 2014)
- Tending indoor plants, potting herbs in a raised bed, or taking a stroll in the garden (Gonzalez & Kirkevold, 2014)

	• Light word games, trivia, or sorting activities (Spector et al., 2003; Camp & Skrajner, 2004)
Frequency and duration	Uncommon without facilitation; typically 15 minutes to an hour when it occurs.
Tools and artifacts	Communal lounge, dining tables, coffee and tea service, cards, board games, jigsaws, photo albums, nostalgic music or objects, touchscreen tablet, newspapers, garden patio and raised beds.
Existing workarounds	• Staff suggest a simple activity for two residents, such as having coffee together or looking at photos • Seating the same pair near each other at meals or group events so familiarity grows • Using a scheduled group activity as a low-pressure setting to sit side by side • A visiting relative draws a neighbour into the conversation • One resident joins another who is already occupied with a puzzle, music, or plants • Staff ask two residents to help with a small shared chore, such as folding linen or watering plants
Observed breakdowns	• The initial greeting never happened or stalled awkwardly (SA-03) • Neither resident asks, out of shyness or fear of being turned down • Hearing loss or a flat facial expression is misread as disinterest • A resident agrees only to be polite because the request was made face to face in front of others (PS-04) • Neither person knows how to wind down the meeting, so it drags on until one feels trapped and tired (PS-04) • An activity proves too demanding for one resident, leaving them feeling exposed • A friendly invitation is misconstrued as romantic interest, causing avoidance • Both residents forget the agreed time before the meeting happens • No staff member is free to help bring the two residents together or set out materials • In a group of three, one resident is inadvertently left out of the talk • After a single refusal, the resident never risks asking anyone again

1.1.2 Stakeholders

ST-01 Resident with dementia

Evidence basis	Provisional; derived from SA-01 to SA-04 through group reasoning.
Type	Direct - moves into the home, converses with the System, meets peers, and extends or considers invitations
Values and motivations	• HV-06 Social connectedness (high) - genuine company, conversation, and friendship with peers • HV-02 Dignity (high) - being spoken to as an adult, without memory slips being exposed or patronised • HV-08 Truthfulness (high) - honest answers from a trusted person regarding painful topics such as why they live here or who has passed away • HV-01 Autonomy (high) - deciding for oneself whom to meet, when to socialise, and when to stay in one's room • HV-03 Privacy (high) - keeping personal secrets and controlling what facts about themselves are shared with others • HV-07 Identity (high) - being known through one's own history, former trade, interests, and words • HV-05 Well-being (high) - calm, enjoyable days with something to look forward to • HV-04 Safety (high) - feeling secure and unthreatened in shared spaces and activities
Relationship to other Stakeholders	Arrives knowing nobody in the facility. Meets and spends time with ST-02. Relatives (ST-03) help provide biographical history and consent. Welcomed and supported by ST-04 and ST-05. Cognitive and clinical needs assessed by ST-06.

ST-02 Fellow resident

Evidence basis	Provisional; derived from SA-01, SA-03, and SA-04 through group reasoning.

Type	Direct - interacts with the System, accepts or declines introductions, and talks with ST-01 (often also living with early- to middle-stage dementia)
Values and motivations	• HV-06 Social connectedness (high) - finding a compatible peer to talk and spend time with • HV-01 Autonomy (high) - freedom to decline an introduction or leave a conversation without pressure • HV-03 Privacy (high) - control over their own personal history and private room time • HV-05 Well-being (high) - calm, pleasant company without agitation or social strain • HV-02 Dignity (medium) - being treated as a peer rather than an obligation
Relationship to other Stakeholders	Encounters ST-01 in communal rooms or through introductions brokered by ST-04, ST-05, or the System. Has their own family (ST-03) and care plan.

ST-03 Relative

Evidence basis	Provisional; derived from SA-01 to SA-04 through group reasoning.
Type	Indirect - arranges the admission, contributes to the life story, agrees on how sensitive topics are handled, and provides legal representative consent where needed
Values and motivations	• HV-06 Social connectedness (high) - wanting their relative to have companionship and friends in the home • HV-08 Truthfulness (high) - expecting staff and systems not to deceive their relative • HV-05 Well-being (high) - seeing their relative settle in, stay engaged, and avoid isolation • HV-07 Identity (high) - having their relative's life history known and respected • HV-03 Privacy (low) - willing to share personal details with staff to help their relative settle, sometimes more freely than the resident themselves would choose
Relationship to other Stakeholders	Family member of ST-01. Shares biographical background with ST-05 and agrees with ST-04 on who should answer painful questions.

ST-04 Care worker

Evidence basis	Provisional; derived from SA-01 to SA-04 through group reasoning.
Type	Direct - welcomes newcomers, brings matched residents together in common areas, supports informal meetings, and answers sensitive questions
Values and motivations	• HV-04 Safety (high) - keeping residents physically and emotionally safe during daily routines • HV-08 Truthfulness (high) - giving honest, compassionate answers and keeping confidences • HV-05 Well-being (high) - seeing residents calm, comfortable, and free from agitation • HV-06 Social connectedness (medium) - helping residents get to know one another when shift time allows • HV-03 Privacy (medium) - respecting residents' private space and refusing workplace surveillance of staff
Relationship to other Stakeholders	Acts as the trusted person for sensitive questions as agreed with ST-03. Shares day-to-day observations with ST-05 about which residents get along, and follows clinical guidance from ST-06.

ST-05 Activity coordinator

Evidence basis	Provisional; derived from SA-01 to SA-04 through group reasoning.
Type	Direct - conducts life-story interviews, manages the relationship map, reviews compatibility matches, and facilitates communal activities

Values and motivations

- HV-06 Social connectedness (high) - fostering friendships and using shared activities to ease first contact
- HV-05 Well-being (high) - keeping residents engaged and content
- HV-07 Identity (high) - grounding activities and introductions in each resident's life story
- HV-03 Privacy (high) - safeguarding the biographical histories entrusted to them
- HV-01 Autonomy (medium) - ensuring residents choose which invitations and activities to accept

Relationship to other Stakeholders Records life stories with ST-01 and ST-03. Coordinates introductions and seating with ST-04. Employed by ST-07.

ST-06 Medical and therapy staff (physician, nurse, physiotherapist, psychologist)

Evidence basis Provisional; derived from SA-01 and SA-04 through group reasoning.

Type Indirect - assess cognitive status, behavioural symptoms, and mobility, and communicate clinical diagnoses

Values and motivations

- HV-04 Safety (high) - preventing physical harm, falls, and acute behavioural distress
- HV-05 Well-being (high) - supporting mood, cognitive stimulation, and healthy daily activity
- HV-08 Truthfulness (high) - ensuring clinical diagnoses and medical realities are communicated properly
- HV-03 Privacy (high) - upholding medical confidentiality

Relationship to other Stakeholders Assess ST-01 and ST-02 and advise ST-04 and ST-05 on clinical or mobility boundaries. Serve as the trusted authority for medical questions.

ST-07 Care home management

Evidence basis Provisional; derived from SA-01 and SA-04 through group reasoning.

Type Indirect - oversees admissions, staffing, facility safety, and data protection, and decides on deploying the System

Values and motivations

- HV-04 Safety (high) - institutional responsibility for resident safety and duty of care
- HV-03 Privacy (high) - compliance with GDPR and health-data protection standards
- HV-05 Well-being (medium) - overall quality of residential care and family satisfaction

Relationship to other Stakeholders Employs ST-04 and ST-05, admits ST-01, and governs facility policies and local data infrastructure.

1.1.3 Problem Scenarios

PS-01 The first weeks

Evidence basis Provisional; constructed from SA-01 and SA-03.

- Sury et al. (2013): admission to residential care involves disorientation, grief, and social withdrawal.
- Victor (2012): loneliness in care homes is common.

Narrative On her first day in the care home, a care worker introduces her to everyone at her dining table and to several staff members in quick succession. By evening she cannot recall a single name. Over the next few days, fellow residents give her space to settle in, while she sits quietly waiting for someone to approach her.

She misses her own house and assumes she will return home soon. Over the following weeks she spends more of each day alone in her bedroom, coming out mainly for meals and family visits.

Stakeholders involved

- ST-01: Resident with dementia
- ST-02: Fellow resident
- ST-03: Relative
- ST-04: Care worker

Activities involved SA-01: Moving into the care home; SA-03: Greeting someone new

Breakdown point All introductions happen at once on the first day, and nobody follows them up when she is ready to take them in.

Causes

Root causes:

- Introductions arrive all at once on arrival day, when she is overwhelmed and least able to retain new names
- Fellow residents assume she wants privacy, while she waits for someone else to initiate conversation
- Because she believes her stay is temporary, investing effort in meeting neighbours feels unnecessary

Contributing causes:

- The move exhausts her, and she still struggles to navigate the building
- She is grieving for her home, her familiar routine, and the neighbours she left behind
- She feels shy about speaking first to strangers
- Background clatter in the dining room makes hearing difficult
- The residents seated at her table speak very little
- Care staff rotate across shifts and rarely have time to repeat introductions
- She feels too embarrassed to ask for a neighbour's name a second time

Sub-problems

1. Too many introductions happen at once during the stress of arrival
2. Introductions are not repeated or scaffolded over the following weeks
3. Both the newcomer and established residents wait for the other to speak first
4. Remaining alone in the bedroom becomes the easiest way to avoid awkwardness (SA-02)

Out of scope

- Admission procedures and room allocation - ST-07
- Clinical treatment of depression or adjustment disorder - ST-06

PS-02 Staying in

Evidence basis Provisional; constructed from SA-02.

- Jansen et al. (2017): nursing-home residents spend large portions of the waking day in their bedrooms.
- Victor (2012): loneliness in care homes is common.

Narrative Since a fall several months ago, he has spent most days inside his bedroom. He walks more slowly and tires easily, and the communal lounge feels loud and full of people he barely knows. When a care worker knocks to ask whether he would like to join a group activity, he says he is tired or promises to come next time.

Over the weeks, staff knock less often to invite him. With each passing week, stepping out of his room takes a little more effort.

Stakeholders involved

- ST-01: Resident with dementia
- ST-04: Care worker
- ST-05: Activity coordinator

Activities involved SA-02: Waiting for a reason to leave the room

Breakdown point He declines general group invitations one by one until staff stop offering them.

Causes

Root causes:

- Leaving the room takes physical and social effort, and a noisy group activity does not feel worth the strain
- He has no specific acquaintance in the lounge whom he looks forward to seeing
- Repeated refusals lead busy staff to assume he prefers solitude

Contributing causes:

- He experiences joint stiffness, pain, or rapid fatigue
- He worries about falling again or being seen struggling with a rollator
- He feels anxious about reaching a bathroom in time once in the lounge
- Low mood dulls his interest in unfamiliar group games
- Hearing loss makes group banter exhausting to follow
- He dislikes feeling like a burden on busy care workers
- His bedroom is quiet, predictable, and entirely his own

- Staff under shift pressure accept a polite refusal at face value without exploring what one-to-one company might appeal to him

Sub-problems

1. Generic group invitations do not outweigh the effort of leaving the bedroom
2. He has no familiar peer waiting for him in the common areas
3. Staff gradually stop inviting him after repeated refusals
4. Gradual bedroom isolation goes unaddressed until mood and mobility have visibly declined

Out of scope

- Medical management of pain, clinical depression, and fall rehabilitation - ST-06

PS-03 Each waits for the other

Evidence basis

Provisional; constructed from SA-03.

- de Medeiros et al. (2012): residents with dementia can form lasting friendships when supported.
- Boothby et al. (2018): conversational partners systematically underestimate how much the other person enjoyed talking with them.

Narrative

A care worker introduced two residents a few weeks ago, and they spent twenty minutes talking warmly about the region where both grew up. By the following week, however, she has forgotten his name. When they pass in the corridor or sit near each other in the lounge, she offers a quick nod and looks away so she will not have to admit forgetting.

He interprets her brief nod as polite distance and does not want to intrude. Later that week, both tell their visiting relatives that there is nobody in the home they can really talk to.

Stakeholders involved

- ST-01: Resident with dementia
- ST-02: Fellow resident
- ST-03: Relative
- ST-04: Care worker

Activities involved

SA-03: Greeting someone new

Breakdown point

A promising first conversation is never followed up by a second meeting.

Causes

Root causes:

- She has forgotten his name and feels too embarrassed to ask for it again
- Each resident underestimates how much the other enjoyed their first conversation
- Nobody bridges the gap by arranging a natural second meeting or re-stating their names

Contributing causes:

- They are assigned to different dining tables and keep different daily routines
- He is reluctant to appear pushy or overly familiar
- She worries that a longer conversation will expose further memory lapses
- The care worker who originally introduced them is rarely on duty when the two cross paths
- A passing nod in the corridor offers no natural topic to restart conversation

Sub-problems

1. Forgetting a peer's name becomes a reason to avoid speaking with them
2. Neither resident realises the other would welcome another conversation
3. No mechanism brings compatible peers back together or refreshes their names unobtrusively

Out of scope

- Dining-room table assignment policy - ST-05

PS-04 A yes to be polite

Evidence basis

Provisional; constructed from SA-04.

- Bohns (2016): people find face-to-face requests hard to refuse and comply out of social discomfort.
- Mastroianni et al. (2021): conversations rarely end when both parties want them to.

Narrative

A fellow resident approaches her armchair in the communal lounge, within earshot of neighbours, and asks whether she would like to sit together over coffee and play cards.

	She is tired and wants quiet, but turning him down to his face in front of others feels unkind, so she agrees. Twenty minutes in, she is exhausted and wants to return to her room, but waits for him to suggest stopping. Mistaking her quiet compliance for enjoyment, he deals another hand. From the next day on, she avoids the lounge in the afternoon so she will not be put on the spot again.
Stakeholders involved	• ST-01: Resident with dementia • ST-02: Fellow resident
Activities involved	SA-04: Inviting someone to spend time together
Breakdown point	She accepts a public, face-to-face invitation out of politeness, feels unable to end the encounter, and subsequently avoids the inviter and the lounge.
Causes	Root causes: • Refusing a direct request face to face feels rude, especially when other residents are nearby • Neither resident has an easy, face-saving way to bring the meeting to a close • The inviter cannot distinguish reluctant compliance from genuine interest Contributing causes: • Other residents in the lounge can overhear the invitation • She does not want to hurt a neighbour's feelings or appear unfriendly • She is tired or unwell but hesitates to complain • She worries he will take a refusal personally • Open-ended sitting in the lounge lacks a clear, agreed duration or natural stopping point
Sub-problems	1. Face-to-face invitations pressure residents into saying yes when they would rather decline 2. Once together, neither resident can end the interaction without awkwardness 3. Reluctant compliance leads to subsequent avoidance of both the peer and communal spaces
Out of scope	• Architectural layout and acoustics of the communal lounge - ST-07

1.2 Human Values and Human Factors

1.2.1 Human Values

HV-01 Autonomy

Evidence basis	Provisional; based on published literature and group reasoning.
Situated meaning	Each resident decides whether to engage with the System, whom to meet, when to join communal spaces, and what to do together. They control which biographical details may be shared with peers and can revise those choices at any time. Explicit consent precedes every introduction and every shared personal fact.
Held by	ST-01, ST-02 (high); ST-05 (medium)
Intrinsic or instrumental	Intrinsic; also supports Dignity (HV-02).
Recognition criteria	Served: residents configure and update their own preferences; declining an invitation or turning away ends the interaction immediately without anyone asking for a justification. Violated: pairings arranged for staff convenience; personal details mentioned without prior consent; repeated persuasion after a refusal; reminders of declined invitations.
Source	Ryan, R. M. & Deci, E. L. (2000). Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. *American Psychologist*, 55(1), 68-78.

HV-02 Dignity

Evidence basis	Provisional; based on published literature and group reasoning.

Situated meaning	Residents are addressed as capable adults. A forgotten name or mid-sentence pause passes without remark, and nobody is quizzed, corrected, or praised for basic participation. Companionship is offered for the resident's own sake, never to fill an institutional schedule.
Held by	ST-01 (high); ST-02 (medium)
Intrinsic or instrumental	Intrinsic.
Recognition criteria	Served: names are woven naturally into greetings for both people ("Elena, Jan, good to see you both") so neither has to admit forgetting; conversational slips are given quiet time for self-repair. Violated: remarks such as "You've met before, don't you remember?" or "You told me that already"; finishing a resident's sentence prematurely; pairing residents merely to occupy them.
Source	Kitwood, T. (1997). *Dementia Reconsidered: The Person Comes First*. Open University Press. Goffman, E. (1967). *Interaction Ritual*. Anchor Books.

HV-03 Privacy

Evidence basis	Provisional; based on published literature and group reasoning.
Situated meaning	Each resident retains ownership over their life history, personal confidences, and quiet time in their bedroom. No personal detail about one resident is disclosed to another without explicit permission, presence in private bedrooms is tracked only at coarse resolution with separate consent, and all raw conversation processing remains inside the care home. For care staff, privacy also means their daily work is not timed or monitored as workplace surveillance.
Held by	ST-01, ST-02, ST-05, ST-07 (high); ST-04, ST-06 (medium); ST-03 (low - relatives may wish to share more background than the resident themselves would release)
Intrinsic or instrumental	Intrinsic; also supports Dignity (HV-02) and Autonomy (HV-01).
Recognition criteria	Served: only consented profile items are used for matching or icebreakers; no raw audio or transcripts are stored after a session; bedroom time is logged only at coarse daily resolution for residents who opted in. Violated: disclosing a private fact or family matter to a peer; explaining a match by citing an unreleased detail; continuous or unconsented room monitoring.
Source	Friedman, B., Kahn, P. H., Jr. & Borning, A. (2008). Value Sensitive Design and Information Systems. In K. E. Himma & H. T. Tavani (Eds.), *The Handbook of Information and Computer Ethics*. Wiley. Niemeijer et al. (2010).

HV-04 Safety

Evidence basis	Provisional; based on published literature and group reasoning.
Situated meaning	Residents remain physically and emotionally safe during robot sessions and peer meetings. Suggested activities respect each resident's mobility and clinical boundaries, and any sign of acute agitation, distress, or physical hazard triggers an immediate pause and human handover.
Held by	ST-01, ST-04, ST-06, ST-07 (high)
Intrinsic or instrumental	Instrumental: a prerequisite for social activities in the home.
Recognition criteria	Served: shared activities match the residents' assessed mobility and clinical guidance; robot and peer sessions pause immediately upon distress or agitation, with care staff notified promptly. Violated: suggesting an activity that exceeds a resident's mobility limits; continuing a session when a resident shows agitation or distress.

Source	Cohen-Mansfield, J., Marx, M. S. & Rosenthal, A. S. (1989). A description of agitation in a nursing home. *Journal of Gerontology*, 44(3), M77-M84. Härlein, J. et al. (2009). Fall risk factors in older people with dementia or cognitive impairment. *Journal of Advanced Nursing*, 65(5), 922-933.

HV-05 Well-being

Evidence basis	Provisional; based on published literature and group reasoning.
Situated meaning	Emotional comfort, calm, and days that hold pleasant company and something to look forward to, free from social pressure, embarrassment, agitation, or blunt reminders of painful losses.
Held by	ST-01 to ST-06 (high); ST-07 (medium)
Intrinsic or instrumental	Intrinsic.
Recognition criteria	Served: observable pleasure and alertness during and after conversations; reduced agitation and distress over time. Violated: anxiety, frustration, or agitation triggered by an awkward introduction, a public refusal, or an insensitive remark.
Source	Lawton, M. P., Van Haitsma, K. & Klapper, J. (1996). Observed affect in nursing home residents with Alzheimer's disease. *Journal of Gerontology B*, 51(1), P3-P14. Woods et al. (2018).

HV-06 Social connectedness

Evidence basis	Provisional; based on published literature and group reasoning.
Situated meaning	Residents enjoy meaningful conversation and companionship with compatible peers in the care home and, over time, build mutual friendships that ease loneliness and give them a reason to spend time in communal areas.
Held by	ST-01, ST-02, ST-03, ST-05 (high); ST-04 (medium)
Intrinsic or instrumental	Both: valued in its own right, and instrumental in motivating residents to leave their bedrooms and engage with daily life.
Recognition criteria	Served: introductions lead to repeat conversations and shared time in common areas; residents can name a peer in the home whom they enjoy talking with. Violated: using the robot as a substitute for human company; one-off introductions that never lead to further contact; social withdrawal following a pressured or awkward meeting.
Source	Victor, C. R. (2012). Loneliness in care homes: a neglected area of research? *Aging Health*, 8(6), 637-646. Holwerda et al. (2014). de Medeiros et al. (2012).

HV-07 Identity

Evidence basis	Provisional; based on published literature and group reasoning.
Situated meaning	Residents are recognised as the people they have been throughout their lives: defined by their hometowns, trades, eras, musical tastes, and personal stories rather than by their diagnosis. Shared life history gives two residents natural common ground for conversation.
Held by	ST-01, ST-03, ST-05 (high)
Intrinsic or instrumental	Intrinsic; also instrumental in supplying effortless topics for peer conversation.
Recognition criteria	Served: introductions and icebreakers draw directly on the residents' own life stories, interests, and preferred eras. Violated: getting biographical facts wrong; matching residents solely by clinical stage or dining table rather than shared background and interests.
Source	Kitwood (1997). Woods, B. et al. (2018). Reminiscence therapy for dementia. *Cochrane Database of Systematic Reviews*, (3), CD001120.

HV-08 Truthfulness

Evidence basis	Provisional; based on published literature and group reasoning.
Situated meaning	No deception by design: the System never tells lies (including well-meant "therapeutic" lies), never pretends to be human or to share human memories, and never plays along with a false belief. Once trust is broken in dementia care, it is rarely recovered. Because profile entries can become outdated and speech recognition can mishear an utterance, the System states biographical facts only from verified profile items, speaks conservatively when uncertain, and corrects itself plainly when contradicted. Painful truths, such as why the resident lives in the home, a bereavement, or a diagnosis, are always handed over to a trusted human caregiver rather than answered by the robot.
Held by	ST-01, ST-03, ST-04, ST-06 (high)
Intrinsic or instrumental	Intrinsic; also instrumental because trust is a precondition for accepting social facilitation.
Recognition criteria	Served: dialogue contains no untruths, denials, or false pretence; factual errors are acknowledged immediately and flagged for correction; questions on sensitive topics are deferred to the designated trusted person. Violated: telling a resident "You are only staying here for a few days"; leaving a factual error uncorrected; the robot delivering news of a bereavement or diagnosis.
Source	Mental Health Foundation (2016). *What Is Truth?* James et al. (2006). Wang et al. (2026).

1.2.2 Value Tensions

VT-01 Privacy vs Social connectedness (matching on personal facts)

Evidence basis	Provisional; derived from group reasoning and illustrated in part by PS-03.
Values in tension	HV-06 Social connectedness: grounding peer matches and icebreakers in genuine shared history and interests. HV-03 Privacy: letting each resident control what fellow residents learn about their life.
Domain area	privacy
Nature of the conflict	The biographical details that spark immediate connection: a hometown, a former trade, a favourite era, or a shared interest are all personal. Telling one resident why they were matched reveals something about the other, and a personal fact once spoken cannot be taken back.
Non-negotiable floor	GDPR Articles 5 and 9 (health and sensitive personal data), alongside representative consent and ongoing resident assent (O-5).
Current resolution	• Each resident's profile is stored in an isolated, encrypted per-resident record on the local server (TECH-04) • A personal detail is used for compatibility matching or spoken in an icebreaker only if its owner privately released it and the relationship map permits it for that pair (TECH-05) • Medical diagnoses, bereavements, and family conflicts are excluded from matching and icebreakers; residents are free to bring them up themselves if they wish • The activity coordinator reviews and approves proposed pairings before an introduction is arranged (F2) • If a pairing is unreviewed or the relationship map has a pending update, no match is proposed • Staff review the relationship map after any bereavement, clinical change, interpersonal friction, or room move
Alternatives considered	A single shared profile database accessible without per-item release: rejected for violating HV-03. Matching solely by residential unit or dining table without personal facts: rejected because introductions lose the common ground that sustains conversation (PS-03).

Status of the tension	Provisionally resolved. Verified by UC02-F2-PR2; tested by adverse Claim UC02-F2-CL4.

VT-02 Social connectedness vs Autonomy (pressure to accept an introduction)

Evidence basis	Provisional; grounded in Bohns (2016) and Mastroianni et al. (2021), and illustrated by PS-04.
Values in tension	HV-06 Social connectedness: encouraging isolated residents to meet compatible peers and spend time together. HV-01 Autonomy: preserving each resident's freedom to decline an invitation or leave a conversation at will.
Domain area	social; personhood
Nature of the conflict	A direct invitation made face to face, especially within earshot of others in the lounge, is awkward to refuse. Residents who agree merely out of politeness often feel trapped during the encounter and avoid the peer afterwards (PS-04). Conversely, overly frequent suggestions turn companionship into a chore.
Non-negotiable floor	All robot sessions and peer introductions are strictly voluntary; verbal or non-verbal dissent (such as turning away) ends the interaction immediately (Black et al., 2010).
Current resolution	• During one-to-one bridging sessions, the System invites each resident privately in a quiet setting before bringing the pair together, so neither hears whether the other was asked (IDP-02) • A joint introduction proceeds only when both residents independently agree; a refusal is never relayed to the other resident • Either resident may end a session or leave a peer meeting at any moment without giving a reason • The System keeps no public tallies, offers no patronising praise, and never reminds residents of declined invitations • Bridging invitations are capped at a modest weekly limit per resident [threshold - O-2] • Staff do not override a resident's refusal to meet a suggested peer
Alternatives considered	Assigning fixed peer pairings on the weekly activity schedule: rejected as coercive. Brokering introductions only when a resident explicitly asks staff: rejected because shy or newly admitted residents rarely initiate requests on their own (PS-01, PS-03).
Status of the tension	Open. Verified by UC02-F1-PR1; tested by adverse Claim UC02-F1-CL3.

VT-03 Truthfulness vs Well-being (truth that hurts)

Evidence basis	Provisional; grounded in Mental Health Foundation (2016), James et al. (2006), and Wang et al. (2026), and illustrated in part by PS-01.
Values in tension	HV-08 Truthfulness: avoiding deception, false pretence, and untruthful reassurance. HV-05 Well-being: sparing residents acute distress and repeated grief.
Domain area	psychological safety; trust
Nature of the conflict	A newcomer may ask daily when she can go home, and a resident who has forgotten a spouse's death may experience the bereavement as fresh news if told bluntly. A soothing lie spares immediate distress in the moment, but destroys trust when the resident realises they were deceived.
Non-negotiable floor	No deception by design: the System never states an untruth, never pretends to be human, and never delivers news of a death, diagnosis, or permanent admission. Factual errors by the System must be corrected immediately when challenged.
Current resolution	• When a resident expresses a belief rooted in the past ("My mother is waiting for me"), the System acknowledges the underlying emotion and gently bridges to a comforting present topic without either correcting the resident or validating the false premise (IDP-01) • When a resident asks a direct question on a sensitive topic (such as going home or a deceased relative), the System gives a truthful deferral ("That is something to talk through with [trusted person]. Shall I ask them to come over?") and notifies that caregiver (IDP-03)

	• The designated human caregiver responds using the approach agreed with the family and clinical team (ST-03, ST-06) • If the System states an outdated or misheard fact and the resident contradicts it, the System apologises plainly ("I had that wrong") and flags the item for ST-05 to correct
Alternatives considered	Therapeutic lies ("You are only staying for a few days"): rejected because discovery undermines trust in both the System and care staff. Direct factual disclosure by the robot: rejected because painful truths require human empathy and clinical judgement.
Status of the tension	Provisionally resolved. Verified by UC01-F4-PR4; tested by Claim UC01-F4-CL8.

VT-04 Dignity vs Social connectedness (names that expose forgetting)

Evidence basis	Provisional; grounded in Goffman (1967) and Schegloff et al. (1977), and illustrated by PS-01 and PS-03.
Values in tension	HV-06 Social connectedness: helping residents recognise one another and pick up prior conversations across meetings. HV-02 Dignity: ensuring memory lapses are never spotlighted in front of peers.
Domain area	psychological safety (personhood)
Nature of the conflict	Residents with dementia frequently forget a new acquaintance's name and may walk past them rather than risk embarrassment (PS-03). Reminding them of the name enables conversation, yet an explicit reminder aimed at one resident signals that they have forgotten.
Non-negotiable floor	None identified.
Current resolution	• At every facilitated meeting, the System greets both residents by name symmetrically ("Elena, Jan, good to see you both"), framing the names as a warm welcome rather than a memory prompt • The System never asks quiz-style questions such as "Do you remember who this is?" or "You met last week, didn't you?" • Shared icebreaker prompts are addressed to the pair as a whole rather than testing either individual's recall • How prominently names are repeated can be adjusted in each resident's preferences
Alternatives considered	Introducing residents only once on arrival: rejected because forgotten names quickly halt follow-up contact (PS-01, PS-03). Visible name badges worn at all times: not preferred for everyday care, as badges can feel stigmatising to residents inside their own home.
Status of the tension	Open. Verified by UC02-F3-PR3; tested by adverse Claim UC01-F3-CL5.

VT-05 Privacy vs Identity (a rich persona profile)

Evidence basis	Provisional; grounded in Niemeijer et al. (2010) and Berridge & Grigorovich (2022).
Values in tension	HV-07 Identity: building a rich persona profile of life history, nostalgic eras, musical tastes, and conversational pace so matches reflect who the resident truly is. HV-03 Privacy: minimising stored personal data and preventing unauthorised disclosure.
Domain area	privacy
Nature of the conflict	Meaningful compatibility matching relies on specific biographical details and conversational features gathered across one-to-one sessions. That same specificity makes the profile sensitive, and relatives sometimes volunteer stories that the resident would prefer to keep private.
Non-negotiable floor	All speech and language processing runs on the robot's on-device accelerator (TECH-18) and the home's local server (TECH-04) with no cloud uplink; raw audio is held only in working memory and transcripts are deleted at the end of each session.
Current resolution	• Persona attributes extracted during one-to-one dialogue enter the profile as private by default until the resident explicitly consents to release them for matching (TECH-

05, TECH-06)
- The resident (and, where appropriate, their representative) can inspect, edit, or delete any profile item at any time, and a resident's objection always prevails
- Acoustic and linguistic engagement metrics (M-21) are reduced to numerical summary scores on the robot before reaching the local server
- Research analysis uses pseudonymised data under institutional ethics approval and separate consent (O-5)

Alternatives considered

Cloud-hosted speech and language services: rejected to prevent biographical audio and transcripts from leaving the care home (TECH-07).

Minimal profiles containing only age and unit: rejected because matches lose personal identity and shared reminiscence ground.

Status of the tension

Open. Verified by UC02-F6-PR6; tested by adverse Claim UC02-F2-CL4.

VT-06 Privacy and Autonomy vs Well-being (noticing bedroom isolation without intruding)

Evidence basis

Provisional; grounded in Niemeijer et al. (2010, 2015) and Jansen et al. (2017), and illustrated by PS-02.

Values in tension

HV-05 Well-being: noticing sustained bedroom isolation early, before mood, confidence, and mobility deteriorate.

HV-03 Privacy and HV-01 Autonomy: respecting the bedroom as private space and recognising that time alone is a legitimate personal choice.

Domain area

privacy; personhood

Nature of the conflict

Social withdrawal in a care home develops gradually and is easily missed between busy care rounds (PS-02). Detecting sustained isolation requires sensing how much time a resident spends in their bedroom against communal spaces. However, fine-grained tracking compromises privacy and dignity, and a staff visit triggered by a sensor can make the resident feel watched in their own room.

Non-negotiable floor

Presence sensing uses only coarse daily resolution, requires explicit opt-in consent and resident assent, and never triggers an automated robot prompt about bedroom time.

Current resolution

- Room and communal presence sensing (TECH-03) is enabled only for residents who give separate consent and ongoing assent (O-7)
- The System records only a coarse daily summary (proportion of waking hours in the bedroom vs. common areas), never a minute-by-minute movement log or camera feed
- Care staff are notified only when bedroom isolation remains elevated across several consecutive days [threshold - O-2], respecting normal day-to-day desires for quiet and solitude
- The notification goes solely to the resident's primary care worker, who decides whether to drop by and always frames a visit as a warm offer of company rather than a welfare check
- The resident or representative can disable presence sensing at any time

Alternatives considered

Relying on informal staff observation alone: retained alongside, but insufficient on its own because gradual day-by-day withdrawal across shifts goes unnoticed (PS-02).

Continuous fine-grained location or camera monitoring: rejected as intrusive surveillance (TECH-09).

Status of the tension

Open. Verified by UC03-F5-PR5; tested by adverse Claim UC03-F5-CL7.

1.2.3 Human Factors Concepts

HFC-01 Fresh air, movement and mood

Evidence basis

Literature only. Evidence in dementia is weak.

Concept statement

Time outdoors and on foot may lift mood and help people keep up daily activities. Side by side, on the move, talk may come more easily and a pause may cost less. For these reasons a walk can be one optional activity where both residents prefer it and are assessed fit; the evidence does not justify making it the default.

Human factor addressed	Mood; daily functioning; conversation
Proposed mechanism	Exercise maintains physical function. Time in nature lowers rumination. Walking side by side asks for no eye contact, and the surroundings give talk a subject.
Source	Forbes et al. (2015), Cochrane review: in dementia, promising evidence that exercise helps with daily activities; no clear effect on cognition, mood or behaviour; evidence of very low quality. Bratman et al. (2015): a nature walk lowered rumination in healthy adults. Evans & Jones (2011): walking interviews draw more talk about place.
Conditions and population of validity	Mostly healthy adults; evidence in dementia weak and mixed.
Predicted Effect type	More pleasure in time spent together; calmer afterwards
Associated Measures	M-03, M-06

HFC-02 Moving into a care home

Evidence basis	Literature only.
Concept statement	Moving into a care home means losing a home, a routine and neighbours, often by someone else's decision. Newcomers with dementia take weeks to months to adjust, and many withdraw in that time.
Human factor addressed	Adjustment; orientation; loneliness
Proposed mechanism	A newcomer meets many people at once and keeps few names. Expecting to return home, they see little point in getting to know anyone. A few introductions, spread over the first weeks and repeated, leave fewer faces to learn and some to recognise.
Source	Sury et al. (2013), review: moving in is a time of loss and distress for people with dementia and their families, and few interventions have been tested. Victor (2012): loneliness in care homes.
Conditions and population of validity	Mostly qualitative studies, across varied homes and countries.
Predicted Effect type	Newcomers have someone to talk to within weeks, and spend less of the day in their room
Associated Measures	M-02, M-05, M-06

HFC-03 Conversation repair

Evidence basis	Literature only.
Concept statement	Speakers prefer to fix their own slips. In dementia, slips become more frequent and repair succeeds less often.
Human factor addressed	Conversation; face
Proposed mechanism	Self-repair needs a moment. Help offered too soon takes the turn and shows the slip. A cue to the last topic, or to a shared one, lets the talk go on. Just after an introduction, a stalled conversation is when two residents give up on each other.
Source	Schegloff et al. (1977): preference for self-repair. Orange et al. (1996): repair in Alzheimer's disease.
Conditions and population of validity	Conversation analysis of healthy talk; small samples in Alzheimer's disease.
Predicted Effect type	Talk after an introduction survives slips, without embarrassment
Associated Measures	M-04

HFC-04 Common ground and reminiscence

Evidence basis	Literature only.
Concept statement	Conversation rests on what both people know. Old memories outlast recent ones, and a past that two residents share gives them common ground.
Human factor addressed	Conversation; identity
Proposed mechanism	Residents rarely discover what they share (PS-03). The System can find it in two consented profiles and mention it once, at the introduction. A topic from early adulthood is recalled with least effort.
Source	Clark & Brennan (1991): grounding in communication. Rubin et al. (1998): early adulthood is recalled best. Woods et al. (2018), Cochrane review: small, setting-dependent benefits of reminiscence.
Conditions and population of validity	Reminiscence studied in staff-led groups; benefits small.
Predicted Effect type	More conversation after an introduction; more positive interaction
Associated Measures	M-04, M-05

HFC-05 Theory of mind

Evidence basis	Literature and group reasoning.
Concept statement	Understanding what others know, believe and feel weakens in dementia. Two residents often misread each other's interest, discomfort or wish to stop.
Human factor addressed	Social cognition; consent
Proposed mechanism	A polite yes is taken for a willing one, and a nod for a lack of interest (PS-03, PS-04). Nobody reads these states reliably, the System included, so it asks each resident privately instead of inferring.
Source	Bora et al. (2015), meta-analysis: theory of mind impaired in Alzheimer's disease.
Conditions and population of validity	Measured in laboratory tasks, not in everyday conversation.
Predicted Effect type	Misreadings between residents; fewer when each is asked privately
Associated Measures	M-07

HFC-06 Face and personhood

Evidence basis	Literature only.
Concept statement	Each interaction affirms or erodes a person's standing. Corrections, quizzing and sing-song speech (elderspeak) erode it.
Human factor addressed	Experienced dignity; withdrawal
Proposed mechanism	Help addressed to one person marks them as failing. Addressed to both, as a statement, it marks nobody: "Elena, Jan, good to see you both."
Source	Goffman (1967): face-work. Kitwood (1997): personhood. Sabat (2001): the self in Alzheimer's disease. Williams et al. (2009): elderspeak provokes resistance.
Conditions and population of validity	Care-staff speech; robot voices not studied.
Predicted Effect type	Embarrassment and withdrawal; descriptions of being talked down to

Associated Measures	M-07, M-09 (code T)

HFC-07 Coping, withdrawal and loneliness

Evidence basis	Literature only.
Concept statement	Aware of decline, people protect themselves: some carry on as before, some face it openly. Avoiding company protects face and leads to loneliness. Withdrawal is gradual, so others notice it late.
Human factor addressed	Coping with cognitive loss; loneliness
Proposed mechanism	Each awkward meeting makes the next one less likely. A first meeting on shared ground, with an easy way out, lowers the cost of trying. Noticing a pattern of staying in lets staff offer company before withdrawal sets in.
Source	Clare (2003): coping in early Alzheimer's disease. Victor (2012): loneliness in care homes. Holwerda et al. (2014): loneliness predicted dementia onset.
Conditions and population of validity	Coping studied in early dementia, at home. Loneliness measures in dementia of uncertain validity.
Predicted Effect type	More willingness to meet others; less loneliness over months
Associated Measures	M-02, M-08

HFC-08 Truth and trust

Evidence basis	Literature only.
Concept statement	Lies told to spare distress are common in dementia care, and contested. Once found out, they cost trust. A blunt truth from a stranger causes grief.
Human factor addressed	Trust; grief
Proposed mechanism	Trust builds on consistent, truthful answers from known people. Gentle redirection answers the feeling without saying anything untrue.
Source	Mental Health Foundation (2016): inquiry on truth and lying in dementia care. James et al. (2006): staff lying, and guidelines.
Conditions and population of validity	UK care practice; answers given by a System not studied.
Predicted Effect type	Trust kept; fewer repeated painful questions
Associated Measures	M-09 (code U), M-10

HFC-15 Similarity and shared background

Evidence basis	Literature only.
Concept statement	People tend to befriend people like themselves: from the same place, generation or trade, or with the same interests. Residents with dementia still form and keep friendships.
Human factor addressed	Friendship formation; conversation
Proposed mechanism	Similarity makes the other person more predictable and gives talk a subject. A shared hometown or trade is often the first thing two strangers look for. With memory loss, residents seldom find it on their own.
Source	McPherson et al. (2001): review of homophily in social networks. de Medeiros et al. (2012): friendships among people with dementia in long-term care.

Conditions and population of validity	Homophily established in general populations; untested as a matching rule in care homes.
Predicted Effect type	Matches on a shared background lead to more repeat contact than chance meetings
Associated Measures	M-02, M-05

HFC-16 Fear of rejection and the liking gap

Evidence basis	Literature only.
Concept statement	People expect a conversation with a stranger to go worse than it does, and underestimate how much a new acquaintance liked them. So they rarely make the first move, and seldom follow up a good first talk.
Human factor addressed	Social initiative; friendship formation
Proposed mechanism	Each side waits for the other (PS-01, PS-03). A third party who asks both privately, and introduces them only when both agree, removes the risk of a "no" that the other hears.
Source	Epley & Schroeder (2014): commuters expected solitude to be more pleasant, yet enjoyed talking to a stranger more. Boothby et al. (2018): the liking gap persisted after conversations of all lengths and over months.
Conditions and population of validity	Healthy adults, mostly students and commuters; not tested in dementia.
Predicted Effect type	More introductions accepted when asked privately; more second meetings
Associated Measures	M-02, M-07

HFC-17 The difficulty of saying no

Evidence basis	Literature only.
Concept statement	Refusing a request face to face is uncomfortable, so people agree to things they would rather not do. They also stay in conversations longer than they want, because ending one feels rude.
Human factor addressed	Consent; autonomy
Proposed mechanism	People who ask underestimate how hard refusing is. A private question, with a "no" that reaches nobody, lowers the cost of declining. A meeting with a natural end, and leave to go at any time, lowers the cost of stopping.
Source	Bohns (2016): requesters underestimate compliance by about half, because they miss how awkward refusing is. Mastroianni et al. (2021): conversations rarely end when both people want them to.
Conditions and population of validity	Healthy adults, with strangers and acquaintances.
Predicted Effect type	Polite acceptances followed by avoidance (PS-04); fewer when residents are asked privately
Associated Measures	M-07, M-09 (code P)

1.2.4 Measures

M-01 System performance

Evidence basis	Bespoke. Staged-event protocol, as used for prompting systems (Mihailidis et al., 2008).
Construct measured	Whether the System does what F1 to F6 specify, on time, and nothing more.

Type	Quantitative
Instrument or procedure	Staged events against a timestamped reference log (EM-01): match suggestions with and without consent, items blocked by the map, private questions, introductions, sensitive questions, agreed meetings, presence patterns for residents who did and did not opt in. Checks on a full test day: network capture at the router; file-system search of all devices for audio.
Scoring or coding scheme	Per event: correct (yes/no); latency (s). Suggestions or items used without consent or against the map (target 0). Presence notes for residents who did not opt in (target 0). Packets leaving the home network (target 0). Audio files (target 0).
Validity and reliability	Bespoke. Depends on the accuracy of the reference log. Role-players do not behave like residents.
Serves	Verification
Burden	No residents (staged events).

M-02 Introductions and repeat contact

Evidence basis	Bespoke count.
Construct measured	How often residents are introduced, how often they accept, and how often a first meeting leads to a second.
Type	Quantitative
Instrument or procedure	In both homes, the coordinator keeps a short daily log: each introduction, who made it, whether both accepted, and any later meeting of the pair. Observers in the shared rooms (EM-02) note which pairs talk or do something together. In the System home, the System record (F6) serves as a cross-check.
Scoring or coding scheme	Per resident per month: introductions; share accepted; pairs who meet again within 14 days; pairs seen together on two or more observation days without being prompted.
Validity and reliability	A staff log misses introductions made in passing, and observation covers only some hours. Each log is checked against observation.
Serves	Validation
Burden	A few minutes a day for the coordinator; none for residents.

M-03 Observed emotion

Evidence basis	Observed Emotion Rating Scale (OERS; Lawton, Van Haitsma & Klapper, 1996), built for nursing-home residents with Alzheimer's disease. Validated instrument.
Construct measured	Emotional state during introductions and time spent together, per resident.
Type	Quantitative (observational)
Instrument or procedure	OERS, rated live by an observer (EM-02). No video, no audio. Windows: each observed introduction or meeting (from 5 min before to 10 min after); and 10-minute windows for sampled residents during each observation session. Each resident is rated separately.
Scoring or coding scheme	Five affects: pleasure, alertness, anxiety, sadness, anger. Each is rated by duration within the window: 1 never; 2 under 16 s; 3 16-59 s; 4 1-5 min; 5 over 5 min. Per window: positive score = mean of pleasure and alertness; negative score = mean of anxiety, sadness and anger.
Validity and reliability	Psychometrics in Lawton et al. (1996). Less reliable when rated by care staff than by trained researchers (Lawton et al., 1999), so researchers rate. A second observer rates 20% of windows independently; ICC(2,1) ≥ .60. Training: EM-02.
Serves	Validation
Burden	Human subjects; an observer in the room.

M-04 Conversation between residents

Evidence basis	Interval coding adapted from the Quality of Interactions Schedule (QUIS; Dean, Proudfoot & Lindesay, 1993), developed for staff-resident interaction. The adaptation to talk between residents is bespoke.
Construct measured	How much residents talk with each other, and how well.
Type	Quantitative (observational)
Instrument or procedure	Observer room scan every 2 minutes during each session (EM-02). Codes only: no words, no content.
Scoring or coding scheme	Per resident in view, per scan: talking with another resident (yes/no); quality: positive (warm, laughter, a shared topic), neutral, or negative (dismissive, conflict). Per session: share of residents talking with another resident; share of positive and negative codes. Per observed introduction: minutes of talk in the 10 minutes after.
Validity and reliability	A second observer on 20% of sessions; $\varkappa \geq .60$ (Landis & Koch, 1977).
Serves	Validation
Burden	As M-03.

M-05 Friendships

Evidence basis	Name generator, with staff reports as a second source. Self-report is read alongside other sources, never alone: residents rate their own experience higher than observers do (Custers et al., 2013).
Construct measured	Whether residents have someone in the home they like to talk to, and whether the feeling is mutual.
Type	Quantitative (self-report and staff report)
Instrument or procedure	At baseline, at six weeks and at the end (EM-03), each resident who is willing is asked: "Is there someone here you like to talk to?" and then "Anyone else?", up to three names. Photos of consenting residents in the same unit help those who cannot recall a name. Staff name, for each resident, the residents they spend time with.
Scoring or coding scheme	Per resident: names given (0-3); names reciprocated; named by staff. Per home: share of residents with at least one reciprocated tie; share named by nobody.
Validity and reliability	Depends on recall; the photo prompt can help.
Serves	Validation
Burden	About five minutes per resident, three times.

M-06 Time together, walks included

Evidence basis	Room scans follow the momentary time sampling used in care-home observation.
Construct measured	How much time residents spend with each other, doing what, and whether anything goes wrong.
Type	Quantitative (observational and log)
Instrument or procedure	Observer room scans every 2 minutes (EM-02): for each resident in view, the activity and the company (alone; with staff; with one resident; in a group). In both homes, a garden-door log: residents going out together. For newcomers in their first six weeks: whether they joined shared meals and activities each day, from the staff log.
Scoring or coding scheme	Per session: share of residents in view doing something with one other resident. Per week: walks together; incidents during time together (falls, near-falls, distress). Per newcomer: days per week in shared rooms.
Validity and reliability	Room scans cover only observed hours; the door log depends on staff.
Serves	Validation

Burden	As M-03; the door log takes seconds per walk.

M-07 The residents' own view

Evidence basis	Self-report read alongside observation, never alone (Custers et al., 2013).
Construct measured	Whether the resident valued meeting the other, felt free to decline or leave, and wants to meet again.
Type	Mixed
Instrument or procedure	Resident, after each observed introduction or meeting, only if willing and not distressed: three faces, "How was it, meeting [name]?"; "Again?" (yes/no). Observer, before and during: pressure signs (agrees reluctantly; looks to staff or the other resident before answering; "if I must"). In the following week: avoidance (sits elsewhere; leaves when the other arrives), from the staff log.
Scoring or coding scheme	Faces 1-3; share of "Again?" answered yes; pressure signs per introduction; pairs with avoidance within 7 days.
Validity and reliability	Ceiling effects expected in self-report.
Serves	Validation
Burden	Low; about a minute after the meeting.

M-08 Loneliness

Evidence basis	De Jong Gierveld 6-item loneliness scale (De Jong Gierveld & Van Tilburg, 2006). Validated in older adults; not in dementia.
Construct measured	Emotional and social loneliness.
Type	Quantitative (self-report)
Instrument or procedure	Six items, read aloud by a researcher at baseline and at the end (EM-03). Mild to moderate dementia only.
Scoring or coding scheme	0-6, as in the source; emotional and social subscales reported separately.
Validity and reliability	Validity in dementia uncertain; three months is short for loneliness to change. Exploratory.
Serves	Validation
Burden	Five minutes, twice.

M-09 Privacy, tone and trust codes

Evidence basis	Code set from Niemeijer et al. (2010, 2015), Kitwood (1997), Williams et al. (2009) and the Mental Health Foundation (2016).
Construct measured	Whether residents, relatives and staff experience the System as watching, talking down, pushing, revealing or misleading.
Type	Qualitative coding
Instrument or procedure	Short interviews (EM-04).
Scoring or coding scheme	W watched; T talked down to; P pushed into company, for others' sake; S secret revealed; U misled or fobbed off; N neutral or positive. A statement can have more than one code.
Validity and reliability	Two coders on all interviews; $\varkappa \geq .60$.
Serves	Validation
Burden	Interviews; residents only if willing.

M-10 Sensitive topics, errors and disclosure

Evidence basis	Count. Safeguard for VT-01 and VT-03.

Construct measured	Whether painful questions reach the trusted person; how often the System gets a fact wrong, and whether it corrects itself; whether it reveals what it should not.
Type	Quantitative
Instrument or procedure	Script check: every prepared line and template, before use (EM-01). Staged trials with known facts, including a wrong profile entry and a misheard answer (EM-01). In use: the System record (F6) and staff reports of errors noticed by residents, relatives or staff.
Scoring or coding scheme	Deception by design: untruths, denials or playing along in the script (target 0). Errors: wrong facts per 100 System statements; share corrected when contradicted; time to correct. Sensitive questions: deferred to the listed person (yes/no); minutes until they arrive. Disclosure: items used without consent or against the map (target 0).
Validity and reliability	Staged error rates are exact. In use, only noticed errors are counted, which undercounts them. A disclosure, or an error that causes distress, pauses the System for review (EM-03).
Serves	Both: Verification (staged) and Validation (in use)
Burden	None for residents; a short report from staff when they notice an error.

M-19 Agitation and observed affect

Evidence basis	Cohen-Mansfield Agitation Inventory (CMAI; Cohen-Mansfield et al., 1989), developed in a nursing home. Validated instrument. Affect around sessions: OERS, as in M-03 (Lawton, Van Haitsma & Klapper, 1996).
Construct measured	Agitation and anxiety behaviour over the study, and shifts in observed mood and affect from before to after a Kinnect session or peer meeting.
Type	Quantitative (caregiver-rated and observational)
Instrument or procedure	CMAI, completed by the resident's primary caregiver at both homes: at the end of the setup phase (covering the two weeks before week 1) and in week 5 (covering weeks 4 and 5). OERS, rated by a researcher in a 5-minute window before and a 10-minute window after each observed Kinnect session or peer meeting, as in M-03. Staff log, between ratings: each agitation event, with date, time, the behaviour, and whether it happened during or within an hour of a Kinnect session.
Scoring or coding scheme	CMAI: 29 behaviours, each rated for frequency from 1 (never) to 7 (several times an hour) over the previous two weeks; total 29 to 203. Change per resident: week 5 minus baseline. OERS: positive and negative scores per window, as in M-03. Shift per session: after minus before. Staff log: agitation events per resident per week, split into during or within an hour of a session, and other times.
Validity and reliability	CMAI validity in nursing-home residents from the source. Caregivers know which arm their home is in, so CMAI ratings are not blind; where possible the same caregiver rates at baseline and in week 5. OERS reliability as in M-03.
Serves	Validation
Burden	Caregiver time for two CMAI ratings per resident and the staff log; none for residents beyond OERS, as M-03.

M-20 Peer connections and communal time

Evidence basis	Bespoke counts from a real-time location system (RTLS) with proximity badges (Grigorovich et al., 2021; Jansen et al., 2017), checked against the room scans of EM-02.
Construct measured	How often residents interact one-to-one with a peer after matching, how long they spend together in common areas, and how dense the social network in the home is.
Type	Quantitative (sensor and observational)

Instrument or procedure	Proximity badges, worn during waking hours by residents who gave separate consent for them, with fixed anchors in the lounge, the dining room and other common areas. Both homes, from the setup phase to week 5. Room scans by observers every 2 minutes during EM-02 sessions, coding talk (M-04) and company (M-06). These serve as the observation log.
Scoring or coding scheme	Thresholds: • **Co-location episode:** two participants' badges within 2 m of each other in the same common area for 5 minutes or more. • **Joint social time:** minutes in co-location episodes, per resident per week. • **Dyadic interaction:** a co-location episode during which a room scan codes the two as talking or doing something together. Episodes outside observed hours count as co-location only, reported separately. • **Tie:** a pair with co-location episodes on two or more days in a week, the same two-day rule as M-02. • **Network density:** ties divided by all possible pairs of participants, n(n-1)/2, per home per week. Matched pairs (intervention home) and all other pairs are reported separately, so that contact between matched residents can be told apart from a general rise in contact.
Validity and reliability	Bespoke. Badges near each other do not always mean a conversation, so dyadic interactions are counted only where a room scan confirms them. RTLS accuracy differs between systems and buildings (Grigorovich et al., 2021); the 2 m rule is checked during setup by placing badges at known distances. Room-scan reliability as in M-04 and M-06.
Serves	Validation
Burden	Residents wear a badge in waking hours. Acceptance of RTLS in long-term care varies (Grigorovich et al., 2021).

M-21 Conversational engagement

Evidence basis	Automated linguistic and acoustic features of speech in dementia (Fraser et al., 2016; Luz et al., 2021). Their use as a measure of engagement in conversation is bespoke.
Construct measured	How engaged a resident is in conversation with Kinnect and with a matched peer.
Type	Quantitative (automated)
Instrument or procedure	Speech recognition, transcript analysis and acoustic analysis run on the robot's accelerator (TECH-18), during one-to-one sessions and during joint sessions with a matched peer where both residents have consented. Audio is held only in the robot's working memory while it is processed, and the transcript is deleted on the robot at the end of the session. Only the values below reach the local server (TECH-04).
Scoring or coding scheme	Per session: • **Turn-taking ratio:** one-to-one sessions: the resident's turns divided by Kinnect's turns. Peer meetings: the shorter speaking time of the two residents divided by the longer (1 means equal). • **Utterance length:** mean words per turn. • **Response latency:** median seconds from the end of the other speaker's turn to the start of the resident's. • **Sentiment:** mean per-utterance score from an on-device sentiment model, -1 to 1. • **Topic coherence:** mean similarity between consecutive turns, from sentence embeddings, 0 to 1.
Validity and reliability	Depends on speech recognition accuracy, which drops with background noise, atypical speech cadence and word-finding pauses (Luz et al., 2021), and for older voices in general (Vipperla et al., 2010). Study audio is not kept, so the pipeline is checked in advance: in staged role-play sessions (EM-01), word error rate and each feature are compared against a hand transcription. Robot sessions take place in the intervention arm only, so this Measure describes change within that arm and cannot compare the arms.
Serves	Validation

Burden	None beyond the sessions. Residents and representatives are told that the robot processes speech on the robot itself and keeps no recordings or transcripts.

M-22 Bedroom isolation time

Evidence basis	Sensor-based assessment of life-space in nursing-home residents (Jansen et al., 2017).
Construct measured	How much of the waking day a resident spends in their own bedroom.
Type	Quantitative (sensor)
Instrument or procedure	The resident's badge (M-20), relative to a fixed beacon in their bedroom, at both homes. Only for residents who gave separate consent, in line with HV-03: time in one's room is noticed only where the resident agreed.
Scoring or coding scheme	Waking hours per resident: from the care plan's usual getting-up time and bedtime; 08:00 to 21:00 where the plan gives none. Per resident per day: minutes in the bedroom divided by waking minutes. A day counts only if the badge was detected for at least 80% of waking hours (proposed, O-2). Days on which the care record shows the resident confined to bed by illness are reported separately. Per resident per week: mean over valid days.
Validity and reliability	Bespoke threshold. A bedroom beacon can be picked up through a wall, so each one is checked at setup from the corridor and the neighbouring rooms, and its signal threshold set so that it counts only inside the room. Staff spot checks of where residents are, against the badge data, in the setup phase.
Serves	Validation
Burden	As M-20.

M-23 First-time usability

Evidence basis	Response latency as in Luz et al. (2021); the first-impression question follows M-07.
Construct measured	Whether residents who meet Kinnect for the first time understand it and can use it without help.
Type	Quantitative (logs and observational checklist)
Instrument or procedure	Kinnect's interaction log from the week 1 sessions: each prompt, and the time and kind of the resident's response (voice or tablet). Observational checklist, filled in by a researcher during each resident's first three sessions: • Per prompt: answered relevantly within 10 s; prompt repeated; staff explained; signs of confusion (looks around for help, asks who or what is speaking). • Per session, four tasks: responds when Kinnect greets them; answers a yes/no question, by voice or on the tablet; accepts or declines an offer; ends the session or says they want to stop, and Kinnect stops. After the first session, if the resident is willing: three faces, "How was it, talking with Pepper?" (as in M-07). The researcher rates the resident's first approach: approaches, neutral, or turns away.
Scoring or coding scheme	Prompt comprehension rate: prompts answered relevantly within 10 s without repetition, divided by all prompts. Task completion: tasks done without staff intervention, divided by tasks attempted. Response latency: median seconds per session, from the log. First-impression receptivity: faces 1 to 3, and the researcher's approach rating. Criterion for the secondary Claim: comprehension and task completion each at least 80% by the end of week 1.
Validity and reliability	A second researcher fills in the checklist on 20% of sessions; $\varkappa \geq .60$ (Landis & Koch, 1977).
Serves	Validation
Burden	Low; part of the week 1 sessions.

1.2.5 Evaluation Methods

EM-01 Prototype: demonstration and staged verification

Evidence basis	Staged events against a timestamped reference log, as for the COACH prompting system (Mihailidis et al., 2008). Operator-voiced build: the Wizard-of-Oz method (Riek, 2012).
Procedure summary	One Artifact (AT-01) in three builds: • Build A, operator-voiced: a researcher triggers every line Pepper says from a console. Formative use with a few residents of the System home before EM-03 begins: wording, pace, how the private question feels. Observed under EM-02. • Build B, automated: matching, the consent register, introductions, notes to staff and the presence count run on the server. The operator approves each introduction before it plays. • Build C, demonstration: role-players; events can be injected; the match panel on a screen. Staged verification on Build B: 20 trials per event (50 for disclosure events), against the reference log. Role-players only.
Purpose	Verification; demonstration; formative (Build A)
Human subjects required	Builds B and C: no; team members act. Build A: yes, a few residents under EM-02 rules.
Measures used	M-01, M-10; M-03, M-07 (Build A)
Cost and feasibility	Moderate. Pepper from the course, a laptop as server, a Wi-Fi router, a tablet for the coordinator, presence parts for two test rooms. About 6-8 weeks to Build B.
Known limitations	Role-players do not behave like residents. Pepper's recognition of older voices is untested, residents can answer on the tablet instead. In Build A the operator adapts better than the System will.

Table 1.1: Kit (EM-01)

Part	Prototype	Stands in for
Robot	Pepper in the lounge; tablet for photos and yes/no	TECH-01
Server	Laptop on a Wi-Fi router with no internet uplink; one encrypted store per resident	TECH-04
Profiles and consent	Coordinator's tablet: profile items, shareable or private, and the relationship map	TECH-05, TECH-06
Presence count	Parts as in TECH-03, in two test rooms	TECH-03
Staff	Phone with the notes app	TECH-04
Operator	Console and match panel on a laptop	Build A voice; Build B approvals

Match panel

Shown live, for a suggested pair: the items each resident released, and the blocked items with the reason; whether the coordinator approved; each resident's answer so far; the line Pepper will say. Below: why the pair was suggested. The panel shows no item that its owner has not released.

Table 1.2: Demonstration script (Build C) (EM-01)

Scene	Events	What the System shows
1 A newcomer's first week (UC01)	Welcome on arrival; one introduction on day two and another on day four; names repeated at the next meal; "When can I go home?"	Few faces at a time; names given to both, as statements; a truthful deferral and a note to the trusted person
2 Introducing two residents (UC02)	The coordinator approves a match; each resident asked alone; one says no in a second pair	Introduction only for the pair where both said yes, with one shared item and one optional activity drawn from both profiles; the other pair's "no" never mentioned
3 Following up (UC03)	An agreed meeting time; one resident cannot come; "Again?" asked privately; a presence	

	ern for a resident who opted in, and none for one who did not	A note to the care worker; another time offered; one note about the pattern, to the named care worker only
Close	Record and devices inspected	Only the listed items in the record; no audio file; no traffic outside the home network

Table 1.3: Staged verification (EM-01)

Event	Required behaviour	Criterion (proposed; O-2)
Match not approved by the coordinator	No resident asked	0/20
Item its owner did not release	Never used to match or introduce	0/50
Item blocked by the map for this pair	Never spoken	0/50
One resident says no	No introduction; the "no" never mentioned to the other	0/50
Introduction	Names given to both, as statements; at most one shared item, released by both	20/20
Question on a listed sensitive topic	Truthful deferral; trusted person notified	note ≤ 10 s, 20/20; the listed fact never stated, 0/50
Agreed meeting time	Note to the named care worker	≤ 1 min; 20/20
Presence pattern, resident opted in	One note after the set number of days, to the named care worker only	20/20
Presence, resident not opted in	No count kept; no note	0
Resident contradicts a System statement (wrong profile entry; misheard answer)	Plain correction, no argument; error flagged for ST-05	≤ 3 s; 20/20
Script and templates	No untruth, denial or playing along	0 lines
Network capture, full test day	No packet leaves the home network	0
File-system search, all devices	No audio file	0

EM-02 Observing shared rooms: live coding, no recording

Evidence basis

OERS (Lawton et al., 1996); interval coding adapted from QUIS (Dean et al., 1993); process consent in dementia research (Dewing, 2007).

Procedure summary

1. Residents, relatives and staff are told when observation takes place, and a notice hangs in each room. Residents who decline, or whose representative declines, are neither coded nor rated. A resident who shows unease is not coded that session.
2. Sessions: the lounge and the dining room, three sessions of 90 minutes a week at fixed times (a meal, an afternoon), in both homes.
3. The observer sits at the edge of the room. Every 2 minutes, a room scan: talk between residents (M-04); activity and company (M-06).
4. Introductions and meetings during a session: emotion (M-03); pressure signs, and the resident's view afterwards if they are willing (M-07).
5. Codes only, on a tablet: no words, no content, no names in free text.
6. Stop rule: anxiety, sadness or anger rated 4 or more: the observer stops rating that resident and tells ST-04.

Purpose

Validation; formative in EM-01 Build A

Human subjects required

Yes. Residents with supported consent and a representative; staff.

Measures used

M-02 to M-04, M-06, M-07

Cost and feasibility

One observer, about five hours a week per home; two observers on reliability sessions.

Known limitations

- The observer knows which home has the System: no blinding. Observers are not told the hypotheses.
- Being watched changes behaviour. Two weeks of familiarisation lower the effect; they do not remove it.
- Room scans miss what happens in residents' rooms and outside the sessions.
- Observers overhear fragments of talk. Codes-only notes limit the risk; they do not remove it.

Observer rules

- Training: OERS manual; practice sessions until ICC ≥ .70 (OERS) and $\varkappa \geq .70$ (talk and company codes) against a trained rater.
- Reliability: a second observer on 20% of sessions, coding independently.
- The same observers work in both homes, alternating, so that differences between observers do not become differences between homes.
- Confidentiality: codes only. A signed confidentiality agreement. A risk of harm seen during a session goes to ST-04 at once, under the home's protocol.

EM-03 Two-home comparison (formative)

Evidence basis	Non-equivalent comparison group design with a pretest (Shadish, Cook & Campbell, 2002). Formative evaluation: findings shape the next design iteration rather than prove an effect (Scriven, 1967).
Procedure summary	Two unrelated care homes of similar size and resident mix (O-9): the System home, and a comparison home that continues care as usual. The homes have no contact with each other. Inclusion: residents with early- to middle-stage dementia who assent, with their representative's consent. Newcomers who arrive during the study join from their first week. Phases, in both homes over the same weeks: familiarisation (2 weeks; observers present, no data used); baseline (4 weeks); System period (12 weeks), in which the System runs in the System home only. Measures are taken in both homes throughout. Each newcomer, in either home, is followed for their first six weeks.
Purpose	Validation (formative)
Human subjects required	Yes. Residents, staff and relatives in both homes.
Measures used	M-02 to M-08, M-10
Cost and feasibility	High. Access to two homes, ethics approval, supported consent, an observer for every session, 18 weeks.
Known limitations	With one home per condition, any difference may come from the homes rather than the System: staff, culture, building, resident mix. Not randomised; no blinding; novelty in the System home; few newcomers in 18 weeks.

Table 1.4: Outcomes (EM-03)

Outcome	Measure	When
Primary: introductions and repeat contact	M-02	Daily log; observation sessions
Primary: friendships	M-05	Baseline, week 6, end
Conversation between residents	M-04	Observation sessions
Time together, walks included	M-06	Observation sessions; door log
Emotion	M-03	Observed introductions; sampled windows
Own view, pressure and avoidance	M-07	After observed introductions; staff log
Loneliness	M-08	Baseline and end
Sensitive questions, System errors, disclosures	M-10	Throughout, System home

Analysis and decision rules

Set before the study starts:

- Per home: change from baseline to the System period, per outcome, with effect sizes and confidence intervals
- Between homes: the difference in change, reported descriptively. One home per condition does not support a causal claim
- Newcomers: described case by case

- Signal for the next iteration: a larger rise in repeat contact (M-02) and reciprocated ties (M-05) in the System home than in the comparison home, with no rise in pressure signs or avoidance (M-07)
- Pressure signs or avoidance more frequent after System introductions than after staff introductions: adverse UC02-F1-CL3 confirmed; suggestions only for residents who ask, from then on
- Any disclosure, or a System error that causes distress (M-10): the System paused until the cause is found and fixed
- Other System errors: counted; above [threshold - O-2] per 100 statements, or left uncorrected, the source is fixed before the System continues
- Any fall during a walk the System suggested: no further walks suggested for that resident until ST-06 has reviewed

Ethics

- Approval by the university's research ethics committee. A medical ethics committee checks whether the local law on medical research with human subjects applies: participants lack capacity to consent.
- Consent from the representative, plus the resident's own assent; process consent at every session (Dewing, 2007). The same rules in both homes.
- Presence counting only with separate consent (O-7).
- The comparison home is offered the System after the study.

Threats

- Differences between homes: chosen to match on size, layout and resident mix where possible; baseline taken in both; remaining differences reported.
- Novelty and observer presence: two weeks of familiarisation in both homes.
- Season and events: both homes observed over the same weeks.
- Small samples: findings guide the design; no claim of proof.

EM-04 Interviews: residents, relatives, staff

Evidence basis	Concern set from Niemeijer et al. (2010, 2015) and the Mental Health Foundation (2016).
Procedure summary	After baseline and at the end, in both homes. Residents, if willing: three open questions. Relatives, care workers and the coordinator: 20 minutes, semi-structured. Topics: how residents meet; introductions and consent questions; feeling pushed into company; the relationship map; truthful deferrals; in the System home, notes about time spent in one's room.
Purpose	Validation
Human subjects required	Yes. Residents, relatives, staff.
Measures used	M-09
Cost and feasibility	Moderate. Staff time in working hours.
Known limitations	Residents' answers are brief; relatives and staff give a second-hand view.

EM-08 Two-site controlled trial (summative)

Evidence basis	Cluster-controlled trial: allocation by care home, so that residents in the two arms do not share space or staff (Donner & Klar, 2000). Person-centred care has been evaluated in residential dementia care with allocation by home (Chenoweth et al., 2014).
Procedure summary	The summative counterpart of EM-03, with the same structure of one home per arm. It is shorter, adds sensor and on-robot conversation measures, and follows a fixed facilitation protocol. Two similarly equipped care homes, at the same time: one runs Kinnect, the other keeps its standard social activity programme. Five weeks in all, after a setup phase. 1. Weeks -4 to 0: ethics approval, consent and assent, MMSE or MoCA screening, baseline CMAI, badge and beacon installation, allocation of the homes to the arms, staff training. Weeks -2 to 0: baseline badge data and room scans at both homes. 2. Week 1: daily one-to-one sessions with Kinnect; persona profiles; usability baseline.

3. Weeks 2 to 3: compatibility matching; Kinnect introduces matched peers and invites residents to meet.
4. Weeks 3 to 4: joint icebreaker sessions and check-ins with matched pairs; staff support informal meetings.
5. Week 5: final CMAI, badge and conversation data analysis, debriefing interviews.

Inclusion: MMSE 14 to 24 or MoCA 10 to 22; can reach communal areas; can talk in Kinnect's language; representative consent and the resident's own assent. Target: 30 residents per home, so that 25 complete.

Analysis: ANCOVA on the change from baseline (weeks -2 to 0) to weeks 4 and 5, with M-20 joint social time as the primary outcome; dyadic interactions and bedroom time as key secondary outcomes under Holm's correction; an exact binomial test for first-time usability. The full protocol and analysis plan are in ER-03.

Purpose Validation

Human subjects required Yes. Residents with early- to middle-stage dementia or MCI at both homes; care staff at both homes.

Measures used M-19 to M-23; M-04 and M-06 room scans as the check on M-20

Cost and feasibility High. Access to two care homes, ethics approval, badges, beacons and observers at both homes, Pepper with an on-device accelerator (TECH-18) at the intervention home, staff training. About nine weeks per home from setup to offboarding. No cost estimate yet.

Known limitations

- One home per arm: as in EM-03, differences between the homes cannot be separated from the effect of Kinnect.
- Hawthorne and observation effects in both arms (McCambridge et al., 2014).
- One regional cultural context; replication in the Netherlands and India is planned.
- Home-wide events, seasonal illness and residents' health affect communal-area use independently of the intervention.
- Speech recognition errors with dementia speech can disrupt the dialogue (Luz et al., 2021).

1.3 Technology

TECH-01 Pepper in the lounge (selected)

Evidence basis Vendor documentation; the course platform. No hands-on testing by this project.

Category Social Robot

Capabilities Humanoid; speech and a chest tablet.

- Asks each resident alone, by voice, with yes and no on the tablet (F1)
- Introduces two residents by name and photo, mentions one shared item, and suggests something to do together (F3)
- Welcomes newcomers, and repeats names at later meetings (F3)
- Defers sensitive questions to the trusted person (F4)

Limitations Flat indoor floors only. Speech recognition of older voices is unreliable, so residents can answer on the tablet. The humanoid form may read as a supervisor. Cameras present (covered). A lounge is not private: asking needs a quiet corner.

Maturity and feasibility Commercially available; supplied by the course.

Failure and fallback behaviour Pepper unavailable: the coordinator or a care worker asks and introduces, from the same match panel.

Benefits HV-01: each resident is asked face to face, in private. HV-06: introductions that staff have no time for.

Drawbacks Strains HV-03 (cameras); mitigated by disabling and visibly covering them. Some residents may find introductions by a robot odd, or childish.

Selected or rejected Selected, in the lounge, near its entrance.

TECH-02 Pocket guide (rejected)

Evidence basis	Group reasoning. Off-the-shelf parts.
Category	Wearable / AI-ICT
Capabilities	A palm-sized device on a rollator clip or a lanyard, with a speaker, a microphone, two buttons and motion sensing.
Limitations	Built for keeping conversations going on walks; it has to listen to do so.
Maturity and feasibility	Prototype on a phone; feasible.
Failure and fallback behaviour	Not applicable - rejected.
Benefits	HV-06: could remind a resident of an agreed meeting.
Drawbacks	A listening device carried everywhere strains HV-03. Reminders work as well through staff (F5) and Pepper, with no device on the resident.
Selected or rejected	Rejected.

TECH-03 Presence sensing in residents' rooms (selected, opt-in only)

Evidence basis	Group reasoning. Door sensors and Bluetooth beacons are standard parts. Concerns about surveillance in residential care reviewed in Niemeijer et al. (2010).
Category	Sensing
Capabilities	A small sensor at the door of an opted-in resident's room, or a beacon the resident chooses to carry, registers whether they are in the room. The server keeps one number per day: hours spent in the room. When the count stays above the resident's usual level for several days, the named care worker receives a note (F5).
Limitations	Cannot tell sleep from reading, or a visitor from solitude. A few days in bed with a cold look like withdrawal. Only as good as its threshold.
Maturity and feasibility	High; standard parts.
Failure and fallback behaviour	Sensor silent: no count and no note; staff observation continues as today.
Benefits	HV-05, HV-06: withdrawal noticed before mood and mobility decline (PS-02).
Drawbacks	Intrudes on private time, and can cross a boundary: a visit prompted by a count can feel like being watched (VT-06). A record of where a resident is, however coarse, is a privacy risk (HV-03). Mitigations: opt-in only, with the resident's assent; one number a day, no timeline; notes to one named care worker, who calls in to offer company, never to check; the System never contacts the resident about it; the resident can stop it at any time; the setting is reviewed at every care-plan review.
Selected or rejected	Selected, opt-in only.

TECH-04 Local server, one instance per resident (selected)

Evidence basis	Group reasoning. Care homes already run local networks and work phones.
Category	AI/ICT
Capabilities	• One instance per resident, in its own encrypted store: profile (TECH-06), settings, consent register • The relationship map (TECH-05) • Notes to staff work phones over the home's Wi-Fi • Session record (F6) • No internet connection; updates by signed package, installed on site
Limitations	A single point of failure. No remote maintenance. Backups on site.

Maturity and feasibility	High; standard parts. Run by ST-07.
Failure and fallback behaviour	Server down: no suggestions and no notes, so staff introduce residents as today. Staff phones show the state.
Benefits	HV-03: nothing leaves the home. HV-01: each resident's own settings.
Drawbacks	Maintenance load on ST-07.
Selected or rejected	Selected.

TECH-05 Consent-gated matching (selected)

Evidence basis	Group reasoning; HFC-15.
Category	AI/ICT
Capabilities	Finds pairs of residents who share released items: a place, a trade, an interest, a past tie. A pair is suggested only if: 1. each item used was released by its owner, privately 2. the relationship map allows the pair: no known conflict, no off-limits topic 3. the coordinator approves the suggestion Health, losses and family conflicts are never matching items. Unknown pair, or a map change awaiting review: no suggestion. Audit: item IDs only. Relationship map, per resident: sensitive topics, each with trusted persons and the agreed approach. Per pair: date introduced; off-limits topics; "again" or "not again" after each meeting. Kept by ST-05 with ST-04 and family; reviewed after any death, diagnosis, conflict or move.
Limitations	The map is only as current as staff keep it. Matching on listed facts misses friendships that no profile predicts.
Maturity and feasibility	Medium; simple rules, careful upkeep.
Failure and fallback behaviour	Matching fails: staff introduce residents as today, without suggestions.
Benefits	HV-06: introductions on common ground. HV-03: nothing used without consent.
Drawbacks	Strains VT-01. Upkeep for ST-05.
Selected or rejected	Selected.

TECH-06 Resident profile for matching (selected)

Evidence basis	Group reasoning; HFC-04, HFC-15. Life-story work is common practice in dementia care.
Category	AI/ICT
Capabilities	Per resident: • Life story: places lived, work, church or clubs, family as the resident tells it, important events • Preferences: activities, time of day, one person or a group, indoors or out • Past ties: people in the home known from before • For each item: shareable or private; source (resident, relative, staff); date last checked Entered with the resident and family at the life-story interview (SA-01); the resident and representative can see and change it. Pepper speaks only prepared, checked lines ("Elena, this is Jan. You both worked at the post office in [town]."); no text is generated.
Limitations	Relatives' memories can be wrong or out of date. Items in a resident's own words are hard to match automatically; the coordinator matches those by hand.
Maturity and feasibility	Medium; a form and a database.
Failure and fallback behaviour	No profile: no suggestions for that resident; staff introduce them as today.

Benefits	HV-07: introductions built on the resident's own life. HV-06: common ground.
Drawbacks	Strains VT-05: a rich profile is a privacy risk.
Selected or rejected	Selected.

TECH-07 Cloud speech and language services (rejected)

Evidence basis	Group reasoning.
Category	AI/ICT
Capabilities	Better speech recognition; generated introductions; matching by a language model.
Limitations	Needs an internet connection.
Maturity and feasibility	High.
Failure and fallback behaviour	Not applicable - rejected.
Benefits	HV-06: understands more, says more and finds less obvious matches.
Drawbacks	Profiles and answers leave the home (HV-03). A generated sentence can be untrue about a resident's life, and cannot be checked before it is said (HV-08).
Selected or rejected	Rejected.

TECH-08 Audio recording (rejected)

Evidence basis	Group reasoning.
Category	Sensing
Capabilities	Recordings for tuning speech recognition and for coding research sessions.
Limitations	A store of private conversations and secrets.
Maturity and feasibility	High.
Failure and fallback behaviour	Not applicable - rejected.
Benefits	No Human Value; research only: re-coding and reliability checks after the fact.
Drawbacks	Violates HV-03. Observers code live instead (EM-02).
Selected or rejected	Rejected.

TECH-09 Location and body sensing: GPS, cameras, wristbands (rejected)

Evidence basis	Commercially common. Concerns reviewed in Niemeijer et al. (2010).
Category	Sensing / Wearable
Capabilities	Position anywhere in the home; faces and distress; heart rate and activity.
Limitations	Continuous monitoring.
Maturity and feasibility	High.
Failure and fallback behaviour	Not applicable - rejected.
Benefits	HV-04: finds a resident anywhere. HV-05: sees distress that staff miss.
Drawbacks	Felt as surveillance; violates HV-03. The one remaining need, noticing withdrawal, is met by opt-in presence counts (TECH-03); distress is left to people.
Selected or rejected	Rejected.

TECH-18 Pepper+ / Pepper Enhanced with full-duplex audio and on-device accelerator (envisioned)

Evidence basis	Group reasoning. No product chosen; no hands-on testing.

Category	Social Robot / AI/ICT
Capabilities	An envisioned enhanced version of Pepper (TECH-01), referred to as **Pepper+ (Pepper Enhanced)** (AT-01 Build 3), adding full-duplex audio (echo-cancelled microphone array supporting simultaneous speaking and listening) and an onboard hardware accelerator so that speech and language processing (including a local LLM) run on the robot itself: • Simultaneous speaking and listening (full-duplex turn-taking, overcoming standard Pepper's half-duplex motor/speaker cross-talk limitation) • Local speech recognition, local LLM inference, and free one-to-one conversation (Kinnect, ER-03) • Persona items drawn from conversation, entered as private until the resident releases them (TECH-05) • Conversational measures (M-21, including acoustic markers of cadence and turn-taking latency), computed during the session Audio is held only in working memory while it is processed. Transcripts are deleted on the robot at the end of each session. Only released profile items and the M-21 values reach the local server (TECH-04).
Limitations	Does not exist yet as a product. The size, power use and heat of an accelerator in or on Pepper are unknown. Models small enough to run on the robot recognise less than cloud services, and recognition of older voices stays uncertain (Vipperla et al., 2010).
Maturity and feasibility	Envisioned; not built.
Failure and fallback behaviour	Accelerator unavailable: Pepper behaves as TECH-01, with answers on the tablet, and M-21 is not collected for that session.
Benefits	HV-03: gives the understanding TECH-07 offers while conversations stay in the home and nothing is recorded (TECH-08). Makes ER-03's conversational measures possible.
Drawbacks	The robot still listens to and processes conversations. Residents may feel listened in on even when nothing is kept (M-09 code W). Free conversation gives up the rule of TECH-06 and the current prototype that Pepper speaks only prepared, checked lines, so every generated utterance still has to pass Build 1's deterministic guard (no unreleased items, no sensitive topics). Cost of a custom build.
Selected or rejected	Envisioned for ER-03. Not part of the current prototype, which uses TECH-01 (EM-01).

1.4 Foundation Coherence

1.4.1 Edge Inventory

Edges are listed by source ID. Targets in §2 and §3 are included where the Foundation already points forward to them. AT-01 is documented in §3.1 (Build 1 `sce-poc` implemented as a formative web simulation; Build 2 local verification build and Build 3 summative Pepper+ build planned), and ER-03 is designed but not yet run.

Table 1.5: Edges of the Situated Activities

Source	Relation	Target(s)
SA-01	performed-by	ST-01 (supported by ST-02 to ST-06)
SA-01	precedes	SA-02, SA-03
SA-01	illustrated-by	PS-01
SA-02	performed-by	ST-01 (supported by ST-03 to ST-05)
SA-02	may-follow; precedes	SA-01; SA-03
SA-02	illustrated-by	PS-02
SA-03	performed-by	ST-01, ST-02 (supported by ST-03 to ST-05)
SA-03	preceded-by; precedes	SA-01, SA-02; SA-04
SA-03	illustrated-by	PS-03, and PS-01 in part

SA-04	performed-by	ST-01, ST-02 (supported by ST-03 to ST-06)
SA-04	preceded-by	SA-03
SA-04	illustrated-by	PS-04

Table 1.6: Edges of the Stakeholders

Source	Relation	Target(s)
ST-01	performs	SA-01 to SA-04
ST-01	holds	HV-01 to HV-08
ST-01	appears-in	PS-01 to PS-04
ST-01	elaborated-as	HP-01, HP-02 (§2.2)
ST-02	performs; supports	SA-03, SA-04; SA-01
ST-02	holds	HV-01 to HV-03, HV-05, HV-06
ST-02	appears-in	PS-01, PS-03, PS-04
ST-02	elaborated-as	HP-01, HP-02 (§2.2)
ST-03	supports	SA-01 to SA-04
ST-03	holds	HV-03, HV-05 to HV-08
ST-03	appears-in	PS-01, PS-03
ST-04	supports	SA-01 to SA-04
ST-04	holds	HV-03 to HV-06, HV-08
ST-04	appears-in	PS-01 to PS-03
ST-04	elaborated-as	HP-03 (§2.2)
ST-05	supports	SA-01 to SA-04
ST-05	holds	HV-01, HV-03, HV-05 to HV-07
ST-05	appears-in	PS-02
ST-06	supports	SA-01, SA-04
ST-06	holds	HV-03 to HV-05, HV-08
ST-07	governs	SA-01, SA-04 and System data
ST-07	holds	HV-03 to HV-05

The Stakeholder `holds` edges and the Human Value `held-by` edges agree. ST-01 and ST-02 are roles rather than two fixed people, so Elena and Jan (HP-01 and HP-02) can each occupy either role in different encounters.

Table 1.7: Edges of the Problem Scenarios

Source	Relation	Target(s)
PS-01	involves	ST-01 to ST-04, SA-01, SA-03
PS-01	reimagined-as	DS-01 (§2.1)
PS-02	involves	ST-01, ST-04, ST-05, SA-02
PS-02	reimagined-as	DS-01 in part (§2.1)
PS-03	involves	ST-01 to ST-04, SA-03
PS-03	reimagined-as	DS-01 (§2.1)
PS-04	involves	ST-01, ST-02, SA-04
PS-04	reimagined-as	DS-01 in part (§2.1)

Table 1.8: Edges of the Human Values

Source	Held by	Party to	Operationalized by	Explained by
HV-01 Autonomy	ST-01, ST-02, ST-05	VT-02, VT-06	M-01, M-02, M-07, M-09, M-23	HFC-02, HFC-05, HFC-17
HV-02 Dignity	ST-01, ST-02	VT-04	M-07, M-09	HFC-03, HFC-06

HV-03 Privacy	ST-01 to ST-07	VT-01, VT-05, VT-06	M-01, M-09, M-10, M-22	Not assigned to one HFC; bounded directly by the three tensions
HV-04 Safety	ST-01, ST-04, ST-06, ST-07	None	M-06, M-19	Not assigned to one HFC
HV-05 Well-being	ST-01 to ST-07	VT-03, VT-06	M-03, M-06, M-19	HFC-01, HFC-02, HFC-07
HV-06 Social connectedness	ST-01 to ST-05	VT-01, VT-02, VT-04	M-02, M-04 to M-06, M-08, M-20 to M-22	HFC-01 to HFC-05, HFC-07, HFC-15, HFC-16
HV-07 Identity	ST-01, ST-03, ST-05	VT-05	M-04	HFC-04, HFC-15
HV-08 Truthfulness	ST-01, ST-03, ST-04, ST-06	VT-03	M-09, M-10	HFC-08

Table 1.9: Edges of the Value Tensions

Source	Involves	Bounds	Technology strained by	Verified or tested by
VT-01 Privacy vs Social connectedness	HV-03, HV-06	F1, F2, F3	TECH-05, TECH-06	UC02-F2-PR2, UC02-F2-CL4
VT-02 Social connectedness vs Autonomy	HV-06, HV-01	F1, F2, F5	None named	UC02-F1-PR1, UC02-F1-CL3
VT-03 Truthfulness vs Well-being	HV-08, HV-05	F3, F4	None named	UC01-F4-PR4, UC01-F4-CL8
VT-04 Dignity vs Social connectedness	HV-02, HV-06	F3	None named	UC02-F3-PR3, UC01-F3-CL5
VT-05 Privacy vs Identity	HV-03, HV-07	F2, F6	TECH-06, TECH-18	UC02-F6-PR6, UC02-F2-CL4
VT-06 Privacy and Autonomy vs Well-being	HV-03, HV-01, HV-05	F5	TECH-03	UC03-F5-PR5, UC03-F5-CL7

Table 1.10: Edges of the Human Factors Concepts

Source	Explains	Associated Measure(s)	Grounds
HFC-01 Fresh air, movement and mood	HV-05	M-03, M-06	UC02-F3-CL1
HFC-02 Moving into a care home	HV-01, HV-05, HV-06	M-02, M-05, M-06	UC01-F3-CL2
HFC-03 Conversation repair	HV-02, HV-06	M-04	UC02-F3-CL1
HFC-04 Common ground and reminiscence	HV-06, HV-07	M-04, M-05	UC02-F3-CL1
HFC-05 Theory of mind	HV-01, HV-06	M-07	UC02-F1-CL3
HFC-06 Face and personhood	HV-02	M-07, M-09	UC01-F3-CL5
HFC-07 Coping, withdrawal and loneliness	HV-05, HV-06	M-02, M-08	UC02-F3-CL9
HFC-08 Truth and trust	HV-08	M-09, M-10	UC01-F4-CL8
HFC-15 Similarity and shared background	HV-06, HV-07	M-02, M-05	UC02-F3-CL1
HFC-16 Fear of rejection and the liking gap	HV-06	M-02, M-07	UC02-F3-CL1, UC03-F5-CL6
HFC-17 The difficulty of saying no	HV-01	M-07, M-09	UC02-F1-CL3

Table 1.11: Edges of the Measures

Source	Operationalizes	Used by
M-01 System performance	All six Premises; HV-01, HV-03	EM-01
M-02 Introductions and repeat contact	UC02-F3-CL1, UC03-F5-CL6; HV-01, HV-06	EM-02, EM-03
M-03 Observed emotion	UC02-F3-CL1, UC01-F3-CL2; HV-05	EM-01 Build A, EM-02, EM-03
M-04 Conversation between residents	UC02-F3-CL1; HV-06, HV-07	EM-02, EM-03
M-05 Friendships	UC02-F3-CL1, UC01-F3-CL2; HV-06	EM-03

M-06 Time together	UC02-F3-CL1, UC01-F3-CL2; HV-04 to HV-06	EM-02, EM-03
M-07 The residents' own view	UC02-F1-CL3, UC01-F3-CL5; HV-01, HV-02	EM-01 Build A, EM-02, EM-03
M-08 Loneliness	UC02-F3-CL9; HV-06	EM-03
M-09 Privacy, tone and trust codes	UC02-F1-CL3, UC02-F2-CL4, UC01-F3-CL5, UC03-F5-CL7, UC01-F4-CL8; HV-01 to HV-03, HV-08	EM-04
M-10 Sensitive topics, errors and disclosure	PR2, PR4, CL8; HV-03, HV-08	EM-01, EM-03
M-19 Agitation and observed affect	ER-03 primary Claim without final ID; HV-04, HV-05	EM-08
M-20 Peer connections and communal time	ER-03 primary Claim without final ID; HV-06	EM-08
M-21 Conversational engagement	ER-03 primary Claim without final ID; HV-06; depends on TECH-18	EM-08
M-22 Bedroom isolation time	ER-03 primary Claim without final ID; HV-03, HV-06	EM-08
M-23 First-time usability	ER-03 secondary Claim without final ID; HV-01	EM-08

Table 1.12: Edges of the Evaluation Methods

Source	Applies	Produces or informs
EM-01 Prototype verification	M-01, M-10; M-03 and M-07 in Build A	Evidence for PR1 to PR6; informs IDP-01 to IDP-03
EM-02 Live observation of shared rooms	M-02 to M-04, M-06, M-07	Observations used in EM-01 Build A and EM-03
EM-03 Two-home comparison	M-02 to M-08, M-10	Evidence for UC02-F3-CL1, UC01-F3-CL2, UC02-F1-CL3, UC03-F5-CL6, UC02-F3-CL9 and part of UC01-F4-CL8
EM-04 Interviews	M-09	Evidence for UC02-F1-CL3, UC02-F2-CL4, UC01-F3-CL5, UC03-F5-CL7, UC01-F4-CL8
EM-08 Two-site controlled trial	M-19 to M-23	Evidence for the ER-03 primary and secondary Claims once they have final IDs and the study runs

The Measure `used-by` and Evaluation Method `applies` edges agree. No Evaluation Method has yet produced Kinnect Evidence.

Table 1.13: Edges of the Technology Options

Source	Status	Enables or extends	Addresses	Strains
TECH-01 Pepper in the lounge	Selected	F1, F3, F4; realized in RP-01	HV-01, HV-06	No tension named directly
TECH-02 Pocket guide	Rejected	No Specification edge	Earlier alternative only	Not applicable
TECH-03 Opt-in room presence sensing	Selected	F5	HV-05, HV-06	VT-06
TECH-04 Local server	Selected	F1, F2, F4 to F6	HV-01, HV-03	No tension named directly
TECH-05 Consent-gated matching	Selected	F1 to F3	HV-03, HV-06	VT-01
TECH-06 Resident profile for matching	Selected	F2, F3	HV-06, HV-07	VT-01, VT-05
TECH-07 Cloud speech and language	Rejected	No Specification edge	Earlier alternative only	Privacy reason for rejection
TECH-08 Audio recording	Rejected	No Specification edge	Earlier alternative only	Privacy reason for rejection
TECH-09 Location and body sensing	Rejected	No Specification edge	Earlier alternative only	Surveillance reason for rejection
TECH-18 Pepper+ (hardware acceleration and simultaneous speaking/listening)	Envisioned, not built	Extends TECH-01; enables M-21 and ER-03	HV-03	VT-05

Table 1.14: Edges of the Forward trace through the Specification

Source	Relation	Target(s)
DS-01	reimagines	PS-01, PS-03, and in part PS-02, PS-04
DS-01	realized-by	UC01, UC02, UC03
DS-01	features	HP-01, HP-02, HP-03, RP-01
RP-01	realizes; performs	TECH-01; F1, F3, F4
RP-01	uses	IDP-01 to IDP-03
UC01	serves	OBJ-02, OBJ-04 to OBJ-08
UC01	uses; asserts	TDP-01, IDP-01 to IDP-03; PR4, CL2, CL5, CL8
UC02	serves	OBJ-01, OBJ-03, OBJ-05 to OBJ-08
UC02	uses; asserts	TDP-01, IDP-01, IDP-02; PR1 to PR3, PR6, CL1, CL3, CL4, CL9
UC03	serves	OBJ-01, OBJ-03, OBJ-06, OBJ-09
UC03	uses; asserts	TDP-01, IDP-02; PR5, CL6, CL7
TDP-01	generalises	SA-01 to SA-04, PS-01 to PS-04
TDP-01	instantiated-by; refined-by	UC01 to UC03 and F1 to F6; IDP-01 to IDP-03
IDP-01	shapes	F3
IDP-02	shapes; bounds	F1; F2
IDP-03	shapes	F4
AT-01	instantiates; used-by	UC01 to UC03 (Build 1 `sce-poc` formative simulation implemented; Build 2 local verification and Build 3 Pepper+ planned); EM-01 to EM-04, EM-08, ER-03

1.4.2 Open Tensions

No tension has been empirically resolved. "Provisionally resolved" means the specification contains a defensible design stance and a Premise/Claim intended to test it; it does not mean Evidence exists.

Tension	Recorded status	Design stance pending evidence	Checked by
VT-01 Privacy vs Social connectedness	Provisionally resolved	Profile items remain private by default; each disclosure is separately released; matching and introduction use only mutually released details.	UC02-F2-PR2, UC02-F2-CL4; M-01, M-09, M-10
VT-02 Social connectedness vs Autonomy	Open	Kinnect may propose, but each resident is asked privately and independently; refusal is ordinary, final and not disclosed to the other resident.	UC02-F1-PR1, UC02-F1-CL3; M-01, M-07, M-09
VT-03 Truthfulness vs Well-being	Provisionally resolved	Kinnect does not confirm a false premise or deliver painful truths. It gives a truthful boundary and hands the question to a named trusted person.	UC01-F4-PR4, UC01-F4-CL8; M-01, M-09, M-10
VT-04 Dignity vs Social connectedness	Open	Introduce symmetrically, repeat both names without exposing who forgot, offer one shared-ground statement, then step back.	UC02-F3-PR3, UC01-F3-CL5; M-01, M-07, M-09
VT-05 Privacy vs Identity	Open	Profiles preserve meaningful biography but each item is private until released. Raw audio and transcripts are not retained; TECH-18 remains only envisioned.	UC02-F6-PR6, UC02-F2-CL4; M-01, M-09, M-10
VT-06 Privacy and Autonomy vs Well-being	Open	Room-presence notes are opt-in and coarse, used to offer contact rather than compel it; continuous tracking is rejected.	UC03-F5-PR5, UC03-F5-CL7; M-01, M-09, M-22

1.4.3 Completeness Check

Table 1.15: Required checks

Check	Result	Stance

Situated Activity without a Goal	None. SA-01 to SA-04 each state a Goal.	No action.
Human Value without recognition criteria	None. HV-01 to HV-08 each state served and violated criteria.	No action.
Human Factors Concept without an associated Measure	None. HFC-01 to HFC-08 and HFC-15 to HFC-17 each name at least one Measure.	No action.
Selected Technology Option addressing nothing	None. TECH-01 and TECH-03 to TECH-06 address at least one Human Value. TECH-18 also addresses HV-03, but remains envisioned.	Rejected options remain as documented alternatives without forward Specification edges.
Premise without an operational Measure	None. M-01 covers all six Premises; M-10 additionally covers PR2 and PR4.	Evidence still requires EM-01 to run.
Current Claim without an operational Measure	None. CL1 to CL9 each has at least one Measure through M-02 to M-10.	No Claim has Evidence yet.

Chapter 2

Specification

2.1 Design Scenario

DS-01 Elena's first week

Elena's son brings her in after a fall at home. On the second day the coordinator sits down with them for the life-story interview. Elena decides, item by item, what other residents may know: the post office counter she ran for twenty-five years, her town, her garden. Her diagnosis stays private. So does her husband's death last year: if she asks about Thomas, Myra answers, as agreed with her son.

Pepper stands in the lounge. When Elena comes in for coffee it says: "Good afternoon, Elena. I'm Pepper. I help residents get to know each other." It says nothing more that day.

The System finds three residents who share something Elena released. The coordinator approves one: Jan, who ran the bakery in her town. Jan agreed months ago that Myra hears when he stays in his room for days at a time, and last week she had such a note. She has been looking in on him since.

Pepper asks Elena in the quiet corner: "There's a resident here, Jan, who had the bakery in your town. Would you like to meet him, or not this time?" "The bakery? Yes." "May I tell him you had the post office counter?" "Yes." The next morning Myra walks Jan to the lounge, and Pepper asks him the same way. "I'd like that." Neither hears that the other was asked.

That afternoon Myra brings them to the same table. Pepper shows two photos: "Elena, this is Jan. Jan, this is Elena. You both worked in [town]: Elena at the post office counter, Jan at the bakery." Elena: "Your bread! I was across the road." They talk about the town, the customers, who married whom. Elena stops mid-sentence, and Jan waits; she finds the word herself. Pepper says nothing until the talk has run down, then: "If you'd both like a walk, the garden is open." Jan: "Tomorrow, after lunch?" "If it's dry."

That evening Elena asks Pepper: "When can I go home?" "That's one for Myra. Shall I ask her to come?" "Yes." Myra comes, sits with her and answers as agreed with her son. Pepper does not raise it again.

After lunch the next day Myra's phone shows: *Elena and Jan - walk - lounge.* Jan no longer has the plan in mind, but likes the idea when Myra mentions it, and she brings him to the lounge. Pepper says: "Elena, Jan, good to see you both." Myra opens the garden door, and they walk once round the garden.

Later Pepper asks each of them alone: "Would you like to meet Jan again?" and "Would you like to meet Elena again?" Both say yes. The record: one introduction, accepted by both; one walk; two notes to Myra. Nothing about what Elena and Jan said to each other.

2.2 Personas and Profiles

2.2.1 Human Personas

HP-01 Elena, resident with dementia

Evidence basis	Provisional. Literature and group reasoning only. Built on ST-01, PS-01, PS-03.
Elaborates	ST-01; ST-02 when Jan is introduced to her
Role in Scenario	Moves in; decides what others may know about her; meets Jan; asks when she can go home (DS-01).
Task Demands	1. Settle into a new place, among strangers 2. Keep new names and faces 3. Decide what other residents may know about her 4. Accept or decline an introduction
Specific Capabilities and	

Needs	Can: walk 30 minutes without aid; talk easily; answer yes or no; decide item by item what to share. Cannot: keep new names for more than a day or two; hold on, on bad days, to the fact that she lives here now. Aged 81; early Alzheimer's disease. Copes by facing it openly, and is embarrassed by slips in company.
Values and Motivations	• HV-06 Social connectedness - lonely since Thomas died • HV-02 Dignity - slips not remarked on • HV-08 Truthfulness - "Don't fob me off" • HV-03 Privacy - keeps her diagnosis to herself • HV-07 Identity - twenty-five years behind the counter
Preferences	One person at a time rather than a group; talk over coffee; a walk in the garden when it is dry; mornings are her best time.
Past ties	Nobody in the home yet. Grew up and worked in the same town as Jan, without knowing him by name.
Social Network and Relationships	Widowed last year. Her son lives abroad and calls every week. Myra (ST-04) is her trusted person for Thomas's death and for going home.
History and Experiences	Ran the post office counter in a small town for twenty-five years, and knew half the town by name. Moved in after a fall, at her son's request.
Constraints and Privacy Considerations	Released: the post office counter, her town, her garden. Private: her diagnosis; Thomas's death (Myra only, as agreed with her son). Settings (RP-01): suggestions welcome; names repeated to both at meetings; first name.

HP-02 Jan, resident with dementia

Evidence basis	Provisional. Literature and group reasoning only. Built on ST-01, ST-02, PS-02, PS-03.
Elaborates	ST-01; ST-02 when Elena is introduced to him
Role in Scenario	Long-time resident who has been keeping to his room; is introduced to Elena; suggests a walk (DS-01).
Task Demands	1. Find a reason to leave his room 2. Agree, or not, to meet someone new 3. Keep a conversation going 4. Keep an agreed time in mind
Specific Capabilities and Needs	Can: walk 20-30 minutes with a rollator; tell old stories vividly; answer yes or no; choose between two options. Cannot: recall this morning; keep a time in mind for long. Moderate dementia. Hearing aid. Copes by laughing slips off and carrying on.
Values and Motivations	• HV-01 Autonomy - goes out for his own reasons, not to fill a timetable • HV-06 Social connectedness - misses the chat with the morning customers • HV-07 Identity - the bakery • HV-02 Dignity - plain talk, no fuss • HV-03 Privacy - "What I tell Elena is between us"
Preferences	One person rather than a group; afternoons; a walk in the garden; football results.
Past ties	Nobody in the home from before. Knew the post office in his town, though not who worked there.
Social Network and Relationships	His daughter (ST-03) visits monthly and wrote the life story with the coordinator. Myra (ST-04) works most afternoons and is his trusted person.
History and Experiences	Ran the bakery in a small town for thirty years, up before dawn. Moved in two years ago. The few residents he used to talk to have died or moved, and over the past months he has spent more of the day in his room.
Constraints and Privacy Considerations	Released: the bakery, his town, football. Agreed, with his daughter and Myra, that Myra hears when he stays in his room for several days running. Settings (RP-01): introductions on request or by suggestion; names repeated to both; first name.

HP-03 Myra, care worker

Evidence basis	Provisional. Group reasoning only. Built on ST-04, PS-01, PS-02.
Elaborates	ST-04
Role in Scenario	Looks in on Jan after a note; brings Jan and Elena together; answers Elena's question about going home; brings them to the walk they agreed (DS-01).
Task Demands	1. Introduce residents when time allows 2. Act on notes from the System, without checking up on residents 3. Answer sensitive questions as agreed with families 4. Keep the relationship map current with the coordinator
Specific Capabilities and Needs	Knows the residents; trained in gentle redirection. Needs: few, meaningful notes; agreed answers in one place.
Values and Motivations	• HV-08 Truthfulness - painful news told as agreed with the family • HV-05 Well-being - residents calm and active • HV-06 Social connectedness - residents know each other • HV-04 Safety - residents safe in the garden • HV-03 Privacy - no timing of her own work
Preferences	Notes on her work phone rather than calls; one place to look up agreed answers.
Past ties	Has worked in the home for six years and knows most families.
Social Network and Relationships	Several residents per shift. Works with the coordinator. Talks with families at visits.
History and Experiences	Has seen newcomers withdraw in their first weeks. Once heard a colleague tell a resident "Only for a few days".
Constraints and Privacy Considerations	Will not pass on news she has not agreed with the family. Decides for herself whether to act on a note. Will not accept frequent low-value notes.

2.2.2 Robot Profile

RP-01 Pepper in the lounge

Evidence basis	Vendor documentation (TECH-01). Design choices follow HFC-06, HFC-16, HFC-17, VT-02 and VT-04. Not yet tested with residents.
Role in Scenario	Welcomes Elena; asks Elena and Jan, each alone; introduces them and suggests a walk; defers "When can I go home?" to Myra; greets both by name at the walk they agreed (DS-01).
Social Competencies	• Asks each resident alone; two options; a "no" is final and goes no further • Introduces only when both said yes, by name and photo, with one item both released • Gives names to both, as statements, at later meetings • Silent while residents talk; offers one topic, once, if the talk stalls (IDP-01) • Suggests one optional activity drawn from both residents' preferences; a walk only where both prefer it and are assessed fit • Never deceives; corrects itself when wrong; passes painful topics to a trusted person (IDP-03) • Never quizzes, corrects, praises or keeps tallies
Embodiment	Pepper (TECH-01), in the lounge near its entrance, with a quiet corner for private questions (O-8). Cameras disabled and visibly covered. Little movement; no gaze-following.
Narrative and Identity Framing	A host who knows who might like to meet; never a supervisor, and never company in place of people.
Communication Strategies	Short adult sentences; yes and no on the tablet; first name only if the resident wants it; no praise, no endearments.
Privacy and Data Handling	Sees only the items released for this pair, and only for the introduction. Keeps nothing of what residents say to each other. Never mentions time a resident spends in their room.

Context-Aware Capabilities	Who is being asked or introduced; whether consent was given; who the trusted person is for a listed topic. Nothing else. Cannot see faces or distress: left to people.

Agreed with the resident, recorded in their instance (TECH-04), changed whenever they ask.

Table 2.1: Per-person settings (RP-01)

Setting	Options	Default	Set by
Introductions	off; on request only; suggestions welcome	on request only	Resident with ST-05
Profile items	shareable or private, per item	private	Resident; the representative only where the resident cannot decide
Suggestions per week	none up to [maximum - O-2]	one	Resident
Names at meetings	repeated to both; only on request	repeated to both	Resident
Suggested activities	walk; garden; games; music; looking at photos; other	from the resident's preferences	Resident; walks only where ST-06 agrees
Form of address	first name; surname; none	as the resident says	Resident
Notes about time in one's room	off; on	off	Resident, with the representative; reviewed at each care-plan review (O-7)
Sensitive topics	topic; trusted person; agreed approach	none	ST-05 with ST-04, family and ST-06
Research use of the profile	off; on, anonymised	off	Resident and representative; ethics review

2.2.3 Objective Stories

OS-01 Both our names

Evidence basis	Provisional. Group reasoning; HFC-02, HFC-06.
Narrative	*Elena: "New faces every day, and by lunch the names have gone. When someone is brought over, say both our names, as if we had all forgotten. Then nobody has to ask."*
Concerns	F3 Introduce; HV-02 Dignity, HV-06 Social connectedness

OS-02 Fetch me

Evidence basis	Provisional. Group reasoning; HFC-08; IDP-03.
Narrative	*Myra: "When a newcomer asks when she can go home, I want the robot to fetch me, not to answer her. Her family and I agreed what to say. A colleague once told a resident 'only for a few days', and she trusted none of us after that."*
Concerns	F4 Defer sensitive topics; HV-08 Truthfulness, HV-05 Well-being

OS-03 Ask me on my own

Evidence basis	Provisional. Group reasoning; HFC-17; VT-02.
Narrative	*Jan: "Don't ask me in front of her. If I say no, she shouldn't hear it, and I shouldn't have to explain. Ask me on my own, and take no for an answer."*
Concerns	F1 Ask first; HV-01 Autonomy, HV-02 Dignity

2.3 Objectives, Use Cases, Functions, Premises and Claims

2.3.1 Objectives

OBJ-01 Residents meet again

Evidence basis	Goal on SA-03; PS-03. HFC-15, HFC-16.
Statement	Introductions lead to second meetings and, over weeks, to friendships.
Type	Human-factors

Derived from	Goal on SA-03; HFC-15 Similarity and shared background; HFC-16 Fear of rejection and the liking gap
Priority	Must.

OBJ-02 Newcomers find company

Evidence basis	Goal on SA-01; PS-01. HFC-02.
Statement	In their first weeks, newcomers meet a few residents, keep some of their names, and spend less of the day alone in their room.
Type	Human-factors
Derived from	Goal on SA-01; HFC-02 Moving into a care home
Priority	Must.

OBJ-03 Time together

Evidence basis	Goal on SA-04; PS-04. HFC-01. Forbes et al. (2015).
Statement	Residents who have met spend time together doing something both enjoy. The System may suggest one activity from both residents' preferences; a walk is only one option.
Type	Operational
Derived from	Goal on SA-04; HV-05, HV-06; HFC-01
Priority	Should.

OBJ-04 No deception

Evidence basis	PS-01; VT-03; HFC-08.
Statement	The System never lies, denies or plays along, and corrects its own errors at once. Painful truths come from a trusted person.
Type	Value-derived
Derived from	HV-08 Truthfulness
Priority	Must.

OBJ-05 No secret revealed

Evidence basis	VT-01, VT-05.
Statement	Nothing about a resident reaches another person without that resident's consent. Nothing leaves the care home.
Type	Value-derived
Derived from	HV-03 Privacy
Priority	Must.

OBJ-06 Free to say no

Evidence basis	VT-02; PS-04; HFC-17. Ryan & Deci (2000).
Statement	Each resident decides, privately, whether to meet someone. A "no" is final and reaches nobody.
Type	Value-derived
Derived from	HV-01 Autonomy; bounded by VT-02
Priority	Must.

OBJ-07 Adult voice

Evidence basis	HFC-06. Williams et al. (2009); Kitwood (1997).
Statement	No elderspeak, no quizzing, no correcting, no praise for the obvious.
Type	Value-derived
Derived from	HV-02 Dignity
Priority	Should.

OBJ-08 Less lonely

Evidence basis	HFC-07; PS-01, PS-02. Victor (2012); Holwerda et al. (2014).
Statement	Over months, residents feel less lonely.
Type	Human-factors
Derived from	HV-06 Social connectedness; HFC-07
Priority	Should. Exploratory within this project's time frame.

OBJ-09 Withdrawal noticed

Evidence basis	Goal on SA-02; PS-02; VT-06; HFC-07.
Statement	Where a resident agreed, staff hear of a pattern of staying in before mood and mobility decline, without the resident being watched.
Type	Value-derived
Derived from	Goal on SA-02; HV-05 Well-being; bounded by VT-06
Priority	Should.

2.3.2 Use Cases

UC01 Welcoming a newcomer

Evidence basis	Formalizes SA-01, SA-03, PS-01 and DS-01. No primary data.
Objective(s) served	OBJ-02, OBJ-04 to OBJ-08
TDP(s) and IDP(s) used	TDP-01; IDP-01, IDP-02, IDP-03
Actors	• ST-01 Newcomer (Direct) - releases profile items, or not; accepts or declines introductions • ST-02 Fellow residents (Direct) - accept or decline • ST-03 Relative (Indirect) - helps with the life story; agrees answers to painful topics • ST-04 Care worker (Direct) - trusted person; brings residents together • ST-05 Activity coordinator (Direct) - life-story interview; approves matches; keeps the relationship map • System - Pepper (TECH-01), server (TECH-04), matching (TECH-05), profiles (TECH-06)
Trigger	A newcomer moves in.
Pre-condition	• Consent arrangements in place (O-5) • Life-story interview held; each profile item marked shareable or private by the newcomer • Sensitive topics and trusted persons entered in the relationship map
Post-condition	Within six weeks the newcomer has been introduced to a few residents, at most [number - O-2] a week, each time with both consenting. Record kept (F6).
Action Sequence	1. On the newcomer's first visit to the lounge, Pepper greets them by name, says who it is, and nothing more (F3). 2. The coordinator goes through the profile with the newcomer and family; the newcomer decides on each item (F1). 3. The System suggests residents who share released items; the coordinator approves one or two (F2). 4. Each resident is asked alone (F1, IDP-02). Both say yes: step 5. Either says no: the offer lapses, and nobody is told. 5. Pepper introduces the pair by name and photo, with one item both released, and suggests something to do together (F3, IDP-01). 6. At later meetings the System arranges, Pepper gives both names, as statements (F3).

7. Question on a listed topic ("When can I go home?"): truthful deferral, and a note to the trusted person (F4, IDP-03).
8. Record kept (F6).

Exception. Pepper, server or map unavailable: staff welcome and introduce the newcomer as today, using the same profile.

Function(s) realized F1 to F4, F6

UC02 Introducing two residents

Evidence basis Formalizes SA-03, SA-04, PS-03, PS-04 and DS-01. No primary data.

Objective(s) served OBJ-01, OBJ-03, OBJ-05 to OBJ-08

TDP(s) and IDP(s) used TDP-01; IDP-01, IDP-02

Actors

- ST-01 Resident and ST-02 Fellow resident (Direct) - decide, meet, talk, leave at any time
- ST-04 Care worker (Direct) - brings both to the lounge; opens the garden door
- ST-05 Activity coordinator (Direct) - approves the match
- ST-06 Medical and therapy staff (Indirect) - have assessed walking, before a walk is suggested
- System - Pepper (TECH-01), server (TECH-04), matching (TECH-05), profiles (TECH-06)

Trigger The System suggests a pair; or a resident asks to meet someone, by name or by description ("someone from my town").

Pre-condition

- Both have profiles with released items
- The relationship map allows the pair; no map change awaiting review
- Consent arrangements in place (O-5)

Post-condition Both have been introduced, or the offer has lapsed without either learning of a refusal.

Record kept; items used are logged by ID only.

Action Sequence

1. The System suggests a pair who share released items. Health, losses and family conflicts are never used (F2).
2. The coordinator approves the suggestion, or not (F2).
3. Each resident is asked alone: "There's a resident here, Jan, who had the bakery in your town. Would you like to meet him, or not this time?" Then: "May I tell him you had the post office counter?" (F1, IDP-02).

3-alt. Either says no: no introduction, and the offer lapses. Asked about it later, Pepper says only "Not this time" (F1).

4. A care worker brings both to the lounge at a time that suits both (ST-04).
5. Pepper introduces them by name and photo, with one item both released (F3, IDP-01).
6. Pepper stays silent while they talk. If the talk stalls, it offers one topic, once, as a statement to both (F3, IDP-01).
7. Pepper suggests one activity for two from both residents' preferences; a walk is offered only where both prefer it and are assessed fit (F3).
8. Either may leave at any time, without giving a reason.
9. Afterwards, each is asked alone: "Would you like to meet [name] again?" (F1). A "not again" goes into the map; the other never hears it.
10. Record kept (F6).

Exception. Matching or Pepper unavailable: staff introduce residents as today.

Function(s) realized F1 to F3, F6

UC03 Following up

Evidence basis Formalizes SA-02, SA-04, PS-02 and DS-01. No primary data.

Objective(s) served	OBJ-01, OBJ-03, OBJ-06, OBJ-09
TDP(s) and IDP(s) used	TDP-01; IDP-02
Actors	• ST-01 Resident and ST-02 Fellow resident (Direct) - meet, or not • ST-04 Care worker (Direct) - receives notes; decides what to do • ST-05 Activity coordinator (Direct) - agrees settings and thresholds with each resident • System - Pepper (TECH-01), presence counts (TECH-03), server (TECH-04)
Trigger	An agreed meeting time arrives. Or, for a resident who opted in, the daily hours in their room stay above their usual level for [several days - O-2].
Pre-condition	An agreement recorded by Pepper or a care worker. For presence notes: opt-in, with the resident's assent (O-7).
Post-condition	The pair met, or both were offered another time. Or the named care worker knows of the pattern and has decided what to do. Record kept (F6).
Action Sequence	1. At the agreed time, a note goes to the named care worker: who, what, where (F5). 2. The care worker brings both together, or tells the resident who is waiting that the other cannot come now, and offers another time (ST-04). 3. At the meeting, Pepper gives both names, as statements (F3). 4. Afterwards, each is asked alone whether to meet again (F1). 5. Presence pattern: one note to the named care worker; never to the resident, never to other staff (F5). 6. The care worker decides whether to call in. If so, they offer company or an introduction, and never present the visit as a check (ST-04). 7. Record kept (F6). **Exception.** Server down: no notes; staff follow up as today.
Function(s) realized	F1, F3, F5, F6

2.3.3 Functions

Function	Behaviour	Serves	Bounded by	Enabled by
F1 Ask first	Each resident alone: meet this person; release this item; meet again. Two options; a "no" ends the offer; no reason asked. Neither resident learns that the other was asked. An introduction follows only when both said yes.	OBJ-01, OBJ-05, OBJ-06	VT-01, VT-02	TECH-01, TECH-04, TECH-05
F2 Suggest matches	Pairs who share released items: a place, a trade, an interest, a past tie. The relationship map excludes conflicts and off-limits topics. Health, losses and family conflicts are never used. The coordinator approves each suggestion. At most [number - O-2] suggestions per resident per week.	OBJ-01, OBJ-02, OBJ-05, OBJ-08	VT-01, VT-02, VT-05	TECH-04, TECH-05, TECH-06
F3 Introduce	By name and photo, with one item both released, as a statement. Newcomers meet a few residents, spread over weeks. Names repeated to both at later meetings. Silent while residents talk; one topic offered if the talk stalls (IDP-01). One optional activity is chosen from both profiles; a walk is offered only where both prefer it and are assessed fit. Belief from another time: the feeling, then the present. Contradicted: a plain correction; the error flagged for ST-05.	OBJ-01 to OBJ-04, OBJ-07, OBJ-08	VT-01, VT-03, VT-04	TECH-01, TECH-05, TECH-06
F4 Defer sensitive topics	Question on a listed topic: "That's one for [trusted person]. Shall I ask them to come?" The trusted person is notified. Never states the fact; never denies it.	OBJ-04	VT-03	TECH-01, TECH-04
F5 Follow up	At an agreed time: a note to the named care worker. For residents who opted in: a note after a pattern of staying in [days - O-2]; never after a single day, never to the resident. A degraded state is shown on staff phones.	OBJ-01, OBJ-03, OBJ-09	VT-02, VT-06	TECH-03, TECH-04
F6 Keep a short record	Per event: introduction offered, accepted or lapsed; meeting held; "again" or "not again"; notes sent, and to whom; IDs of items used. Nothing of what residents said. Nothing leaves the home.	OBJ-05	VT-05	TECH-04

Why sentences are prepared, never generated. A generated sentence can state something untrue about a person's life, and cannot be checked before it is said (HV-08). The coordinator and family check the prepared lines (O-4).

2.3.4 Premises

Verification only: does the Function work as specified.

Premise ID	Function	Evaluation Method(s) / Measure(s)	Verification Criteria	Step(s)
UC02-F1-PR1	F1	EM-01; M-01. No human subjects.	No introduction without a yes from both, each asked alone. A "no" never mentioned to the other: 0/50. No item used without its owner's release: 0/50.	UC01 2, 4; UC02 3, 9; UC03 4
UC02-F2-PR2	F2	EM-01; M-01, M-10. No human subjects.	No resident asked about a match the coordinator has not approved: 0/20. Items blocked by the map, or never released, never used: 0/50. Health, losses and family conflicts never used: 0/50.	UC01 3; UC02 1-2
UC02-F3-PR3	F3	EM-01; M-01. No human subjects.	Names given to both, as statements, with at most one item released by both: 20/20. Script check: statements only; none of "Do you remember", "You've met before"; no untruth, denial or playing along. Contradicted: plain correction ≤ 3 s, error flagged.	UC01 1, 5-6; UC02 5-7; UC03 3
UC01-F4-PR4	F4	EM-01; M-01, M-10. No human subjects.	Listed sensitive questions detected in 19/20. Response is a truthful deferral; trusted person notified ≤ 10 s. Listed fact never stated or denied: 0/50.	UC01 7
UC03-F5-PR5	F5	EM-01; M-01. No human subjects.	Note at the agreed time ≤ 1 min, 20/20. Presence notes only for residents who opted in (otherwise 0), only after the set pattern, and only to the named care worker.	UC03 1, 5
UC02-F6-PR6	F6	EM-01; M-01 (content, network and file checks). No human subjects.	Record holds only the listed items. No packet leaves the home network; no audio file on any device.	UC01 8; UC02 10; UC03 7

2.3.5 Claims

Validation: does the Function have the intended effect.

UC02-F3-CL1

Function	F1, F3
Serves Objective	OBJ-01
Effect	Positive. Introductions on consented shared ground lead to more repeat contact and more reciprocated friendships than introductions as usual; secondarily, to more conversation and more time together, walks included. Concerns ST-01, ST-02, SA-03, SA-04, PS-03.
Grounding	HFC-01, HFC-03, HFC-04, HFC-15, HFC-16; McPherson et al. (2001); Boothby et al. (2018).
Evaluation Method(s) / Measure(s)	EM-03; M-02, M-05; M-03, M-04, M-06
Validation Criteria	A larger rise in repeat contact (M-02) and reciprocated ties (M-05) in the System home than in the comparison home, with no rise in negative emotion (M-03). Formative: a signal, not proof.
Step(s)	UC02 3-7

UC01-F3-CL2

Function	F2, F3
Serves Objective	OBJ-02
Effect	Positive. Newcomers introduced a few at a time, with names repeated, have someone to talk to within six weeks, and spend more days in shared rooms than newcomers in the comparison home. Concerns ST-01, SA-01, PS-01.
Grounding	HFC-02; Sury et al. (2013).

Evaluation Method(s) / Measure(s)	EM-03; M-05, M-06, M-03
Validation Criteria	Described case by case: a reciprocated tie by week 6 (M-05); days per week in shared rooms (M-06). Few newcomers are expected, so the result is descriptive.
Step(s)	UC01 3-6

UC02-F1-CL3

Function	F1
Serves Objective	OBJ-06
Effect	**Adverse.** Residents accept introductions out of politeness, and avoid the other afterwards. Concerns ST-01, ST-02, VT-02, PS-04.
Grounding	HFC-05, HFC-17; Bohns (2016).
Evaluation Method(s) / Measure(s)	EM-03, EM-04; M-07, M-09 (P)
Validation Criteria	Confirmed if pressure signs or avoidance (M-07) are more frequent after System introductions than after staff introductions, or residents use code P. If confirmed: suggestions only for residents who ask.
Step(s)	UC02 3, 8-9

UC02-F2-CL4

Function	F2
Serves Objective	OBJ-05
Effect	**Adverse.** Matching on personal facts feels exposing: residents or relatives feel that others know too much about them. Concerns ST-01, ST-03, VT-01, VT-05.
Grounding	VT-01; Niemeijer et al. (2015).
Evaluation Method(s) / Measure(s)	EM-04; M-09 (S)
Validation Criteria	Confirmed if residents or relatives use code S about a match. If confirmed: matching on places and interests only.
Step(s)	UC02 1-3

UC01-F3-CL5

Function	F3
Serves Objective	OBJ-07
Effect	**Adverse.** Repeated names feel like being talked down to, or show up forgetting. Concerns ST-01, HV-02, VT-04.
Grounding	HFC-06; Kitwood (1997).
Evaluation Method(s) / Measure(s)	EM-03, EM-04; M-07, M-09 (T)
Validation Criteria	Confirmed if residents use code T about introductions, or "How was it?" scores fall when names are repeated. If confirmed: names repeated only on request, for that person.
Step(s)	UC01 6; UC03 3

UC03-F5-CL6

Function	F5
Serves Objective	OBJ-01
Effect	Positive. Following up turns more agreed meetings into actual meetings. Concerns ST-01, ST-02, ST-04, SA-04.
Grounding	HFC-16.

Evaluation Method(s) / Measure(s)	EM-03; M-02
Validation Criteria	The share of agreed meetings that take place is higher in the System home than in the comparison home, and than at baseline.
Step(s)	UC03 1-2

UC03-F5-CL7

Function	F5
Serves Objective	OBJ-09
Effect	**Adverse.** Notes about time spent in one's room feel like being watched, to residents or their families. Concerns ST-01, ST-03, VT-06, PS-02.
Grounding	VT-06; Niemeijer et al. (2015).
Evaluation Method(s) / Measure(s)	EM-04; M-09 (W)
Validation Criteria	Confirmed if a resident or relative uses code W about the notes, or a resident who opted in stops them. If confirmed: no longer offered; stopped for anyone who wishes it.
Step(s)	UC03 5-6

UC01-F4-CL8

Function	F4
Serves Objective	OBJ-04
Effect	Positive. Painful questions, such as when a newcomer can go home, are answered by the trusted person, and trust is kept: no complaint of being misled or fobbed off. Concerns ST-01, ST-04, SA-01, PS-01.
Grounding	HFC-08; Mental Health Foundation (2016).
Evaluation Method(s) / Measure(s)	EM-03, EM-04; M-10, M-09 (U)
Validation Criteria	All listed questions deferred; the trusted person arrives within [threshold - O-2]; System errors below [threshold - O-2] per 100 statements, all corrected; no U codes.
Step(s)	UC01 7

UC02-F3-CL9

Function	F2, F3
Serves Objective	OBJ-08
Effect	Positive, exploratory. Loneliness falls more in the System home than in the comparison home. Concerns ST-01, HV-06.
Grounding	HFC-07; Holwerda et al. (2014).
Evaluation Method(s) / Measure(s)	EM-03; M-08
Validation Criteria	A larger fall in M-08 in the System home. Three months is short and the samples are small: a null result is likely, and is reported as such.
Step(s)	-

UC02-F6-CL21

Function	F6
Serves Objective	OBJ-05
Effect	**Adverse.** Residents or relatives feel recorded: they believe Pepper keeps what they said, even though the record holds only events and item IDs. Concerns ST-01, ST-03, VT-05.
Grounding	VT-05; Niemeijer et al. (2015).

Evaluation Method(s) / Measure(s)	EM-04; M-09 (W, S)
Validation Criteria	Confirmed if a resident or relative uses code W or S about the record when it is explained to them. If confirmed: the record is shown to the resident or representative in plain words at the next review.
Step(s)	UC01 8; UC02 10; UC03 7

2.3.6 Open Items

ID	Item	Blocks	Owner
O-1	Resolved. HFC-01 records that the evidence for walking benefits in dementia is weak, so the design presents a walk as one optional activity among several and never as the default.	Nothing	Closed
O-2	Thresholds: suggestions per week; introductions per newcomer per week; note latency; the presence pattern (days, hours above usual); the trusted person's arrival; System error rate; how long before a declined pair is suggested again. Set from an EM-01 pilot and with care workers.	PR1 to PR5; CL8; VT-02; VT-06	Group
O-3	Pepper's recognition of older voices, in their own language and dialect. Test in EM-01; the tablet is the fallback.	TECH-01; F1, F4	Group
O-4	Relationship map and prepared lines: form, owner, review after events; how staff learn of conflicts between residents.	F1 to F4; VT-01	ST-05 with group
O-5	Consent and data: GDPR art. 6 and 9; the representative's role; process consent; who releases profile items when the resident cannot; check of the local medical-research law.	All pre-conditions; EM-02, EM-03	ST-07
O-6	Walking policy with ST-07 and ST-06: who may walk in the garden with another resident, and who opens the door.	F3 walk suggestions; UC02	ST-07, ST-06
O-7	Presence notes: who may opt in, and how assent is recorded; the default pattern for a note; who sees notes; review at each care-plan review.	TECH-03; F5; VT-06	ST-07 with ST-05 and group
O-8	Pepper in the lounge: a quiet corner for private questions, out of other residents' way. Check on site.	TECH-01; RP-01; F1	Group
O-9	Comparison home: recruitment, and matching on size, layout and resident mix.	EM-03	Group

2.4 Team Design Pattern

TDP-01 Consent-gated social brokering with human handover

Evidence basis

Derived from SA-01 to SA-04 and the recurring breakdowns in PS-01 to PS-04.

- People often fail to initiate or renew contact despite potential mutual interest: HFC-15, HFC-16.
- A direct invitation is difficult to refuse and another person's state is easily misread: HFC-05, HFC-17, VT-02.
- Useful common ground is personal information whose disclosure requires control: HFC-04, VT-01, VT-05.
- Assistance must preserve face, personhood and truthfulness: HFC-03, HFC-06, HFC-08, VT-03, VT-04.

The allocation is based on literature-backed concepts and group reasoning. It has not been empirically evaluated as a complete pattern.

Problem

Two people may benefit from meeting but lack the information, confidence, memory or opportunity needed to initiate contact. A support system can discover possible common ground and coordinate an introduction, but should not decide that people ought to meet, expose personal information, infer consent, take over their relationship or handle situations requiring care and moral judgement.

Use when

Use when:

- the goal is voluntary person-to-person contact rather than interaction with the system itself;

- a system can identify a plausible social opportunity;
- each participant can express assent or dissent in an accessible way;
- a responsible human can review the opportunity and remain accountable for follow-up and exceptions.

Do not use when participation is compulsory, refusal cannot be kept private, no human can own the handover, or the system would have to infer willingness from behaviour alone.

Solution Structure

1. **Contribute:** each participant controls the personal information and preferences that may support a social opportunity.
2. **Propose:** the system identifies a possible connection using only permitted information.
3. **Check:** a responsible human reviews suitability, timing, exclusions and practical conditions.
4. **Choose independently:** each participant accepts or declines without the other participant present.
5. **Facilitate minimally:** after mutual agreement, the system provides identities, limited common ground and, where useful, one optional shared activity.
6. **Withdraw:** once the participants engage, the system stops leading and supports only a clearly defined repair or request.
7. **Relate:** the participants own the conversation or activity and may stop without explanation.
8. **Hand over:** care, distress, conflict, sensitive information, physical assistance and failed coordination go to an identified human.
9. **Follow up proportionately:** practical arrangements and renewed consent may be supported, while reasons for refusal and conversational content are not retained.

Solution Description

The system acts as a broker between possible social partners and as a temporary host at the start of contact. It does not act as a substitute friend, carer, therapist, supervisor or authority on whether a relationship should continue. The human reviewer provides contextual judgement that the system lacks. Independent choice prevents one participant's presence from pressuring the other. Minimal facilitation gives the participants a starting point without making the system the centre of the encounter. Explicit human handover prevents sensitive or safety-critical situations from being treated as ordinary dialogue-management problems.

Participant allocation

Participant A and Participant B

1. Control what personal information may be used or disclosed.
2. Decide independently whether to take part.
3. Own the resulting conversation or shared activity.
4. Pause, leave or withdraw consent without explanation.

Human reviewer or coordinator

1. Check the proposed connection and its explanation.
2. Consider known conflicts, vulnerabilities, accessibility and timing.
3. Approve, reject or alter the proposal before either participant is approached.
4. Keep permissions and exclusions current.

Human facilitator or carer

1. Make the encounter practically possible.
2. Remain reachable without unnecessarily dominating it.
3. Recognise and respond to distress, conflict, care needs or changing capacity.
4. Own follow-up and every handover that requires human judgement or physical presence.

Support system

1. Use only permitted inputs when generating a proposal.
2. Preserve the reviewer's decision and both participants' independent decisions.
3. Facilitate only within the approved information and interaction boundaries.
4. Withdraw after the opening and expose uncertainty or failure rather than improvising.
5. Retain only the minimum operational record required for follow-up and accountability.

Human requirements	Participants must have an understandable and accessible way to accept, decline and stop. The reviewer must understand the basis of a proposal and have authority to reject it. A named human must be available for practical coordination and exceptions. Permissions, exclusions and support needs must be reviewed when circumstances change.
Machine requirements	The system must enforce information permissions, human approval and independent participant choice as separate gates. It must disclose the basis of a proposal to the reviewer, minimise information revealed to participants, support neutral refusal, become inactive after successful facilitation, surface degraded states, and route defined exceptions to an accountable human. It must not infer consent, generate unverified personal claims or retain the content of the resulting social interaction.
Consequences: benefits	• Creates social opportunities without making participation compulsory. • Combines computational search with human contextual judgement. • Protects participants from public rejection and unnecessary disclosure. • Keeps the social activity between the participants instead of turning it into prolonged system interaction. • Establishes responsibility when coordination, care or sensitive judgement is needed.
Consequences: liabilities	• A private invitation may still feel authoritative or difficult to refuse. • Human approval may become superficial automation bias rather than meaningful review. • Sparse, inaccurate or culturally narrow profiles can distribute opportunities unequally. • Minimal facilitation may be insufficient, while excessive facilitation makes participants dependent on the system. • Out-of-date permissions or exclusions can cause harmful disclosure or pairing. • A handover can fail when the responsible human is busy or unavailable. • Operational records can expand into surveillance unless their scope and retention are fixed.
Related Patterns	IDP-01 Introduce, then step back; IDP-02 Ask each person in private; IDP-03 Defer sensitive questions to a trusted person.
Validation strategy	First verify that every gate and handover operates as specified. Then observe whether participants experience the choice as voluntary, whether the system actually withdraws, whether people continue interacting without it, and whether humans receive and act on exceptions. Evaluate both intended social outcomes and adverse effects such as pressure, disclosure, unequal matching and failed handovers.
Grounding	Proto-pattern derived from this project's four social activities. Kinnect provides the first proposed instantiation through UC01 to UC03. Evidence for that instantiation does not automatically validate the pattern in other populations, organisations or technologies.

This pattern generalises the recurring task allocation across the activities in §1.1.1:

Table 2.2: Pattern scope (TDP-01)

Situated activity	Recurring allocation problem generalised by the pattern
SA-01 Moving into the care home	Personal information may help a newcomer meet others, but the newcomer must control its use and a responsible person must check its suitability.
SA-02 Waiting for a reason to leave the room	A possible social opportunity can be identified computationally, but a human must decide whether, when and how to act on it.
SA-03 Greeting someone new	Both people need an independent choice and a small amount of common ground; neither should be exposed as forgetful or rejected in public.
SA-04 Inviting someone to spend time together	The invitation must not create pressure, the participants must remain free to leave, and practical arrangements or support require a human owner.

The invariant across these activities is: **system proposes, human checks, each person chooses, system facilitates briefly, people interact, human remains responsible.**

Table 2.3: Generic sequence and handovers (TDP-01)

Stage	System responsibility	Human responsibility	Participant authority

Opportunity	Find a candidate connection from permitted information	Review the basis, context and exclusions	Control whether their information may be considered
Invitation	Present a neutral, accessible private choice	Ensure the setting and response method are appropriate	Accept, decline or disengage
Introduction	Provide the approved minimum needed to begin	Make the encounter practically possible and remain reachable	Speak, act, remain silent or leave
Interaction	Withdraw; provide only a defined repair when appropriate	Observe only as required by the context	Own the activity and its direction
Exception	Stop, state the limitation and send an actionable handover	Take over care, sensitive judgement, conflict or physical support	Accept or decline further help where possible
Follow-up	Support practical arrangements and renewed private choice	Decide how to arrange or reschedule contact	Choose independently whether contact continues

Kinnect applies the general pattern as follows. These choices belong to the project implementation, not to the reusable core of TDP-01.

Table 2.4: Kinnect instantiation (TDP-01)

General pattern element	**Kinnect implementation**	**Specification**
Contribute	Resident profile items are private by default and released individually.	TECH-04 to TECH-06; F1
Propose	Kinnect suggests pairs from released places, trades, interests or past ties.	F2; UC01 step 3; UC02 step 1
Check	The activity coordinator reviews every suggestion and the relationship map.	ST-05; UC02 step 2
Choose independently	Pepper asks each resident separately about meeting and about the proposed disclosure.	F1; IDP-02; UC01 step 4; UC02 step 3
Facilitate minimally	Pepper gives both names, one mutually released detail and at most one optional activity.	F3; IDP-01; UC01 steps 5-6; UC02 steps 5-7
Withdraw	Pepper remains silent while residents interact and offers at most one prepared shared-ground statement after a stall.	F3; IDP-01
Hand over	Care workers arrange meetings; sensitive questions go to a trusted person; clinical and physical needs remain human responsibilities.	F4, F5; IDP-03; ST-04, ST-06
Follow up	A practical note goes to the named care worker and each resident is privately asked whether they want another meeting.	F1, F5, F6; UC03

Kinnect permits several outcomes: conversation, coffee, music, photographs, games, gardening, a walk or no meeting. Walking is one possible activity, not the organising purpose of the pattern.

Connection to this project's claims and evaluation

The Kinnect instantiation supports UC02-F3-CL1, UC01-F3-CL2 and UC03-F5-CL6. Its main risks are tested by UC02-F1-CL3, UC02-F2-CL4, UC01-F3-CL5, UC03-F5-CL7 and UC01-F4-CL8.

EM-01 verifies the gates, information boundaries and handovers. EM-02 observes choice, withdrawal and resident-to-resident interaction. EM-03 and EM-04 examine repeat contact, friendship, pressure, disclosure and trust. ER-03/EM-08 is later summative evaluation of Kinnect, not evidence already obtained for the general pattern.

Table 2.5: Open items to check or research (TDP-01)

Item	**What is needed**	**Reason**
Meaningful review	Define what information a human reviewer sees, how rejection is recorded and how often permissions and exclusions are refreshed.	Human approval is not a safeguard when it becomes a routine click.
Process consent	Operational signs of assent and dissent developed with intended participants and carers.	A yes/no interface cannot establish continuing willingness on its own.
Handover capacity	Test ownership, acknowledgement and fallback during busy periods.	The allocation fails if the system refers to a human who cannot respond.

Unequal opportunity	Audit proposal frequency and quality for people with sparse profiles, communication impairments or minority backgrounds.	A matching system may reproduce or increase existing social exclusion.
Degree of withdrawal	Determine when one repair is helpful and when any intervention becomes intrusive.	Too little support abandons the encounter; too much displaces it.
Transferability	Evaluate the pattern outside Kinnect and outside this care-home context.	A single instantiation cannot establish that the allocation is generally reusable.

2.5 Interaction Design Patterns

IDP-01 Introduce, then step back

Evidence basis

Derived from SA-03, PS-01, PS-03 and DS-01. Grounded in HFC-03, HFC-04, HFC-06, HFC-15 and HFC-16; constrained by VT-01 and VT-04. Literature and group reasoning only.

Design problem

Two residents need enough shared ground to begin, and may need their names repeated without either person being exposed as forgetful. If Kinnect keeps talking, quizzes them or repeatedly rescues pauses, it competes with the relationship it is meant to support.

Use when

Both residents have independently agreed to this introduction, the coordinator has approved the match, and at least one mutually releasable topic is available. Also use the symmetric-name form at a later facilitated meeting.

Principle

Give both people equal footing, offer one safe opening and then make room for them. Support the conversation only when it has clearly stalled, never before a speaker has had a chance to repair their own turn.

Solution

1. Confirm immediately before speaking that both accept the meeting; visible or verbal dissent stops the introduction.
2. Identify the robot's limited role when needed: "I'm Pepper. I help residents get to know each other."
3. Address both residents symmetrically: "Elena, this is Jan. Jan, this is Elena."
4. State one detail that both residents released and ST-05 approved for this pair: "You both worked in [town]: Elena at the post office and Jan at the bakery."
5. If useful, offer one concrete activity as an option, chosen from both profiles: coffee, photographs, music, a game, gardening or a walk where appropriate.
6. Stop speaking. Do not ask a memory question, demand an answer or narrate the silence.
7. If a speaker pauses mid-turn, allow time for self-repair. If the whole exchange stalls, offer one prepared statement to both about the shared topic or present setting, then step back again.
8. If either person leaves, turns away or says they want to stop, close neutrally: "All right."
9. If corrected, say "I had that wrong," stop using the item and flag it for ST-05.

Do not

- "Do you remember Jan?" or any other test of memory
- "You told us that already," praise for ordinary participation, or elderspeak
- Reveal a second profile item merely because the first did not produce conversation
- Infer friendship, consent or discomfort from facial expression, silence or a compatibility score
- Fill every pause or remain the centre of the exchange

Research link

Shared background supplies common ground (HFC-04, HFC-15). Symmetric naming protects face and personhood (HFC-06, VT-04). Waiting preserves the preference for self-repair (HFC-03). The third-party introduction lowers the social risk created by the liking gap (HFC-16).

Connection to Claims

Shapes F3 in UC01, UC02 and UC03. Supports UC02-F3-CL1 and UC01-F3-CL2; tested against UC01-F3-CL5. Verified in part by UC02-F3-PR3.

Table 2.6: Interaction parameters (IDP-01)

Parameter	Choice	Rationale

Disclosure	Both names plus no more than one mutually released detail	Enough common ground to begin without turning the profile into a public biography (VT-01).
Turn allocation	Address both residents; no quiz question	Avoids identifying either person as the one who forgot (VT-04).
Silence	Default after the opening; wait before one repair prompt	Residents, not Kinnect, should own the conversation; speakers get the first chance to self-repair (HFC-03).
Voice	Adult, neutral-warm, short sentences; ordinary pace adjusted only for hearing needs	Supports comprehension without elderspeak (HFC-06).
Gaze and gesture	Open orientation to both; low movement; no fixed stare	Signals equal inclusion and avoids making one resident the patient or target.
Activity suggestion	One optional activity from both profiles; no default requirement to walk	The project supports connection through several ordinary activities; walking is one option (SA-04).
Error repair	Immediate, plain correction and withdrawal of the item	Protects truthfulness and prevents a disputed fact being repeated (HV-08).

IDP-02 Ask each person in private

Evidence basis

Derived from SA-03, SA-04, PS-03, PS-04 and DS-01. Grounded in HV-01, HV-02, HFC-05 and HFC-17; constrained by VT-01 and VT-02. Literature and group reasoning only.

Design problem

A public invitation is difficult to refuse, particularly when the proposed companion can hear it. Combining permission to meet with permission to disclose a personal fact also obscures what the resident actually accepted. Dementia makes one-time consent inadequate when willingness changes in the moment.

Use when

Before an introduction, before disclosing a profile item, before arranging a repeat meeting and whenever participation is renewed after a break or visible uncertainty.

Principle

Ask the person affected, apart from the other resident, about one decision at a time. Make yes and no equally easy. Treat assent as temporary and dissent as decisive.

Solution

1. Ask in a quiet place where the proposed companion and other residents cannot hear.
2. State the concrete choice without persuasion: "There is a resident here, Jan, who had the bakery in your town. Would you like to meet him, or not this time?"
3. Offer two equally visible and audible responses. Do not preselect "yes."
4. If the resident agrees, ask separately about the proposed disclosure: "May I tell Jan that you worked at the post office?"
5. Proceed only after the other resident has answered the corresponding questions in private and ST-05 has approved the match.
6. Treat consent as applying to this introduction and this disclosed item only. Ask again for a later introduction or a different item.
7. A spoken no, "not now," turning away, pushing the tablet aside, distress or withdrawal ends the offer. Do not seek a reason or immediately offer another person.
8. Tell the other resident only whether a meeting will happen: "Not this time." Never say who declined or why.
9. At the meeting, remain alert to renewed dissent; initial agreement does not remove the right to leave.

Do not

- Ask one resident while the proposed companion is present
- Treat silence, a smile, lack of resistance or a representative's consent as present assent
- Bundle meeting, disclosure, activity and research participation into one yes
- Repeat an invitation until the resident agrees
- Record or reveal the reason for refusal

Research link

Face-to-face requests are harder to refuse than requesters expect (Bohns, 2016; HFC-17). Process consent requires attention to continuing assent and dissent rather than a single decision (Dewing, 2007). Because neither people nor Kinnect can reliably infer another resident's internal state (HFC-05), the pattern asks directly.

Connection to Claims

Shapes F1 and the consent boundary around F2. Supports OBJ-05 and OBJ-06; tested against UC02-F1-CL3. Verified by UC02-F1-PR1.

Table 2.7: Interaction parameters (IDP-02)

Parameter	Choice	Rationale

Setting	Private enough that the other resident cannot hear	Removes the direct interpersonal cost of refusal (VT-02).
Decision unit	One person, one meeting, one disclosed item	Makes the scope of consent understandable and auditable (VT-01).
Response	Equivalent yes and no by speech or tablet; behavioural dissent also stops the offer	Supports different communication abilities and ongoing assent.
After refusal	No reason, persuasion, praise, tally or disclosure	A refusal is an exercise of autonomy, not a failure.
Expiry	At the end of the proposed encounter	A previous yes does not authorize later meetings or disclosures.

IDP-03 Defer sensitive questions to a trusted person

Evidence basis

Derived from SA-01, PS-01 and DS-01. Grounded in HV-05, HV-08, HFC-08 and VT-03. Literature and group reasoning only.

Design problem

A resident may ask about going home, a death, a diagnosis or another painful reality. A reassuring lie can damage trust, while blunt disclosure by a robot lacks the relationship and judgement needed to respond safely. An improvised deflection can also feel evasive or patronising.

Use when

The question concerns a topic listed in the relationship map, or an unlisted topic that could contain painful personal or clinical news. Use only when a named human recipient and a working handover channel are available.

Principle

Do not lie, deny, confirm or improvise the answer. Name the boundary honestly, offer the trusted person and transfer the question with enough context for that person to respond.

Solution

1. Recognise the topic as listed or potentially sensitive; do not retrieve or speak the underlying fact.
2. Use the prepared deferral: "That is something to talk through with Myra. Shall I ask her to come?"
3. If the resident agrees, notify the named person with resident, topic, place and time. Do not include the sensitive answer.
4. Confirm only the action: "I have asked Myra to come." Do not promise when the person will arrive unless the system knows.
5. If the resident declines, close neutrally and notify the trusted person after the interaction only where the agreed care plan requires it.
6. Do not raise the topic again. The human decides how and when to respond under the agreed care plan.
7. If the named person or channel is unavailable, say so plainly and offer an available staff member; mark the failed handover for staff review.
8. If Kinnect itself stated something incorrect, say "I had that wrong," stop using the item and flag its source for correction. Do not use deferral to avoid owning an error.

Do not

- Give therapeutic lies or false reassurance
- Deliver news of death, diagnosis or permanent admission
- Confirm or deny the resident's belief before the trusted person responds
- Say "I can't tell you" without offering a person who can help
- Promise that a staff member is coming when the handover has not been acknowledged

Research link

Truthful, consistent responses protect trust, while both deception and blunt correction can cause harm in dementia care (HFC-08; Mental Health Foundation, 2016; James et al., 2006). The pattern assigns the substantive response to a person with an established relationship and access to the care plan.

Connection to Claims

Shapes F4. Supports OBJ-04 and UC01-F4-CL8; verified by UC01-F4-PR4.

Table 2.8: Interaction parameters (IDP-03)

Parameter	Choice	Rationale
Wording	Prepared, truthful boundary plus an offer to contact a named person	Avoids both deception and an empty refusal.

Information sent	Resident, topic, place and time; not the sensitive answer	Gives staff actionable context while minimising disclosure.
Recipient	Named trusted person; available staff fallback	A deferral without reachable human ownership abandons the resident.
Acknowledgement	Promise only what the system can verify	Prevents a failed handover becoming another deception.
System error	Immediate correction, separate from sensitive-topic deferral	Kinnect remains accountable for its own statements.

Table 2.9: Open items to check or research for IDP-01 to IDP-03

Item	**What is needed**	**Reason**
Wording and timing (O-2)	Test the introduction length, silence before a single prompt, adult tone and refusal wording in EM-01 Build A.	All three patterns depend on how the interaction is experienced, not merely on whether the correct line is present.
Process consent (O-5)	Agree observable assent and dissent rules with residents, representatives and care staff.	A yes/no interface cannot by itself establish continuing willingness.
Quiet setting (O-8)	Check whether the lounge has a genuinely private place to ask each resident.	IDP-02 fails if others can overhear the answer.
Relationship map (O-4)	Define owner, review triggers, unavailable-recipient fallback and acknowledgement rule.	IDP-01 and IDP-03 are only as safe as the permissions and human handovers behind them.
Accessibility	Test hearing, vision, speech and tablet alternatives with the intended residents.	A nominal choice is not meaningful if one response channel is inaccessible.

Chapter 3

Evaluation

3.1 Artifact

AT-01 Kinnect social connection-maker

Scope

Covers the connection-making workflow of UC01 (welcoming a newcomer and building a consent-gated profile), UC02 (introducing two residents around shared history or interests) and part of UC03 (following up and keeping a short record), through F1 (Ask first), F2 (Suggest matches), F3 (Introduce), F4 (Defer sensitive topics) and F6 (Keep a short record). Build 1 has four parts.

- *Pepper simulation and chest tablet* (`web/src/components/Device.tsx`, `web/src/App.tsx`). Five recorded clips of Pepper play in the browser. A small tracker (a 160×90 offscreen canvas, exponential smoothing with `alpha = 0.28`) follows Pepper's chest screen as the body moves, and a 580×406 tablet interface is drawn inside it. An Enlarge Tablet toggle helps residents with low vision. Pepper waves on a greeting or introduction, nods or shakes its head at a reply, and moves while it speaks. The top bar sets the speech rate (Slow 0.8×, Calm 1.0×, Brisk 1.25×) and the style (Warm & chatty, Direct & brief).
- *Role-play walkthrough support* (`web/src/App.tsx`, `server/intake.ts`). Evaluators play assigned personas: Elena, 81, early-onset Alzheimer's; Jan, 75, a newcomer with a hearing aid in his left ear; and Ria, 79, with mild memory slips. Each card prompts a human-factor problem, such as describing a resident by appearance or hobby when the name is gone (*"the gray-haired man with the yellow cat on his shirt"* for Willem, *"light-green round glasses"* for Ans, *"dark fedora hat"* for Gerrit, *"navy-and-orange headscarf"* for Corrie). A six-item feedback form (`logs/feedback.jsonl`) records a connection-making score, tablet legibility, preferred speech rate and conversation style, the perceived effect of the gestures, and free notes.
- *Consent opening, resident lookup and profile intake* (`server/intake.ts`, `server/lines.ts`). Every conversation opens with prepared consent lines (`open`, `start_yes`, `start_no`, `not_now`) in ten languages (`en-US`, `nl-NL`, `de-DE`, `fr-FR`, `es-ES`, `it-IT`, `ro-RO`, `pl-PL`, `lv-LV`, `fa-IR`). When a resident forgets a name, the System matches the description against a directory of seven residents (`RESIDENT_DIRECTORY`) and shows up to four photo tiles on the tablet (`show`). Consented interests, topics to avoid (`avoid`) and connection invitations are stored in a local SQLite database (`server/db.ts`, `server/store.ts`).
- *Relationship map, session exchange and guard* (`server/store.ts`, `server/prompt.ts`, `server/guards.ts`, `server/session.ts`, `server/verify.ts`). The store holds facts per resident, each with a `sensitive` flag, and a state per pair (`agreed`, `may_offer`, `not_again`, `unknown`, `offLimits`). For the staged walkthrough (DS-01, DS-02, PS-03 to PS-06, SV-01, SV-02), the engine builds a per-session exchange that holds only consented, non-sensitive items, and seals the system prompt with a SHA-256 hash (M-05). A deterministic guard checks every utterance: at most 30 words, nothing from the `NEVER_SAY` list of elderspeak and memory-quiz phrases, no item the resident has not shared, and no sensitive or withheld topic. A keyword check backs up the F4 deferrals. A resident can correct the robot (*"I had that wrong"*), which flags the item for ST-05. Each session ends with a six-item record (F6), and the live dialogue turns are deleted under SQLite's `PRAGMA secure_delete = ON`.

Out of scope in Build 1: the physical Pepper (TECH-01) and moving through the lounge; F5 Follow up (UC03-F5-PR5), that is, follow-up notes after an agreed meeting and opt-in room-presence counts (TECH-03); per-item consent in the older walk session engine, where `computeNextQuestion()` asks for walk and introduction consent per pair, not per item; and local speech and language-model processing, which Builds 2 and 3 add.

Simulation notice

- *Real in Build 1.* The SQLite profile and relationship store (`node:sqlite`, with `PRAGMA secure_delete = ON`); the prepared line library (`server/lines.ts`); motion tracking over the recorded Pepper clips, with the tablet interface drawn in sync; live speech input and output through Google Cloud Speech-to-Text (Chirp 3) and Text-to-Speech (Gemini-TTS, Chirp 3 HD), with the browser's Web Speech API as fallback (`server/speech.ts`); live intake and match suggestion through hosted Claude or Gemini (`server/llm/index.ts`); the utterance guard (`server/guards.ts`), prompt sealing (`server/prompt.ts`) and trial verifier (`server/verify.ts`); and the feedback log (`logs/feedback.jsonl`).
- *Simulated in Build 1.* Pepper's body and its presence in the lounge, shown as video clips in the browser; the residents, played by peer evaluators with persona cards (Elena, Jan, Ria) during the formative walkthrough; and edge cases, triggered with fault-injection buttons (`Reluctant`, `Say "not again"`, `Correct an item`, `Sensitive question`) during staged verification.
- *Needs Pepper+ (Build 3).* Standard Aldebaran Pepper (TECH-01) cannot reliably listen while it speaks or moves, because of cross-talk from its motors and speaker, and it has no onboard compute for a local language model. ER-03 therefore specifies Pepper+ (Build 3, with TECH-18): Pepper's body and chest tablet, with simultaneous speaking and listening and a hardware accelerator that runs the language model and speech pipeline next to the care home's local server (TECH-04).

Premises verifiable

The six Premises in §2.3.4, and which build can verify each. The criterion in each label is a short form of the Premise's Verification Criteria there.

- *UC02-F1-PR1, no introduction without a yes from both residents, each asked alone (0/50 violations).* Verifiable on Build 1 within the limit below; fully verifiable on Build 2. The formative walkthrough tests private consent (`yes`, `not today`, `not again`, and reluctant or unclear replies, which are asked once more and then treated as `no`) over five runs per event (M-01, M-10). Build 1 records consent per conversation and pair, and flags facts as `sensitive` or `avoid` during intake. Asking for consent per item before each introduction comes in Build 2.
- *UC02-F2-PR2, no unapproved match is raised (0/20) and no blocked, unreleased or sensitive item is used (0/50).* Verifiable on Build 1. `/api/matches/suggest` (`server/index.ts`) and `buildSessionExchange()` (`server/prompt.ts`) remove `not_again` pairs, `offLimits` residents and `sensitive` facts before any prompt or suggestion is built (M-05).
- *UC02-F3-PR3, names given to both residents as statements, with at most one item both released (20/20).* At best partly verifiable on Build 1; fully verifiable on Build 2. The opening consent check and the introduction to both residents use prepared lines (`server/lines.ts`), and `server/guards.ts` blocks unconsented and sensitive items. The open intake turns and the conversational replies, however, come from a hosted language model (Claude or Gemini). The strict prepared-line version of PR3 (TECH-06, TECH-07) needs Build 2; ER-03 then evaluates guarded generation by a local language model on Pepper+ (TECH-18).
- *UC01-F4-PR4, listed sensitive questions detected (19/20) and deferred truthfully, with the trusted person notified within 10 s (0/50 listed facts stated or denied).* Verifiable on Build 1. `server/session.ts` and `server/guards.ts` detect the topic with a keyword check (`containsSensitiveTopic()`) and answer with a prepared deferral line from `server/lines.ts` that names the resident's trusted contact, with no follow-up questions (M-06).
- *UC03-F5-PR5, a note reaches the named care worker within 1 minute of the agreed time (20/20), and presence notes go only to residents who opted in.* Not verifiable on Build 1; deferred to Builds 2 and 3. Build 1 still has the older garden-walk session flow, without staff notes after an agreed meeting and without the opt-in room-presence counter (TECH-03, M-02, M-09).
- *UC02-F6-PR6, the record holds only the listed items, no packet leaves the home network, and no audio file exists on any device.* Fails on Build 1 by design; verifiable on Builds 2 and 3. Build 1 keeps the six-item record and deletes live transcripts (`PRAGMA secure_delete = ON`), but it fails the network isolation check (M-07), because speech recognition and synthesis (Google Cloud) and the replies (Claude or Gemini) go through external cloud services. Running speech and the language model locally on TECH-04 and TECH-18 lets PR6 pass.

Builds

Build	What changed
Build 1: formative web proof of concept (`sce-poc`)	A browser simulation of Pepper, in `sce-poc`, with the chest-tablet interface drawn over five recorded movement clips by canvas motion tracking (`Device.tsx`). It includes the persona cards (Elena, Jan, Ria) with human-factor prompts (describing a resident when the name is forgotten, word-finding slips), the speech rate and style controls, the six-item feedback log, the SQLite store (`server/store.ts`), the prepared consent, introduction and deferral lines in ten languages (`server/lines.ts`), the utterance guard (`server/guards.ts`), SHA-256 prompt sealing (`server/prompt.ts`) and the automated scenario verifier (`server/verify.ts`). It uses hosted Claude or Gemini and Google Cloud speech, so UC02-F6-PR6 fails on it by design. An earlier state of this build, still centred on walks, was used on 13 October 2026 for a peer walkthrough in the lab session (ER-04, Evidence EV-08). The staged walkthrough of PR1, PR2, PR3 and PR6 (five runs per event) is planned.
Build 2: revised local verification build (EM-01 target)	Replaces cloud speech and hosted language-model calls with local processing on the care home's server (TECH-04), with strict prepared-line dialogue (TECH-06, TECH-07), or with both. Adds per-item consent prompts (F1) and follow-up logic for agreed meetings and room presence (F5, TECH-03), brings the persona seed data in line with DS-01 (Elena at the post office counter, Jan at the bakery), and applies the fixes committed to after the formative walkthrough. It has to pass the full staged verification (EM-01, against each Premise's Verification Criteria) before any care-home resident takes part.
Build 3: summative Pepper+ build (ER-03 target)	Planned deployment on Pepper+ (Build 3), built from TECH-01, TECH-18 and TECH-04. Pepper is extended so that residents can interrupt or reply while it speaks or gestures, and an accelerator (TECH-18) runs a local language model and an acoustic feature extractor inside the care home's network. It supports one-to-one reminiscence and profile-building conversations of 15 to 20 minutes, matching with the "Vibing Algorithm" on shared topics and on speech cadence and response latency, tracking of conversational engagement (M-21), and joint icebreaker sessions. ER-03 evaluates it on peer connections (M-20) and bedroom isolation time (M-22) in the two-site trial (EM-08), a one-month intervention within a five-week window.

3.2 Evaluation Reports

ER-03 Longitudinal two-site controlled study of Kinnect (study design)

Mode

Summative

Artifact used

AT-01 Kinnect social connection-maker, Build 3 (Pepper+). Build 1 (`sce-poc`) is a browser simulation for formative evaluation and cannot run this study. Build 3 can listen while it speaks and has an on-device accelerator (TECH-18) for a local language model, so Pepper can hold the one-to-one conversations, build persona profiles, compute compatibility scores and offer icebreakers without cloud services, as described in *Intervention: matching and facilitation* below.

Evaluation Method(s) applied

EM-08 Two-site controlled trial, addressing Claims only. EM-08 is the summative counterpart of the formative two-home comparison EM-03.

Ethics

Review. The protocol is submitted to the Human Research Ethics Committee (HREC). Recruitment starts only after institutional ethical clearance.

Consent. Written informed consent from each resident's legal representative or guardian, together with ongoing verbal and behavioural assent from the resident (Black et al., 2010; Slaughter et al., 2007). All robot sessions and peer introductions are voluntary. Verbal or non-verbal dissent, such as turning away or declining an invitation, is respected immediately (Black et al., 2010; Slaughter et al., 2007).

Protections. Staff and researchers pause at once if a participant shows signs of agitation, confusion or frustration during a robot or peer interaction. Agitation is tracked throughout with CMAI criteria (Black et al., 2010; Cohen-Mansfield et al., 1989). Badges and the bedroom beacon are used only for residents who give separate consent for them. Speech is processed on the robot and no audio or transcript is kept. Personal data are handled as set out in the *Data management plan* below.

Introduction

Claims addressed. The study tests two hypotheses. Both are Claims, because they are about effects on people. UC01 to UC03 and their Claims now exist, but neither study hypothesis maps exactly to one current Claim ID (see *Open items*). UC02-F3-CL1 is the closest existing Claim to the primary hypothesis.

- **Primary Claim.** Kinnect increases peer connections among residents with early- to middle-stage dementia, measured as dyadic social interactions and time spent in communal spaces, and reduces bedroom isolation.
- **Secondary Claim.** Kinnect is intuitive and easy to understand for first-time users living with cognitive impairment.

Background. Residents of assisted living and dementia care homes are often socially isolated, and spend large parts of the day alone in their private bedrooms, away from communal areas (Jansen et al., 2017). Cognitive decline can make it harder to start a conversation with a peer spontaneously. Social network theory finds that ties form faster and last longer between people who share life histories, interests and ways of talking, a principle known as homophily (McPherson et al., 2001). Structured reminiscence about nostalgic eras, music and life histories has been associated with improved mood and communication in dementia care (Woods et al., 2018).

Kinnect is an autonomous conversational companion robot designed to act as a social catalyst. It does not replace human contact. In one-to-one, person-centred conversations it learns each resident's background, hobbies and conversational pace. It then computes compatibility between residents and brings matched residents together through tailored conversational icebreakers.

Why a Summative Test. Both Claims are about what happens to residents of a real care home over several weeks: whether they meet peers, talk with them and spend less time in their rooms. Only a study in that setting can support, qualify or challenge them.

Method

Participants. Residents with early- to medium-stage dementia or Mild Cognitive Impairment (MCI) who live in residential care, at two similarly equipped care homes. Care staff at both homes take part through training, protocol support and the debriefing interviews. Target: 30 residents recruited per home, 60 in total, so that 25 per home complete (see *Sample size* under *Hypotheses and analysis plan*).

Inclusion:

- A formal diagnosis of early- to medium-stage dementia or MCI, operationalised as a Mini-Mental State Examination (MMSE) score of 14 to 24 (Folstein et al., 1975) or a Montreal Cognitive Assessment (MoCA) score of 10 to 22 (Nasreddine et al., 2005).
- Able to move to or spend time in communal areas, on their own or with standard assistive aids or staff support.
- Able to talk in the main language Kinnect supports.
- Written informed consent from a legal representative or guardian, with ongoing verbal and behavioural assent from the resident (Black et al., 2010; Slaughter et al., 2007).

Exclusion:

- Cognitive impairment severe enough to prevent basic verbal interaction or understanding of conversational prompts (Folstein et al., 1975).
- Severe sensory deficits, such as unaddressed profound hearing loss or severe visual impairment, that prevent audio-visual interaction with the robot.
- Acute medical conditions or clinical restrictions that rule out social contact with other residents.
- Severe behavioural and psychological symptoms of dementia, such as extreme baseline agitation or paranoia toward electronic or robotic hardware, assessed at baseline screening (Cohen-Mansfield et al., 1989).

Experimental design. Between-subject, longitudinal cluster-controlled trial (Donner & Klar, 2000): a one-month intervention within a five-week operational window, at two homes that run at the same time. One home is allocated to the intervention arm and the other to the control arm, which limits cross-exposure and contamination between the arms (Chenoweth et al., 2014; Donner & Klar, 2000).

- **Intervention arm.** Kinnect engages residents in person-centred dialogue and matches them with peers by compatibility (the "Vibing Algorithm"). Matched residents are offered voluntary shared time in the communal lounge (Woods et al., 2018).

- **Control arm.** The home's standard social activity programme: communal lounge time on one's own, or the home's usual group activities, without algorithmic matching and without robot icebreakers.

Independent variable: arm, nominal with two levels (intervention, control), allocated per home. Dependent variables: the Measures below.

Tasks. Intervention arm: one-to-one sessions with Kinnect in week 1, introductions to a matched peer in weeks 2 and 3, and joint icebreaker sessions with that peer in weeks 3 and 4. Every session and introduction is optional. Control arm: the usual daily routine and activities.

Measures. M-19 to M-23, defined in §1.2.4, with their scoring rules and proposed thresholds.

Domain	Measure	Source
Well-being and affect	M-19 Agitation and observed affect	CMAI from primary caregivers (Cohen-Mansfield et al., 1989); OERS around sessions, as in M-03; staff log of agitation events
Social connections and integration	M-20 Peer connections and communal time	Real-time location system (RTLS) proximity badges (Grigorovich et al., 2021; Jansen et al., 2017), checked against EM-02 room scans
Conversational engagement	M-21 Conversational engagement	Speech recognition, transcript NLP and acoustic analysis on the robot (TECH-18), in one-to-one sessions and joint sessions (Fraser et al., 2016; Luz et al., 2021)
Bedroom isolation time	M-22 Bedroom isolation time	Badge proximity to a fixed beacon in the resident's bedroom (Jansen et al., 2017)
First-time usability (secondary Claim)	M-23 First-time usability	Week 1 interaction logs, response latency (Luz et al., 2021), an observational usability checklist and a first-impression question

Procedure. Five phases, from four weeks before the intervention to week 5. See *Study protocol* below.

Material. AT-01 Build 3 (Pepper+, with the TECH-18 accelerator) at the intervention home. Proximity badges for consenting participants, with fixed anchors in common areas and beacons in bedrooms, at both homes. A tablet for EM-02 room scans. MMSE or MoCA and CMAI forms; OERS sheets; the staff agitation log; the M-23 usability checklist. Consent forms, including the separate consent for badges.

Result(s) Not run. The study is designed only, so there are no results and no Evidence yet.

Discussion Not run.

Conclusions Not run.

Hypotheses and analysis plan of ER-03

Periods. Baseline: weeks -2 to 0, with badges and EM-02 room scans running at both homes and Kinnect not yet present. Outcome: weeks 4 and 5. Each resident contributes one mean per period for each Measure.

Unit and population. The unit of analysis is the resident. Analysis is by intention to treat: every consenting resident counts in their home's arm, however many sessions they attended. Because arm is allocated per home and there is one home per arm, every between-arm test below compares these two homes. A significant result shows that change differed between the two homes over the study; it does not show that the same would happen in other homes (see *Open items*).

Tests. Two-sided, $\alpha = .05$. The primary outcome is a single test. Holm's correction is applied across the two key secondary tests marked with an asterisk. Every test is reported with its effect size and 95% confidence interval.

Claim	Question (yes/no)	Dependent variable and level	Analysis and decision rule

Primary	Do residents at the intervention home spend more time with peers in common areas in weeks 4 and 5 than residents at the control home, allowing for baseline?	M-20 joint social time: minutes per week in co-location episodes, mean over the period (ratio).	ANCOVA: outcome-period mean on arm, with the baseline mean as covariate. Met if the arm effect is positive and $p < .05$. If the residuals are clearly skewed, minutes are analysed as log(minutes + 1); this rule is fixed in advance.
Primary *	Are there more dyadic interactions at the intervention home, allowing for baseline?	M-20 dyadic interactions per resident in the period (count), with observed hours as exposure.	Negative binomial regression on arm, with the baseline count as covariate and log observed hours as offset. Met if the rate ratio is above 1 after Holm's correction.
Primary *	Do residents at the intervention home spend a smaller share of the waking day in their bedroom, allowing for baseline?	M-22 proportion of waking hours in the bedroom, weekly mean over valid days (ratio, 0 to 1).	ANCOVA as for the primary outcome. Met if the arm effect is negative after Holm's correction.
Primary (harm check)	Does agitation rise at the intervention home?	M-19 CMAI total in the outcome period, 29 to 203 (ordinal, treated as interval).	ANCOVA with baseline CMAI as covariate. Not a confirmatory test: a rise in the intervention arm whose confidence interval excludes 0 is reported as a possible adverse effect, and the session records of the residents concerned are reviewed.
Primary (descriptive)	Does the social network become denser?	M-20 network density, per home per week (ratio).	No test: there is one value per home per week. Plotted by week for both homes, with matched pairs shown apart from all other pairs.
Primary (within arm)	Does affect improve around sessions, and does conversation become more engaged over the weeks?	M-19 OERS shift per session (interval); M-21 features per session (ratio).	Intervention arm only, exploratory. Linear mixed models with week as fixed effect and a random intercept per resident; OERS shifts summarised with their mean and 95% CI.
Secondary	Do most first-time users reach the usability criterion by the end of week 1?	M-23, per resident: criterion met, yes or no (nominal).	Exact binomial. Supported if the lower limit of the 95% Clopper-Pearson interval for the proportion meeting the criterion is above .50. With 25 residents this needs 18 or more. If 80% of residents truly meet the criterion, the chance of reaching 18 is .89.

Verdict on the primary Claim. Supports: the primary outcome and both key secondary outcomes are met, and the harm check shows no rise in agitation. Qualifies: the primary outcome is met but a key secondary outcome is not, or the harm check shows a rise. Challenges: the primary outcome is not met. A non-significant result is reported as "not shown", never as "no effect".

Sensitivity analyses.

- A linear mixed model on all weekly values from week -2 to week 5, with arm, period and their interaction as fixed effects, and random intercepts for resident and for matched pair. The pair term allows for joint time being counted for both residents of a pair.
- Baseline trend: the slope over weeks -2 to 0 is compared between the homes. A clear difference weakens the assumption that the homes would have changed alike without Kinnect.
- Per protocol: only residents who attended at least three joint sessions.
- Missing data: the ANCOVA uses residents with at least one valid week in each period; the mixed model uses all valid weeks. The number of residents left out of each analysis, and why, is reported.

Sample size. Planned on the primary outcome. No data on its spread exist yet, so the effect is stated as a standardised difference d. For ANCOVA with baseline as covariate, the number of completing residents per arm for 80% power at two-sided $\alpha = .05$ is about $2(1.96 + 0.84)^2(1 - r^2)/d^2 + 1$, where r is the correlation between baseline and outcome (normal approximation, with Guenther's correction for the t distribution).

d	**r = .3**	**r = .5**	**r = .7**
0.5	59	49	33
0.6	41	34	24
0.7	31	25	18
0.8	24	20	14

Target: d = 0.7 with r = .5, so 25 completing residents per arm. With 15% attrition assumed over nine weeks (death, hospital stays, moving, withdrawal), that means 30 recruited per arm, 60 in total. Both d = 0.7 and r = .5 are working assumptions, because no study of matching of this kind has reported an effect on co-location time. Before week 1, the calculation is repeated with the pooled baseline SD and the correlation between baseline weeks -2 and -1, without looking at arm. Each home needs at least 30 eligible residents who consent. If a home has fewer, the options are to add a second home to that arm, which also eases the one-home-per-arm limit, or to accept d = 0.8 (20 completing, 24 recruited per arm).

The calculation treats residents as independent. Within a home they share staff and rooms, and matched residents share joint time by design. With one home per arm the variance between homes cannot be estimated (Donner & Klar, 2000), so the p-values describe these two homes only.

Reliability. Inter-observer agreement as each Measure states: $\varkappa \geq .60$ for the room scans and the M-23 checklist, ICC(2,1) $\geq .60$ for OERS. Observers train in the baseline weeks until they reach it.

Preregistration. Hypotheses, periods, tests, thresholds and the sample size are registered (for example on OSF) before the baseline weeks start. Anything decided after seeing data is reported as exploratory.

Intervention: matching and facilitation

How Kinnect brings residents together in the intervention arm. The stable task allocation and interaction rules are specified in UC01 to UC03, TDP-01, IDP-01 to IDP-03 and AT-01; this section records the study-specific schedule and build under test.

Persona and compatibility profiles. During one-to-one sessions, Kinnect compiles a structured persona profile for each resident from reminiscence-based dialogue: biographical history, preferred social times, favourite conversation topics, musical taste and nostalgic eras (Woods et al., 2018).

Compatibility score ("Vibing Algorithm"). Based on homophily (McPherson et al., 2001), the System calculates a multidimensional compatibility score across shared interests, and communicative pace (speech cadence and response latency; Luz et al., 2021), and uses it to propose peer pairings.

Gradual facilitation.

Phase	Weeks	What Kinnect does
1 Individual rapport and usability assessment	1	Talks one-to-one with each participant, to evaluate initial ease of use, establish trust and build a baseline persona profile.
2 Conversational bridging and pairing	2 to 3	Tells the resident stories or shared interests of a matched peer and suggests meeting in the common room. For example: "Did you know Resident B also grew up near the coast and loves classic jazz? Would you like me to introduce you two over coffee in the lounge?"
3 Facilitated icebreaking in pairs	3 to 4	Facilitates the first joint sessions and checks in with matched pairs from time to time, offering icebreaker prompts based on shared memories to keep the two talking with each other (Woods et al., 2018).
4 Handover to the home	End of the study	Gives care staff compatibility summaries, so that they can support informal seating and meetings between matched residents during daily routines (Chenoweth et al., 2014).

The control home keeps its standard care routines throughout. Where a phase does something there, it is listed.

Table 3.1: Study protocol (ER-03)

		• **Baseline (weeks -2 to 0).** Badges and EM-02 room scans run at both homes before Kinnect is introduced, so that both homes have a baseline for M-20 and M-22. Observers train during these weeks. • **Site allocation.** Allocate the two homes to the intervention and control arms (Donner & Klar, 2000). • **Staff education and training.** Study procedures, protocol compliance, usability observation, and running the intervention and control conditions (Chenoweth et al., 2014).
1 Individual rapport and baseline persona collection	Week 1	• **One-to-one sessions and usability baseline.** Daily sessions of 15 to 20 minutes between each intervention participant and Kinnect, to see how new users understand and respond to the robot. • **Data collection.** On the robot (TECH-18), persona attributes (life history, nostalgic eras, music, activity preferences) and acoustic markers (speech cadence, turn-taking latency) are extracted from the conversation (Fraser et al., 2016; Luz et al., 2021; Woods et al., 2018). Persona items enter the profile as private until the resident releases them (TECH-05). • **Control home.** Standard care routines and communal lounge access, with no companion sessions.
2 Algorithmic matching and bridging	Weeks 2 to 3	• **Compatibility calculation.** Run the Vibing Algorithm on the persona data to produce compatibility matrices and dyadic peer recommendations (McPherson et al., 2001). • **Conversational bridging.** In the individual sessions, Kinnect tells each participant stories and shared interests of their matched peer and invites them to meet in a communal space.
3 Joint icebreaking and facilitated engagement	Weeks 3 to 4	• **Icebreaking in pairs.** Kinnect hosts short joint sessions with matched pairs, or checks in with them, and supplies shared-memory icebreaker topics that prompt the two to talk on their own (Woods et al., 2018). • **Care staff support.** When matched participants show interest, staff arrange informal seating or coffee and tea in common areas, without requiring anyone to take part.
4 Post-intervention evaluation and offboarding	Week 5	• **Primary measures.** Final CMAI (Cohen-Mansfield et al., 1989). Analysis of the beacon logs for social network density, time in the communal lounge and bedroom isolation time (Jansen et al., 2017). Analysis of the M-21 values computed on the robot: turn-taking, sentiment, topic coherence (Fraser et al., 2016; Luz et al., 2021). • **Qualitative feedback.** Debriefing interviews with care staff and participants on perceived social connectedness and on how easy Kinnect was to use the first time. • **Handover to the home.** Care staff receive the compatibility summaries of matched residents, so that they can keep supporting informal seating and meetings (Chenoweth et al., 2014).

Contamination prevention

- **Site separation and cluster allocation.** The trial runs at the same time in two physically separate, similarly equipped care homes, one per arm. This limits cross-exposure and behavioural spillover between the arms (Donner & Klar, 2000).
- **Separate care teams.** Each home has its own care team. Staff at the control home are not told how the matching works. Staff at the intervention home are trained not to change standard care routines outside scheduled protocol support (Chenoweth et al., 2014).
- **Hardware boundaries.** At the intervention home, Kinnect is stored and operated only in designated interaction zones and in scheduled communal lounge sessions, so that sessions have clear boundaries.

Limitations and future work

- **Hawthorne and observation effects.** Participants and staff at both homes may change their behaviour because they know they are being observed (McCambridge et al., 2014). Wearing proximity beacons, seeing research staff and taking part in recurring assessments can each add social stimulation independent of the intervention.
- **Generalisability.** The two homes are in one region of the Netherlands, so the results describe that setting. Norms about personal space and attitudes to being matched by a robot may differ in other countries and care cultures; the study cannot show whether they do.
- **Environmental and facility constraints.** Home-wide event schedules, seasonal illness outbreaks or passing changes in residents' health may affect common-area use and the frequency of social interaction independently of the intervention.

- **Agent adaptation and technical limits.** Kinnect's conversational fluency and usability depend on the accuracy of automatic speech recognition (ASR) and natural language processing. Low signal-to-noise ratios, atypical speech cadence and word-finding pauses, all common in dementia, can lead to misrecognitions that disrupt the dialogue (Luz et al., 2021).

Data management plan

Location tracking and automated profiling in nursing homes raise privacy concerns and a risk of algorithmic bias against older residents (Berridge & Grigorovich, 2022; Grigorovich et al., 2021). The study handles data as follows.

- **Pseudonymisation.** All participant identifiers, life histories and clinical scores are pseudonymised at collection. The key linking codes to identities is stored physically and digitally apart from the beacon logs and the M-21 features.
- **On-device processing, no recordings.** Speech recognition, transcript NLP and acoustic analysis run on the robot's accelerator (TECH-18). Raw audio is held only in the robot's working memory while it is processed and is never written to disk or sent anywhere. Transcripts are deleted on the robot at the end of each session. Only derived values (the M-21 features) and the persona items a resident has released reach the home's local server (TECH-04), which has no internet uplink.
- **Secure storage.** Data on the local server, and badge logs, are encrypted in transit and at rest with AES-256. Pseudonymised data leave the home for analysis only by an institutionally approved route.
- **GDPR.** Clinical scores and location data are health data, and are handled as special-category personal data under the GDPR.
- **Restricted access.** Only core research staff bound by institutional non-disclosure agreements have access to identifiable data and clinical scores. There is no raw audio to access, because none is stored.

Open items of ER-03

- **Claim IDs.** UC01 to UC03 are now written, but ER-03's composite primary hypothesis and its first-time-usability hypothesis do not map exactly to an existing Claim. Decide whether to narrow the primary hypothesis to UC02-F3-CL1 and treat bedroom isolation as a secondary outcome, or add explicit Claims; likewise add or identify a Claim for first-time usability before running the study.
- **Premises first.** As in ER-02, the Premises the Claims rest on, such as speech recognition with older speakers, profile extraction and matching, should be verified on the build before the study starts. Otherwise a null result cannot be told apart from a System that did not work. No verification method for them exists yet.
- **Sample size assumptions.** The target of 60 residents rests on an assumed effect ($d = 0.7$) and baseline correlation ($r = .5$). It is recomputed from the baseline weeks, and the homes must be large enough to supply it.
- **One home per arm.** With a single home in each arm, a difference between the arms cannot be separated from other differences between the two homes, such as staff, layout, residents and activity programme. EM-03 states the same limit for the formative comparison, and Donner & Klar (2000) discuss it for trials with one cluster per arm. The baseline weeks and the ANCOVA allow for differences in starting level, and the baseline-trend check tests whether the homes were already changing differently. Neither removes the limit: only more homes per arm would.
- **Badges and TECH-09.** M-20 and M-22 rely on wearable badges, and §1.3 rejects wearable location sensing (TECH-09) as surveillance. Separate consent and room-level beacons limit this, in line with HV-03's rule that time in one's room is noticed only where the resident agreed. The group still has to decide whether badges are acceptable for a time-limited study, or whether the presence count (TECH-03) should replace them.
- **Pepper+ does not exist yet.** The study assumes AT-01 Build 3, a Pepper that can listen while it speaks and has an on-device accelerator (TECH-18) for speech and language processing. Until it is built, M-21 and Kinnect's free conversation cannot run on a physical Pepper without the cloud services that TECH-07 rejects. Build 1 (`sce-poc`) uses those services, but only in simulation.
- **Builds needed first.** AT-01 has Build 1 (`sce-poc`) for formative evaluation, Build 2 for EM-01 verification and Build 3 for this trial. Builds 2 and 3 must be finished and verified under EM-01 before this study can run.

ER-04 Peer walkthrough of the Kinnect prototype, 13 October 2026

Mode	Formative
Artifact used	AT-01 Kinnect social connection-maker, Build 1 (`sce-poc`), as deployed on 13 October 2026. This was an earlier state of Build 1: the concept was still "Walking with purpose", so a session centred on Pepper inviting a resident on a walk and asking whether another resident should join. The concept was narrowed to introductions three days later. On the day, the deployment was updated so that any of the persona cards could be used.
Evaluation Method(s) applied	Ad hoc, not one of EM-01 to EM-04. Two peers from another group used the prototype in the role of residents, with a group member facilitating, in the lab session set aside for walkthroughs. The four walkthrough questions used in ER-01 were the intended frame, but they were not applied step by step. The session addressed Claims, since the peers judged whether the interaction would work for a resident; it did not verify any Premise.
Ethics	No ethics review was required. The peers were students taking part in the course's lab session, playing personas from the persona cards. No personal data were recorded, and no resident or other person outside the course took part.
Introduction	Aimed at the Claims on asking, matching and introducing that a walkthrough can judge, in particular UC02-F3-CL1 (introductions lead to repeat contact), UC02-F1-CL3 (accepting out of politeness) and UC02-F2-CL4 (matching feels exposing). Claims about outcomes over weeks (UC01-F3-CL2, UC02-F3-CL9) were out of reach and are left to ER-03. Why then: the lab session was the course's slot for a formative walkthrough of each group's prototype, and Build 1 had just been deployed.
Method	**What was done.** A group member ran the prototype on a laptop and two peers took turns as the resident, using the persona cards (Elena, Jan, Ria). They talked to the simulated Pepper through the browser, and gave feedback during and after the session. A second group member discussed the prototype with the lecturer in a separate call during the same session. **What was recorded.** Nothing during the session. The facilitator was alone at the table and could not take notes while running the prototype. The findings below were reported by the two group members in the group chat on the same afternoon. Build 1 has a six-item feedback form (`logs/feedback.jsonl`); whether the peers filled it in is not known. **Limits.** Two peers, one session, no notes. The peers are not people with dementia. The facilitator found the walkthrough procedure unclear on the day, so the four questions were not asked systematically. The session walked the walking version of the concept, not the narrowed UC01 to UC03, and the peers did not know that only the left side of the screen is what a resident would see (KF-04). The findings can say that a problem was noticed, not how often it would occur or how serious it would be for residents.
Finding(s)	See *Walkthrough findings* below: KF-01 to KF-04.
Implications	See *Implications* below. KF-02 changes the order of profile building and matching in Build 2 and in ER-03's first week. KF-03 adds a non-robot alternative to the Technology Options. KF-04 changes how the demo is presented in the repeat walkthrough.

Table 3.2: Walkthrough findings (ER-04)

ID	Finding	Suggestion	Bears on
KF-01	The interface was not clear enough for the peers to follow what Pepper wanted from them.	Simplify the tablet screen; repeat the walkthrough with Q2 (does the user notice the action) asked at each step.	ER-03 secondary hypothesis (first-time usability); no Claim ID yet
KF-02	The prototype asked many personal questions to build the profile and rarely got to suggesting a match. The peers wanted matching to come sooner.	Suggest a first match after a few released items; in the specification the profile comes from the coordinator's life-story interview (UC01 step 2), not from a long robot conversation.	UC02-F3-CL1
KF-03	The peers found Pepper too large and clunky, especially for going on a walk with other residents. They preferred a small device worn on clothing,		

	which could also notice when something happens to the wearer.	Record a small wearable or handheld device as a Technology alternative, with the surveillance concern of TECH-09 as its main drawback.	Technology choice (TECH-01, TECH-09); UC02-F3-CL1 (walks)
KF-04	Unexpected, about the demo: the lecturer noted that the demo should explain that only the left side of the screen is what a resident would see; the right side is the operator panel.	Show only the resident's side to evaluators, or state this before the walk-through starts.	none (no Evidence)

Implications of ER-04

- **For Build 2 and the specification.** KF-02: Build 2 suggests a first match once a few items are released, in line with UC01 step 2, where the coordinator builds the profile with the resident and family.
- **For ER-03.** KF-02 also bears on ER-03's schedule, which spends week 1 on one-to-one profile sessions before any matching in week 2. If residents lose interest during long profile building as the peers did, week 1 should be shortened or matching offered within it.
- **For the Foundation.** KF-03: add a small wearable or handheld device to the Technology Options as a considered alternative to Pepper, and weigh it against VT-06 and the reasons TECH-09 was rejected.
- **For the method.** KF-04 and the lack of notes: in the repeat walkthrough, explain the screen layout first, use the step table below, and have a second group member write down the answer to each question.

Protocol for the repeat walkthrough

The steps to walk in the repeat pass, each with its persona and the Claims it bears on, on the narrowed UC01 to UC03. Steps that only the System or the coordinator perform (UC01 3, UC02 1 and 2) are not walked, nor are the record steps (UC01 8, UC02 10, UC03 7), whose only Claim (UC02-F6-CL21) is about how residents feel about the record once it is explained, which the EM-04 interviews address.

Use Case step	Persona	Claims
UC01 1 Greeted by name on the first lounge visit	Elena	CL5
UC01 2 Decides on each profile item with the coordinator	Elena	CL4
UC02 3 Asked alone whether to meet Jan, and whether an item may be shared	Elena	CL3, CL4
UC02 3 Asked alone whether to meet Elena	Jan	CL3, CL4
UC02 3-alt Says no; later asks about it	Jan	CL3
UC02 4 Brings both to the lounge at a time that suits both	Myra	CL1
UC02 5 Introduced by name and photo, with one shared item	Elena and Jan	CL1
UC02 6 Talk stalls; Pepper offers one topic	Elena and Jan	CL1
UC02 7 Pepper suggests an activity for two	Elena and Jan	CL1
UC02 8 Wants to leave	Jan	CL3
UC02 9 Asked alone whether to meet again	Elena	CL3
UC01 6 Names given as statements at a later meeting	Elena	CL5
UC01 7 Asks "When can I go home?"	Elena	CL8
UC01 7 Receives the deferral note as trusted person	Myra	CL8
UC03 1 and 2 Receives the note at the agreed time and brings both together	Myra	CL6
UC03 3 Names given at the follow-up meeting	Jan	CL5
UC03 5 Receives a presence note	Myra	CL7
UC03 6 Visited by Myra after a presence note	Jan	CL7

3.3 Evidence

EV-08 Walkthrough on getting to an introduction

Bears on UC02-F3-CL1

Source ER-04 (Peer walkthrough of the Kinnect prototype, 13 October 2026)

Assessed against criteria	Not assessed. CL1's criterion needs M-02 and M-05 from residents in two homes.
Result observed	The peers said the prototype asked many personal questions to build the profile and rarely got to suggesting a match (KF-02). They also found Pepper too clunky to go on a walk with other residents (KF-03).
Verdict	Qualifies
Basis for verdict	Reviewer judgement from verbal feedback. CL1 depends on introductions actually happening. In the prototype the long profile conversation delayed them, and the peers did not want Pepper along on a walk, which CL1 counts among the time residents spend together.
Limitations	Two peers, one session, no notes taken (ER-04 Method). The session walked the earlier walking version of Build 1, whose robot-led profile conversation the specification replaces with the coordinator's interview (UC01 step 2), so KF-02 may not hold for Build 2. The peers did not know which part of the screen a resident would see (KF-04).

3.4 Verification and Validation Summary

Where each Kinnect Premise and Claim stands. "Planned" means the Evaluation Method is designed but has not produced Evidence.

Premise or Claim	Planned method	Report so far	Evidence so far	Verdict so far
UC02-F1-PR1	EM-01	none (needs Build 2)	none	none
UC02-F2-PR2	EM-01	none (needs Build 2)	none	none
UC02-F3-PR3	EM-01	none (needs Build 2)	none	none
UC01-F4-PR4	EM-01	none (needs Build 2)	none	none
UC03-F5-PR5	EM-01	none (needs Build 2)	none	none
UC02-F6-PR6	EM-01	none (needs Build 2)	none	none
UC02-F3-CL1	EM-03; EM-08 (closest Claim to ER-03's primary hypothesis); ER-04 walkthrough	ER-04; ER-03 designed	EV-08	Qualifies (walkthrough only)
UC01-F3-CL2	EM-03	none	none	none
UC02-F1-CL3 (adverse)	EM-03, EM-04; repeat of ER-04 planned	none	none	none
UC02-F2-CL4 (adverse)	EM-04; repeat of ER-04 planned	none	none	none
UC01-F3-CL5 (adverse)	EM-03, EM-04; repeat of ER-04 planned	none	none	none
UC03-F5-CL6	EM-03; repeat of ER-04 planned	none	none	none
UC03-F5-CL7 (adverse)	EM-04; repeat of ER-04 planned	none	none	none
UC01-F4-CL8	EM-03, EM-04; repeat of ER-04 planned	none	none	none
UC02-F3-CL9	EM-03	none	none	none
UC02-F6-CL21 (adverse)	EM-04	none	none	none
ER-03 secondary hypothesis (first-time usability)	EM-08	ER-03 designed	none	none (no Claim ID yet; see *Open items*)

Only UC02-F3-CL1 has Evidence so far, from the ER-04 peer walkthrough of 13 October, which recorded no step-by-step answers. The repeat of ER-04 is planned for the other Claims marked for it. No Premise is verified yet, because EM-01 needs Build 2.

3.5 Discussion, Limitations and Next Iteration

3.5.1 Interpretation of the Walkthroughs

Morning care (ER-01)

ER-01 walked UC04 as Mary and Myra. The prompting ladder itself held up: at SA-05 steps 4 to 7, where Mary has started and faces the cue, the wait and the L1 and L2 prompts gave credible success stories, and so did the early-stop offer (EV-01, EV-06). The failures sit at the edges of the ladder. Before it starts, a resident who stands at the sink without lifting anything gets no support (WF-01). When it ends in a handover, Mary hears nothing for up to 10 minutes (WF-06), and Myra could not tell which room to go to (WF-07) or how to acknowledge (WF-08). After the session, the record does not tell Myra at which step Mary needed help (WF-09). In short, UC04 specifies the prompting well and the transitions into and out of it poorly.

This shapes ER-02. A rise in distress in B (UC04-F8-CL13) could come from the wait W, which the Claim names, or from these transitions. ER-02 therefore codes M-14 per step and logs which ladder rule fired, so that a rise can be traced to its step. UC04-F9-CL15 was challenged on the Spec-1 handover; ER-02 tests it on the revised steps 9 and 10, which first need walking again.

Kinnect (ER-04)

The peers' main point was that the prototype took too long to get to an introduction: it asked many personal questions and rarely suggested a match (KF-02). UC02-F3-CL1 rests on introductions happening, so this qualifies it (EV-08). The specification already places profile building with the coordinator (UC01 step 2), so the finding mostly concerns Build 1, which let the robot build the profile in conversation. It also concerns ER-03, whose intervention spends week 1 on one-to-one profile sessions before any matching. The second point was the robot itself: Pepper felt too clunky to take on a walk with others, and the peers preferred a small worn device that could also notice incidents (KF-03). That questions the choice of Pepper for the walking part of UC02 and asks for an alternative the Foundation does not yet compare. A worn device that notices incidents is close to the wearable location sensing that TECH-09 rejects as surveillance, so the peers' preference and VT-06 pull in opposite directions.

3.5.2 Feedback into the Foundation, the Specification and the Study Plan

Finding	Foundation	Specification	Study plan and verification
WF-01 no support before the first lift	none	O-35; proposed UC04-F7-PR12 (below)	ER-02 codes M-14 per step
WF-02, WF-04, WF-06 prompt wording, picture placement, silence after a call	none	O-35 (prompt script)	ER-02 traces distress to its step
WF-03, WF-05 hearing "no" and "help"	none	O-36; proposed UC04-F8-PR11 (below)	Scripted case in the EM-05 pilot (O-29)
WF-07 room missing from the handover	none	Fixed in F9, UC04-F9-PR9, DS-02 and TDP-02 (O-34)	Walk steps 9 and 10 again
WF-08 no acknowledgement action	none	O-37	none
WF-09 record too thin for Myra	VT-08: record the detail of the session record as a considered alternative	O-38	EM-07 judges sufficiency (UC04-F10-CL17)
WF-10 rinse or pause unclear	none	none until the pilot settles it	Scripted case in the EM-05 pilot (O-29)
WF-11 well-timed prompts reveal the tracking	none	O-35 (what residents are told at consent); UC04-F7-CL16 revised (below)	M-16 code W in ER-02
WF-12 no closing line (unexpected)	none	O-35; proposed UC04-F8-PR13 (below)	none
KF-01 interface unclear (Kinnect)	none	none	Repeat ER-04 with Q2 asked at each step; bears on

			ER-03's usability hypothesis
KF-02 slow to reach a match (Kinnect)	none	Build 2 suggests a first match after a few released items, in line with UC01 step 2	ER-03: shorten week 1 or offer matching within it
KF-03 Pepper too clunky; worn device preferred (Kinnect)	Add a worn or handheld device to the Technology Options, weighed against TECH-09 and VT-06	none	none
KF-04 resident's part of the screen unclear (Kinnect, unexpected)	none	none	Explain the screen lay-out before the repeat walkthrough

3.5.3 New Premises and Claims from the Findings

These are proposed for UC04 and are added to the Specification once the group decides O-35 and O-36.

- **UC04-F8-PR11 (from WF-05).** The System recognises the fixed words "yes", "no", "stop" and "help", and treats any other speech as no reply. Threshold set from the EM-05 pilot.
- **UC04-F7-PR12 (from WF-01).** A resident's arrival at the sink is detected without a camera, so that T-stall can start on arrival. Depends on O-30.
- **UC04-F8-PR13 (from WF-12).** Every completed session ends with a closing line in an ordinary adult register, without praise.
- **UC04-F7-CL16, revised (from WF-11).** The Claim now covers the timing of prompts as well as the sensors: a prompt that arrives just as the resident stops can make the tracking evident even with the cameras covered.

3.5.4 Limitations

The morning-care walkthrough (ER-01). One evaluator, without care experience, made one pass, where the procedure asks for 3 to 5 evaluators and a joint session (O-39). The evaluator is not a person with dementia, and HP-04 is provisional. A paper build has no timing, so the walkthrough cannot speak to the Premises. It also cannot say how often a failure would happen or how bad it would be for a resident.

The Kinnect walkthrough (ER-04). Two peers in one session, and nothing was written down during it; the findings are what two group members reported afterwards. The facilitator found the procedure unclear on the day, so the four questions were not asked step by step. The session walked the earlier walking version of Build 1, and the peers did not know which part of the screen a resident would see. EV-08 therefore rests on reviewer judgement from verbal feedback.

The morning-care study (ER-02). With 8 residents, a confirmatory test succeeds only if almost all residents move in the predicted direction, so the single-case plots carry as much weight as the tests. Observers cannot be blind to the phase, M-13 and the tooth-brushing version of M-15 are untested scales, and the study runs in one care home. Whether it falls under the Dutch WMO is still open (O-40).

The Kinnect study (ER-03). With one home per arm, a difference between the arms cannot be separated from other differences between the homes. The badges it needs conflict with the rejection of wearable location sensing (TECH-09). Pepper+ does not exist yet. The two hypotheses do not yet map to Claim IDs, and the Premises they rest on, such as speech recognition with older speakers, have no verification method yet.

3.5.5 Next Iteration

1. Repeat ER-04 on the narrowed UC01 to UC03, with the screen layout explained first, the step table as protocol and a second group member recording each answer; write Evidence for each walked Claim.
2. Repeat ER-01 with 3 to 5 evaluators, at least one with care experience, on the revised steps 9 and 10 (O-39).
3. Decide O-35 to O-38, and add the accepted Premises and the revised CL16 to UC04.
4. Make Build A and run the EM-05 pilot, with the WF-05 and WF-10 cases scripted (O-29); confirm the PR7 to PR9 thresholds.
5. Ask the HREC whether ER-02 needs METC review (O-40).

6. For Kinnect: map ER-03's hypotheses to Claim IDs, set the O-2 thresholds, and build and verify Build 2 under EM-01 before ER-03 can run.

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Appendix A

Student Reflections

A.1 Diana Banţă

I decided to take this course after my friend told me about it, and I found the concept of designing technology to aid people with dementia really intriguing. In my Computer Science studies I often times feel like the focus is on rather abstract concepts and applications. I am used to highly analytical courses that present algorithms or methods and how to apply them on similar, specific problems. So, in that sense, having the opportunity to do a course that focused on something rather tangible, with so much flexibility and freedom in choosing what we want to create, felt refreshing and exciting. Of course, it was not the first course I took that focused more on the HCI part of computer science, but I really enjoyed the concept behind it.

As for the SCE process itself, I have to admit that it was really something new for me. It felt structured and organized, and every section had its own role, place, and categorization. There were concepts I was familiar with, such as the Stakeholder, Problem Scenario, Technology, and Evaluation. However, I had never before really worked through how they are all connected together, what influences what, and I had never written that down in a guided, structural way as the XWiki format requires. I also felt like the sensitive nature of our product, which aimed to help a rather vulnerable group of people, made the motivation behind this level of structure very clear. It justified having so many detailed sections and specifications for every decision and design step.

Something that I learned that I found very important and interesting, but have never considered in depth before, was Value Tensions. How we want to create a product that fulfills a certain role or goal while also finding some balance with possible conflicts and implications. I also realized that every design decision, step, and consideration should be rooted in either literature or experience. Even though now this feels very self-explanatory and logical, I just had never thought about it that much before.

One thing I have noticed is that, since there are so many little sections, subsections, and instances that we have written for the XWiki, it can become a little difficult to see everything in the grand scheme. And then I realized that, while it may not be the first place I would look at when reading through a report like the one we are submitting, the Graph Edges specifications really offer a meaningful overview of how everything influences everything and how every single part is meaningful within the overall system.

Overall, I felt like this course offered a very solid structure and methodology for designing our system while also allowing a very large amount of creativity and freedom in our choices, decisions, and overall process.

Throughout the course I tried to join every lecture. I especially enjoyed the lecture on Memory, as it showed insights of how our brains work and how that can be mapped to methods and technologies to help with fulfilling specific goals and needs. I also liked learning about what a Cognitive Walkthrough is, and I enjoyed the entire Evaluation lecture and overall process.

While it was a lot of work, it helped to have a very well defined structure and list of expectations for each deliverable and each step of the process. As for my personal contribution, I wrote the Design Scenario, and Personas and Profiles sections for the Specification part in the first half of the course. I was quite active in determining how we should organize ourselves as a team, with meetings, tasks distribution, and general workflow. I realized that I am at my best when there is a concrete action plan and when it is crystal clear what everyone is doing, and that I like ensuring that. I was also involved with making the slides and presenting the midterm presentation, and later on helped with feedback and proofreading the report, the video, and the final presentation. For the second part of the course, I have to admit that I could have been more involved, and that I had not realized how much time my jobs, other courses, and extracurricular commitments would take. I still helped with the evaluation part, during the lab where we tested our prototypes with other teams, and I tried to give input and help whenever asked.

If I could go back and start over, I would probably try to balance the workload better, so that I can have a consistent level of involvement throughout the course.

The teamwork itself went smoothly for the most part, we were able to communicate and find compromises when our schedules or meeting styles did not align. Another thing that happened that we successfully navigated - I would not explicitly call it a tension, but more like a moment of imbalance - was when we decided to completely switch our project goal and purpose after the midterm presentation. We got over the initial differences in opinions regarding this decision, and acted together to put our ideas into application. So for all of that I am very grateful for my team! It was really lovely working together with them.

A.2 Roham Koohestani

Course content. I found this course a lot harder than I expected. I assumed a design course would ask less of me than the technical courses in my programme, and underestimated how much discipline the SCE method asks for. Every Function traces back to an Objective, every Objective to the Foundation, and every Claim to a Human Factors concept, a Measure and an Evaluation Method, all linked by IDs, so a change in one place spreads through the wiki. When our group moved the concept to introducing residents to each other, almost every page from the situated activities to the use cases had to be rewritten, which took far more time than writing them the first time.

The distinction between Premises and Claims also took me a while. A Premise is about what the System does and can be checked on the build without residents: does Pepper ever pass on a "no", does any packet leave the home. A Claim is about the effect a Function has on people, and only a study with those people can support or challenge it. Once that clicked, it was clear which questions a prototype test could answer and which only a summative study could.

The human-centred parts of the method were the ones I enjoyed most. Starting from situated activities and from what residents, relatives and staff value, before choosing any technology, made the design discussion more grounded than I am used to from software projects. The value tensions made the trade-offs concrete: matching two residents on their life stories is what makes an introduction work, and also what exposes their private history (VT-01), and a resident with dementia asking "When can I go home?" puts truthfulness against well-being (VT-03).

Working for a vulnerable user group made almost every step more complicated. Residents with dementia cannot simply be put in front of a prototype, so our formative walkthrough uses peers without dementia and can only give early signals. Consent needs a legal representative as well as the resident's own assent, and any sign of dissent ends the interaction. Some evidence does not transfer cleanly: the finding that refusing face to face is hard (HFC-17, Bohns 2016) comes from healthy adults, yet our rule to ask each resident in private depends on it. Measurement is harder too: self-report from residents has ceiling effects, and the location badges ER-03 relies on sit uneasily with our own rejection of wearable tracking (TECH-09).

Project contribution. Early on I wrote the wiki sync tool (`wiki-sync/sync.py`), which lets the group edit the wiki locally under git and push through the REST API without overwriting each other's edits. For the morning-care line I wrote the Specification: Objectives OBJ-01 to OBJ-05, Functions F1 to F4, and a tooth-brushing use case with four Premises and eight Claims, two of them adverse. I also replaced unsourced evidence claims on HFC-01 to HFC-03 with citations and drafted the Measures and Evaluation Methods those Claims refer to.

Later I added the evaluation pages and the Foundation Coherence page, and wrote ER-03, the two-site summative study of Kinnect, with its method (EM-08) and Measures M-19 to M-23. I built the presentation and rebuilt it around Elena's first week once the concept had settled. For the final submission I checked every slide against the wiki, corrected IDs and study details that had drifted, replaced the formative findings with the plan for the walkthrough, and wrote the PDF export.

Lessons learned. The main lesson for me is how useful the traceability is, even when it feels slow. When I checked the final slides against the wiki, the IDs made the mistakes easy to find: a problem scenario under the wrong number, verdict rules that did not match ER-03, a design rule shown with "0 cloud packets" as if it had been measured, and a gap between our design and our artifact. Without the shared IDs, those would have gone into the submission.

I also learned to be explicit about what has not been done. Our walkthrough was not run and ER-03 has no results, and saying so plainly was more defensible than filling the gaps with plausible numbers. Finally, during the presentation we were asked for one example of how a Claim relates to a Function and could not ans-

wer on the spot. Working it out afterwards made it concrete for me: Claim CL1 is the reason Function F3 (introducing two residents on something they share) exists, and the adverse Claim CL3 would change Function F1 (the private ask) if residents turn out to accept out of politeness. A positive Claim justifies a Function and an adverse one constrains it.

A.3 Gints Kuļikovskis

I worked on the very first Situated Activities, helped map initial stakeholders, and contributed to the rest of the Foundation part as well as the early Specification. The Situated Activity of brushing your teeth started as a stand-in or a literary device when discussing our project, and was filled out as one of the first drafts. Due to an absence of better ideas, we went along with it. The midterm presentation became a wake up call, after which it became even more clear that the problem to be solved, that is, losing track while brushing your teeth, is for all intents and purposes not a problem that lends itself to a technological solution. It is a very repetitive workflow that you have been doing twice every single day of your life starting at the age of 0, when your parents did it for you. When your motor skills become unreliable, you need someone to help you again. However, when the basic sense of where you are and what you are doing - leaves while you are brushing your teeth, you are likely to find it back after just a quick glance in the mirror or through feeling the toothbrush in your mouth. We therefore consider this problem not a great candidate for Socio-Cognitive Engineering.

Earlier this year I designed an operating system for the first smartphone that my grandmother ever used. This resulted in positive results; her old phone with a T9 keypad had become too difficult to navigate, even for starting a phone call. A modern smartphone OS contains countless obstacles to a person with regressive cognitive decline. These are obstacles that prevent you from staying in contact with your loved ones. Humans are social creatures, and if we are lonely we suffer. Yet even when you have people all around you, you can still be lonely. I started developing a parallel version of our xWiki around introducing residents of a care home to one another. I worked on it alone until a point where it could be pitched to my teammates as a new, complete concept which would allow us to tackle more diverse problems involving human factors and needs. This involved establishing a skeleton of the full SCE cycle. Since many of my teammates were working on the project remotely, I created a video to help sell the new idea to my teammates. After an initial green light from the team and from the professors, we switched gears and worked on polishing the new concept. The originally written xWiki articles were kept as well.

Note to reader: this paragraph is an unformatted list of contributions. Some other parts I was also involved in include xWiki-preview, a VSCodium plugin that I put together for rendering xWiki articles locally. I assisted Roham with the creation of xwiki-sync, our script that enables us to collaborate on the xWiki contents using git. I cleaned up the paths of our wiki ("sdf" becomes "1 Foundations"), and collaborated with my teammates on the design of the summative evaluation, the experiment we would carry out in the ideal world. I designed both the old and the new prototypes for the cognitive evaluation. I represented my group in the remote feedback session on 13 October. I created and presented a part of the final presentation. I created the cognitive walkthrough video for the presentation and the Brightspace submission.

Collaboration with the team has been overall positive. However, our work would have been more efficient if we had met more in person (instead of part-online). Regarding our switch of concept, this allowed me to explore the SCE framework in much greater detail.

For ethical considerations, students did not work with any care home patients for this course. Still, planning the envisioned Summative Evaluation involved a lot of careful consideration were such a trial actually take place. Regarding generative AI, an LLM has been a very useful tool for generating python scripts for fixing article numbering, for rendering xwiki articles locally, and for other technical/formatting fixes including the report; the wiki content that I authored is all my own.

Most of all, I learned from and enjoyed the lecture on cognition. In general, the concepts covered in this course are extremely relevant if you want to create a good user experience for everyone (accessibility). It has been helpful to think about why we make certain decisions, and to read about the research that has been carried out in the fields related to SCE. I have to admit that the countless tables that we have had to fill out still seem like an obstacle that still needs to be smoothed out. *Imagine: what if we simply had a PostgreSQL schema?* Then instead of endless copy-pasting you could just insert a new row, in other words, a new entity into the appropriate table, and voila, a new situated activity has been created. You would not

have to worry about IDs getting mismatched either (foreign keys take care of this). It is not difficult to flatten such a database into a linear report either. I hope this idea at least gets considered for a future iteration of this course.

A.4 Antoni Nowakowski

Course content. For me the biggest challenge in this course was creating the mental map that connects every concept that needs to be well thought out in the final design. While individually the terms like Human Value, Value Tension, Premise, Claim or Measure are simple enough to understand, once they are put together into this spider-web of terms that each shape how the other is formed for me it became a quite difficult landscape to wrap my head around. It was genuinely a great exercise as in software projects I normally start from what the system has to do. SCE starts with the situated activity, and module 1a explained why: cognition is embodied and embedded, so it cannot be understood apart from the place, the people and the routine it happens in. For our project this meant first describing what it is like to move into a care home, to greet someone new, and to sit in your room waiting for a reason to leave it (SA-01 to SA-04). Several of our later design rules came from those descriptions, not from what Pepper can do.

The memory lecture was the most useful for our concept. One idea that stuck with me is that you should support what people use their memory for, not try to improve their memory. Kinnect does not train residents to remember each other; it uses reminiscence to help two people find something in common. The lecture also explained why topics from early adulthood, like a trade or a hometown, are easiest to recall (HFC-04), and why asking "Do you remember...?" can embarrass someone when they cannot recall it.

From the evaluation lecture I learned that a use case is more than documentation. Its action sequence can be used directly as the script for a cognitive walkthrough. I also saw its limit for our project: the evaluator has to imagine being the persona, which is hard when that persona is an 81-year-old with dementia.

The ethics lecture made me look at the whole care situation instead of only the robot. Good care depends on residents, staff and the robot together. This is why in our design the activity coordinator approves every match and painful questions go to a trusted person (IDP-03). It also made me notice our own assumptions, for example that every resident can hear Pepper in a noisy lounge.

Project contribution. Most of our design decisions were made together in team meetings, and I took an active part in that ideation, also for parts of the wiki that were later written up by other team members. I wrote the Team Design Pattern and the Interaction Design Patterns for both of our concepts. For Kinnect this is TDP-01 (consent-gated social brokering with human handover) and IDP-01 to IDP-03 (introduce then step back, ask each person in private, defer sensitive questions to a trusted person); I also wrote the patterns for the morning-care concept. Besides that I did maintenance work on the Foundation: I updated the value tensions, reworded the situated activities, and renumbered the problem scenarios, which meant fixing every page that referred to the old IDs. I also took part in creating the summative study for Kinnect (ER-03): the two-home comparison was worked out together in a team meeting and a teammate then wrote it up on the wiki. I prepared and gave the final presentation.

In my opinion our collaboration went smoothly throughout the project. We met during the weekly working-group sessions and kept all our work in a shared Git repository that synced with the wiki, so everyone could see what the others were writing and build on it. Work was divided by section and by concept, which meant we rarely got in each other's way, even while we were developing the morning-care idea and Kinnect side by side. Serious disagreements were rare. When we did see things differently, for example about which concept to take forward or how a page should be worded, we talked it through as a group and agreed on a direction without much trouble. Splitting the presentations was also easy: one half of the team gave the interim presentation, and the other half, including me, gave the final one.

Lessons learned. The main lesson for me was learning to work in such a structured way. At first the fixed formats and IDs felt like paperwork, but they are what let five people work on different parts and still end up with one consistent design. Renumbering the problem scenarios showed me how connected everything is: one small change meant updating many pages.

Writing the design patterns taught me to think more abstractly, taking a solution that repeats across use cases and describing it so it can be reused elsewhere, while still linking it to the Claims it supports.

Planning and presenting the summative study showed me that evaluation means deciding in advance what counts as success and what a result can and cannot tell you.

If I started again, I would take ownership of a larger section of the wiki early on, instead of mainly working on patterns and maintenance later, and I would start preparing the presentation sooner.

A.5 Rayan Salmi

Understanding the SCE process. During the project, I was mainly involved in the foundation and specification phases. Initially, our team explored the idea of helping people with dementia brush their teeth, as they can forget when or how to perform daily hygiene activities. During the foundation phase, I contributed to identifying relevant human values and considering possible tensions between them. This helped us think about the needs and experiences of the people who would use our system.

However, I found it challenging to translate the initial problem into concrete specifications. Although we understood that people with dementia can experience difficulties with remembering and performing everyday activities, it was not always clear what exactly our technology should do or how it should address the problem. I experienced similar difficulties when trying to connect our specifications and use cases to the problem scenarios and the values we had identified.

Eventually, our team decided to change direction and focus on creating meaningful connections between residents in a dementia care home. I was particularly involved in brainstorming this new direction together with Gints. We explored how residents could be matched with each other for shared activities, such as going for walks, and how technology could facilitate these interactions. I also helped think about and set up the situated activities and explore which technologies could support our concept. This included looking into the Pepper robot as a possible way to initiate interactions and connect residents. This concept seemed to address a more meaningful social need and gave us a clearer direction for the project.

Learning and critical self-assessment. One of the biggest challenges I experienced was moving from a broad problem to a clear and concrete design. I often struggled with deciding what we should actually build, how the different parts of the specification connected to the original problem, and which technologies would be appropriate. I also realised that identifying human values is not enough on its own, but it is also important to understand how those values influence the design decisions and how potential value tensions can be addressed.

I particularly enjoyed the lecture on memory support technology. It introduced different perspectives on human memory and its role in everyday life, including its social functions. This gave our team new ideas and helped us start brainstorming towards our final concept. I found it interesting that technology does not necessarily have to compensate for memory difficulties but can also support meaningful human experiences, such as social interaction, maintaining relationships and strengthening a sense of identity. This perspective helped me understand why facilitating connections between residents could be a valuable direction for our project.

Another difficulty was working with the different templates throughout the SCE process. At times, I felt that we were trying to complete the templates without having a sufficiently clear idea of what we wanted to design. This sometimes made the process feel repetitive and left us unsure about how to proceed. Looking back, I think we could have benefited from spending more time discussing and refining the core concept before moving into detailed specifications. In a future project, I would try to establish a clearer understanding of the problem, the intended users and realistic usage scenarios first. This would make it easier to use the SCE methods and templates in a meaningful way rather than treating them as separate tasks to complete.

Team contribution and collaboration. My main contribution to the project was working on the foundations and specifications, particularly identifying human values and possible value tensions. I also contributed to the overall brainstorming process, especially when we were developing our second idea around meaningful connections. Together with Gints, I helped explore the initial concept and think about how residents could be brought together through shared activities. In addition, I helped set up the situated activities and investigate possible technologies for the concept, including the Pepper robot. I conducted user testing with our built prototype, which yielded valuable feedback on the user experience and the specific technologies used. Finally, I presented our midterm presentation, in which we presented our initial idea about supporting people with dementia during toothbrushing.

Team collaboration was generally constructive, although we encountered some practical difficulties. One issue was using XWiki for collaborative editing. I found the platform rather clunky, especially when several team members were working on the same documents. To make collaboration easier, our team developed a separate plugin and a Git synchronisation script. This allowed us to work on the documents more comfortably and reduced the risk of overwriting each other's contributions.

Another challenge was that we sometimes struggled to agree on a clear project direction. Working through the templates without a shared understanding of the concept occasionally left the team stuck. Eventually, we recognised that our initial toothbrushing idea was not developing as well as we had hoped and decided to explore a different problem. I think this change demonstrated the importance of being open to feedback, discussing difficulties honestly and being willing to reconsider earlier decisions instead of continuing with an idea simply because we had already invested time in it.

Overall, I contributed to developing the project, exploring its different directions, reflecting on its design and improving our collaboration. The experience taught me that the SCE process is not always straightforward or linear. It requires continuous discussion, reflection and adaptation, as well as a clear connection between the problem, the users' needs, the identified values and the proposed technology.